Spin labelling via metabolic glycoengineering for studying post-translational modification by electron paramagnetic resonance spectroscopy

Dissertation zur Erlangung des

Akademischen Grades eines

Doktors der Naturwissenschaften (Dr. rer. nat.)

vorgelegt von

Anna Rubailo

an der



Mathematisch-Naturwissenschaftliche Sektion

Fachbereich Chemie

Konstanz, 2023

Konstanzer Online-Publikations-System (KOPS)
URL: http://nbn-resolving.de/urn:nbn:de:bsz:352-2-1jj0uhjrj72xw0

Tag der mündlichen Prüfung: 15.09.2023

1. Referent: Prof. Dr. Malte Drescher
2. Referentin: Prof. Dr. Karin Hauser

“À la guerre comme à la guerre”

A good French proverb that accompanied me

in the end of my PhD

Acknowledgements

I am very grateful to my supervisor Prof. Dr. Malte Drescher to accept me as a PhD student to his research group and for giving me such an interesting and perspective project. The beginning of my PhD was different from other students in our group due to the absence of my biological skills. I am very thankful that you trusted me and gave a time to learn new methods and technique required for my project. Dear Malte, thanks a lot for your support with the questions concerning both scientific aspects and non-scientific life problems solution finding. Unfortunately, all my term of the PhD study accompanied by various kinds of difficulties… like a COVID pandemic and the biggest problem was the war in my country. Regarding to the current situation in Ukraine, I fully recognize how crucial for my future career will be the contribution of my PhD experience. Till the end of life, I will be thankful that you accepted me to your group and gave me a brilliant ticket not just to increase my knowledge but also to have better quality of my future and my family‘s life. Especially grateful for unbelievable trip to Marseille for the EPR summer school, the 52nd Annual Meeting of the ESR spectroscopy group of the Royal Society of Chemistry in Glasgow, trip to our collaboration partner of the Cuernavaca University in Mexico.

I acknowledge Prof. Dr. Karin Hauser to be the second reviewer of my PhD thesis and to Prof. Dr. Valentin Wittmann to make the corrections of our manuscript and helpful discussions and my project. I acknowledge Prof. Dr. Karin Hauser and Prof. Dr. Christoph Hauk to be a part of the thesis committee and for your brilliant advices concerning the strategy of the project during the term of my PhD study.

I am strongly believing that everything in this world happened for some reason like our meet with Dmytro Sysoiev in the hall of the Institute for Single Crystals National Academy of Sciences of Ukraine in Kharkiv. He advised me to send my CV to the University of Konstanz for a PhD position.

Thanks to the German Academic Exchange Service (DAAD) for the grant that supported me during first four years. Warmest thanks to my curator Elena Schmid from DAAD. My sincere gratitude to Konstanz Research Chemical Biology School and to the whole EPR spectroscopy group of Konstanz University for the warm, friendly, and very motivated atmosphere. Thanks for their support and help during the most difficult period of my project, retreats and organisation of conferences. I especially want to thank Dr. Mykhailo Azarkh for his

professional instructions and assistance in the use of spectrometers, helping to understand EPR spectroscopy theory and maintaining a friendly atmosphere in the AG Drescher. My work in the lab was made much easier when Martina joined the group. She was like a laboratory fairy who opened up to me the world of biology and saved me in the most critical situations in the biolab.

Special thanks to Lina Rivillas-Acevedo, her husband Rodrigo, and the whole Lina's research group in Cuernavaca. I will never forget our time together in Konstanz. The Christmas protein expression in 2022 in Mexico for a long time will be in my sweet memories.

PhD project implementation was a big challenge in my biography and has been complicated by events in Ukraine in 2022. I express my special gratitude to Dr. Andreas Marquardt and Dr. Anna Sladewska- Marquardt from proteomic centre for their nor just scientific but also to help me with my private difficulties.

Special thanks to Sonja Tischlik for her permanent moral support and participation in the meeting gains Russian aggression in Ukraine, to Anandi Kugele for our night trip to Strasburg to pick up refugee from Ukraine. Thanks to Dr. Annalisa for the support in my crises emotional moments and an amazing soup when I had COVID. I will never forget our group trip to Italy and miracle sweet garden full of cakes at Annalisa's wedding.

I am grateful to Alice Kornilovska for her creative imagination and inspiration. Thanks to Johannes Mause for the private training and sport motivation at the gym. I have enjoyed communication with Mariia Molchanova throughout my time at the University of Konstanz. Our conversations have always been motivating and supportive at every stage of our academic journey.

When I did't know how to move part of my family and how to locate them in Konstanz the queue of questions appeared: "How to move on and if to move on - how? Where to start when it is already over? How to finish what has already been started? True lies or false truths? How to make sense of the mess? How to get out of the mess?".

I didn't know how to give the answers on that life's questions in the most difficult part of my Ph.D when the war started. I am grateful to Konstantin Afanasiev, Peter Truschkow, Sebastian Höfel, Gesa and Hauke Hansen family, Hanna Prater and Sebastian Vettel who unbelievably helped me to find the solution on all that questions. I am extremely thankful to Olga Andreeva, Gonzalo Milosevich Fernandez and family of Birgitt and Detlef Repplinger

that helped me with adaptation of my mother in Konstanz. Particular gratitude for the emotional support from Prof. Dr. Oksana Zhukova.

From the bottom of my heart I would like to thanks my parents. They gave me the most precious things: their absolute love, great memories from the childhood, education and understanding that in this world you always need to remain “human”.

I know, it was not the easiest time for my parents to be a long time on distance during my Master study in France and my long term of PhD study in Germany. Especially, when we were extremely limited to see each other, during COVID time and when the war in Ukraine started. My biggest dream would be to see my healthy family all together under safe, enjoyable circumstances how it was before the war started.

It is impossible for me to describe all the pain that I experienced during the war in Ukraine. Was difficult to realize that dearest people in this world, my parents, are under bombardment in my lovely Kharkiv city. But even in these moments, you tried to hide many troubles to not hurt me.

Summary

The biological functions of protein are responsible for all cellular tasks and are intimately dependent on their conformational dynamics. The proper folding of proteins determines the correct fashion of its interaction with other molecules and myriad of vital cellular processes in biological systems. Proteins fold up into specific shapes according to the sequence of amino acids in the polypeptide chain. If the specific 3D structure is disrupted, for which the protein is said to be denatured, the protein loses its activity. In a word, the protein function is directly dependent on the protein structure, that cannot be carried out once the structure is disturbed. The elucidation of the basic cellular processes is important and required the complete understanding of protein interactions, including protein's post-translational modifications (PTMs).

PTMs are modifications of protein's canonical amino-acids site chains with different functional groups, such as phosphate or sugar derivatives and several others. Post-translational modifications are involved in many cellular processes. Any kind of PTMs affect protein's structure, function, stability, and localization. In many cases, their origin boils down to protein misfolding. Malfunctions in these lead to manifold disease patterns such as cancer or neurodegenerative diseases. The addition of O-linked β-N-acetylglucosamine (O-GlcNAc) is a ubiquitous post-translational modification, essential for mammalian cells. O-GlcNAcylation is a reversible serine/threonine sites glycosylation for regulating protein activity and availability inside cells. O-GlcNAcylation regulates plenty of physiological and pathological processes. However, understanding the diverse functions of O-GlcNAcylation is often challenging due to the difficulty of detecting and quantifying the modification. Thus, robust methods to study O-GlcNAcylation are crucial to elucidate its key roles in the regulation of individual proteins, complex cellular processes, and disease. The crucial techniques for PTMs study are mass spectrometry, immunoprecipitation, western blotting, fluorescent microscopy. The marking of the PTM proteins plays critical role since PTM is the secondary gene product and cannot be expressed with fluorescent label (protein tag) in cells. Many biochemical approaches were used for the fluorescence labelling strategy and PTM detection by microscopy. Due to limited number of glycosylation sites, just a few methods are available for the detection of post-translationally glycosylated states of the particular proteins.

The present thesis deals with the development and application of EPR spectroscopy for the detection and quantitative study of the protein O-linked glycosylation modification. In that way, chemically stable paramagnetic species can be attached to the site of interest, in our case it is O-glycosylated site of the protein. To achieve that goal, spin labelling specific for glycosylation sites in combination with EPR spectroscopy was utilized to probe glycoprotein structural changes under O-glycosylation.

Metabolic glycosaccharide engineering (MGE) approach was used to investigate the sugar moieties with corresponding chemical reporters prone for further spin labelling. O-GlcNAcylated proteins in cooperation with bioortogonal nitroxide spin-labeling (SL) was utilized to enable mapping of O-GlcNAc to specific serine/threonine residues within proteins of interest to facilitate functional studies. The approach exploited the incorporation of azide or carbamatelinked methylcyclopropene of the sugar moeities of modified proteins of interest followed by site-directed spin labelling technique. The application of the copper-catalyzed azide-alkyne cycloaddition “click” reaction was discussed in relation to attachment of alkyne-containing chemical spin label to GlcNAz and it was demonstrated how this functionalization of O-GlcNAz-modified proteins can be used to realize identification of the protein modification by electron-paramagnetic spectroscopy. The Ac_4GlcNCyoc compound discussed later in this thesis reacts faster than terminal alkenes in the inverted Diels-Alder reaction (DA_{inv}) with tetrazines and also was identified by EPR spectroscopy. Overall, this technique, which utilizes commercially available reagents and *O*-GlcNAc transferase enzyme as an individual protein model, has a potential to serve as a protein model to study functions of O-GlcNAcylation PTM on the protein structure in mammalian cells in a complete *in cellular* fashion as well as with an *in vitro* spin labelling step.

The developing of methodology for EPR spectroscopy studies of post-translational O-GlcNAcylation established possibility to perform qualitative and quantitative study of the PTM levels under influence of different inhibitors. O-GlcNAc enzymatic labelling allows to consider O-GlcNAcylation process that is going in parallel to MGE.

Zusammenfassung

Die biologischen Funktionen von Proteinen sind für alle zellulären Aufgaben verantwortlich und hängen eng mit ihrer Konformationsdynamik zusammen. Die richtige Faltung von Proteinen bestimmt die korrekte Art und Weise ihrer Interaktion mit anderen Molekülen und unzählige lebenswichtige zelluläre Prozesse in biologischen Systemen. Proteine falten sich entsprechend der Abfolge der Aminosäuren in der Polypeptidkette in spezifische Formen. Wenn die spezifische 3D-Struktur gestört ist, was als Denaturierung des Proteins bezeichnet wird, verliert das Protein seine Aktivität. Kurzum, die Funktion des Proteins hängt direkt von der Proteinstruktur ab, die nicht mehr ausgeführt werden kann, wenn die Struktur gestört ist. Die Aufklärung der grundlegenden zellulären Prozesse ist wichtig und erfordert ein vollständiges Verständnis der Proteininteraktionen, einschließlich der post-translationalen Modifikationen (PTM) von Proteinen.

PTMs sind Modifikationen der kanonischen Aminosäureketten von Proteinen mit verschiedenen funktionellen Gruppen, wie Phosphat- oder Zuckerderivaten und anderen. Post-translationale Modifikationen sind an vielen zellulären Prozessen beteiligt. Jede Art von PTM wirkt sich auf die Struktur, Funktion, Stabilität und Lokalisierung von Proteinen aus. In vielen Fällen sind sie auf die Fehlfaltung von Proteinen zurückzuführen. Fehlfunktionen in diesen führen zu vielfältigen Krankheitsbildern wie Krebs oder neurodegenerativen Erkrankungen. Die Anlagerung von O-verknüpftem β-N-Acetylglucosamin (O-GlcNAc) ist eine ubiquitäre post-translationale Modifikation, die für Säugetierzellen essenziell ist. Bei der O-GlcNAcylierung handelt es sich um eine reversible Glykosylierung an Serin/Threonin-Stellen zur Regulierung der Proteinaktivität und -verfügbarkeit in Zellen. Die O-GlcNAcylierung reguliert eine Vielzahl von physiologischen und pathologischen Prozessen. Das Verständnis der vielfältigen Funktionen der O-GlcNAcylierung ist jedoch oft schwierig, da es den Nachweis und die Quantifizierung der Modifikation voraussetzt. Daher sind robuste Methoden zur Untersuchung der O-GlcNAcylierung von entscheidender Bedeutung, um ihre Schlüsselrolle bei der Regulierung einzelner Proteine, komplexer zellulärer Prozesse und Krankheiten zu klären. Die wichtigsten Techniken zur Untersuchung von PTMs sind Massenspektrometrie, Immunpräzipitation, Western Blotting und Fluoreszenzmikroskopie. Die Markierung der PTM-Proteine spielt eine entscheidende Rolle, da es sich bei PTM um sekundäre Genprodukte handelt, die nicht mit fluoreszierenden Markierungen (Protein-

Tags) in Zellen exprimiert werden können. Für die Fluoreszenzmarkierungsstrategie und den mikroskopischen PTM-Nachweis wurden viele biochemische Ansätze verwendet. Aufgrund der begrenzten Anzahl von Glykosylierungsstellen sind nur wenige Methoden gut genug für den Nachweis des post-translationalen glykosylierten Zustands der jeweiligen Proteine.

Die vorliegende Arbeit befasst sich mit der Entwicklung und Anwendung der EPR-Spektroskopie für den Nachweis und die quantitative Untersuchung der O-verknüpften Glykosylierungsmodifikation von Proteinen. Auf diese Weise können chemisch stabile paramagnetische Spezies an die interessierende Stelle, in unserem Fall die O-glykosylierte Stelle des Proteins, gebunden werden. Um dieses Ziel zu erreichen, wurde die ortsgerichtete Spin-Markierung in Kombination mit der EPR-Spektroskopie eingesetzt, um die strukturellen Veränderungen von Glykoproteinen unter O-Glykosylierung zu untersuchen.

Der Ansatz des Metabolic Glycosaccharide Engineering (MGE) wurde verwendet, um die Zuckergruppen mit entsprechenden chemischen Reportern zu untersuchen, die für eine weitere Spin-Markierung anfällig sind. O-GlcNAcylierte Proteine wurden zusammen mit bioortogonaler Nitroxid-Spinmarkierung (SL) verwendet, um die Zuordnung von O-GlcNAc zu spezifischen Ser-/Thr- Aminosäuren in Proteinen von Interessezu ermöglichen und funktionelle Studien zu erleichtern. Der Ansatz nutzte den Einbau von Azid- oder Carbamat-verknüpften Methylcyclopropenen in die Zuckermoleküle der modifizierten Proteine, gefolgt von einer ortsbezogenen Spin-Markierungstechnik. Wir erörtern die Anwendung der kupferkatalysierten Azid-Alkin-Cycloaddition "Click"-Reaktion, um Alkin-haltige chemische Spin-Marker an GlcNAz zu binden, und zeigen, wie diese Funktionalisierung von O-GlcNAz-modifizierten Proteinen zur Identifizierung der Proteinmodifikation durch elektronenparamagnetische Spektroskopie verwendet werden kann. Die Ac4GlcNCyoc-Verbindung, die später in dieser Arbeit besprochen wird, reagiert schneller als terminale Alkene in der umgekehrten Diels-Alder-Reaktion (DAinv) mit Tetrazinen und wurde ebenfalls durch EPR-Spektroskopie identifiziert. Insgesamt hat diese Technik, bei der handelsübliche Reagenzien und das O-GlcNAc-Transferase-Enzym als individuelle Proteinmodelle verwendet werden, das Potenzial, als Proteinmodell zu dienen, um die Funktionen der O-GlcNAcylierungs-PTM auf die Proteinstruktur in Säugetierzellen in einer vollständigen Zelle sowie mit einem In-vitro-Spin-Markierungsschritt zu untersuchen.

Durch die Entwicklung einer Methodik für EPR-Studien zur post-translationalen O-GlcNAcylierung wurde die Möglichkeit geschaffen, qualitative und quantitative Studien zum

PTM-Niveau unter dem Einfluss verschiedener Inhibitoren durchzuführen. Die enzymatische Markierung von O-GlcNAc ermöglicht die Berücksichtigung des O-GlcNAcylierungsprozesses, der parallel zur MGE abläuft.

Table of contents

CHAPTER I 1
1. Introduction 3
CHAPTER II 7
2. State of knowledge relevant for protein glycosylation, metabolic glycoengineering, and EPR studies 9
2.1. Post-translational modification of proteins 9
2.1.1. Biological importance of glycans 10
2.1.1.1. Architecture and functions of glycans 12
2.1.1.2. Cellular organization of glycosylation 14
2.1.1.3. Major classes of glycoproteins: N-glycans and O-glycans 15
2.1.2. Biosynthesis of O-linked glycans 16
2.1.2.1. O-linked GlcNAcylation of proteins 18
2.1.2.2. O-linked GlcNAcylation on the cell membranes 23
2.1.3. Interplay between O-GlcNAcylation and phosphorylation PTMs 23
2.2. O-GlcNAc online resources and computational modelling 24
2.3. EPR spectroscopy 26
2.3.1. Theoretical background of EPR spectroscopy 26
2.3.2. Spin interactions and spin Hamiltonian 26
2.3.3. Basic principles of magnetic resonance 29
2.3.3.1. Magnetic moment of the electron spin, Zeeman effect and Larmor theorem 29
2.3.3.3. Relaxation, saturation and EPR line shapes 30
2.3.4. Continuous wave EPR spectroscopy and rotational dynamics detection 32
2.3.4.1. Determination of spin Hamiltonian parameters from CW EPR spectra 32
2.3.4.2. Spin label dynamics determination in cell and *in vitro* 33
2.3.4.3. Spin counting and spin labelling efficiency determination 35
2.3.5. Rapid-scan spectroscopy 35
2.3.6. Double electron-electron resonance (DEER) spectroscopy 36
2.3.6.1. Distance distribution determination 36
2.3.6.2. DEER data processing 37
2.3.6.3. Some examples of DEER applications 39
2.4. Site directed spin labelling 40

2.4.1. The stability of the radicals in biological systems....42
2.4.2. Spin-labelling approaches for glycoproteins & chemical reporters on the sugars....44
2.4.2.1. Cu(I)-catalyzed azide-alkyne 1,3-dipolar cycloaddition (CuAAC)....44
2.4.2.2. The Diels–Alder reaction with inverse electron demand....45
2.5. Fluorescent labelling spectroscopy for understanding the biological processes in glycoproteins....47
CHAPTER III....49
3. Designing strategy for the biological sample preparation....51
3.1. The general strategy of O-GlcNAcylated protein structure study by EPR....51
3.2. Optimisation of the biological sample preparation....53
3.2.1. Suitable mammalian cell line and modified sugar selection....53
3.2.2. Protein expression methods....54
3.2.3. Metabolic glycan engineering....56
3.2.4. Protein Purification Tag....58
3.2.5. Lysation and protein purification....59
3.2.6. Spin labelling by Click chemistry azide-alkyne cycloaddition....59
3.2.7. Isolation of O-GlcNAcylated protein of interest after CuAAC reaction....60
3.3. Selection of a protein model for EPR measurements....61
3.3.1. Alpha-synuclein....62
3.3.2. OGT protein....64
3.3.3. P53 protein....66
3.3.4. Alpha-crystallin protein....67
3.4. Optimisation of the protocol for the sample preparation for EPR measurements....68
3.4.1. Regular protocol for the similar biological samples preparation....68
3.4.2. Step-by-step improvement of the EPR sample preparation protocol....69
3.4.3. In vitro O-GlcNAcylation....75
3.4.4. Intermediate detection of O-linked glycosylated OGT protein on purification matrix....77
3.4.5. Final strategy of EPR sample preparation....79
3.4.6. Spectra of modification by Ac4GalNAz and Ac4GlcNAz....81
3.5. Intermediate conclusions regarding optimisation of the biological sample preparation protocol....82

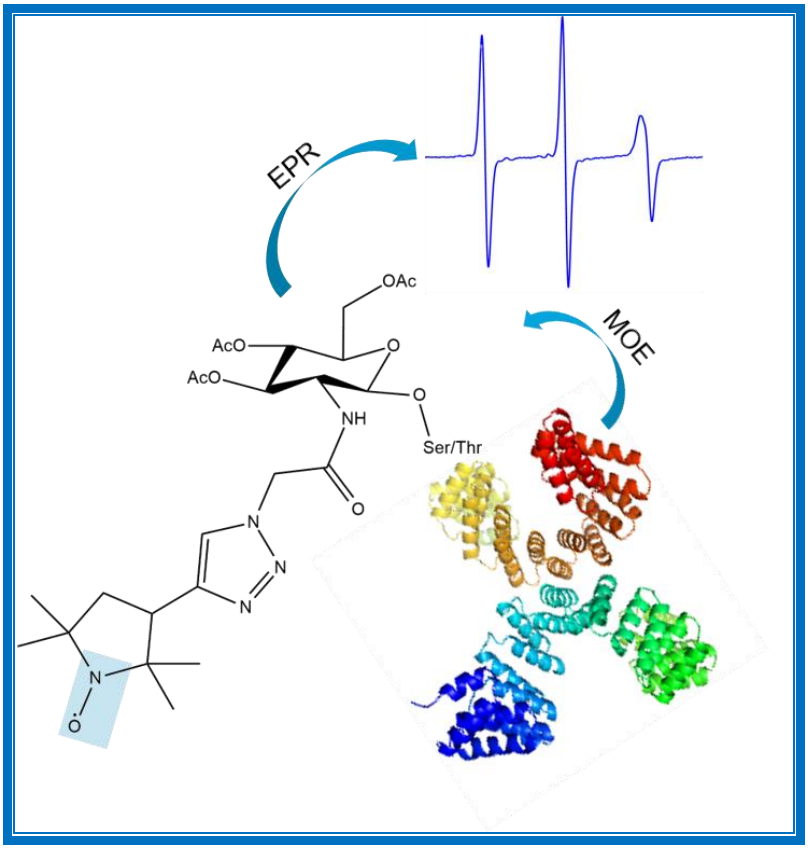
CHAPTER IV 85
4. Results and discussion 87
4.1. Application of the developed methodology for another protein model 87
4.1.1. P53 protein model 87
4.2. Conclusions and outlook for the EPR methodology application to different protein models 92
4.3. Influence of the molecule's size and the solvent on the nitroxide radical's mobility 92
4.4. CW-EPR spectra analysis elucidates main steps in the glycan spin labelling 95
4.4.1. EPR signal intensities and CuAAC reaction 99
4.4.2. Inhibitors can influence the interplay between O-GlcNAcylation and phosphorylation 99
4.5. Rapid scan measurements 101
4.6. Identification of the native O-GlcNAcylated sites and the O-GlcNAzylated sites 104
4.7. PaNDA spin labelling 108
4.8. DEER measurements for the detection of O-GlcNAc modification 112
4.9. DEER spectroscopy application for exploring the binding site, sugar relative, fucose-functionalized precision glycomacromolecules on the structure of human norovirus capsid protein. 119
3.2. Conclusions 123
4.3. Project perspectives 125
CHAPTER V 127
5. Matherial and methods 129
5.1. Chemical synthesis 129
5.2. Molecular biology 129
5.2.1. Prokaryotic cells 129
5.2.2. Eukaryotic cells 129
5.2.3. DNA plasmids 130
5.3. Reagents 130
5.4. Enzymes 132
5.5. Kits 132
5.6. Equipments 132
5.7. Biochemical methods for prokaryotic cells 133
5.7.1. Preparing chemically competent *E.coli* 133

5.7.2. Transformation of *E. coli* 134
5.7.3. DNA plasmid preparation 134
5.7.4. Cloning 134
5.7.5. Glycerol-stock preparation 135
5.7.6. Protein expression in *E.coli* 135
5.8. Buffers and solutions 136
5.8.1. Commercially available buffers and solutions 136
5.8.2. Self-made buffers and solution 136
5.9. Biochemical methods for mammalian cells 140
5.9.1. Cultivation of mammalian cells 140
5.9.2. DNA plasmid transfection into HEK 293T cells 140
5.9.2.1. By calcium phosphate co-precipitation 140
5.9.2.2. Lipofectamine-mediated transfection 141
5.9.3. Metabolic Glycoengineering of the protein *in cellula* 141
5.9.4. HEK293T cell lysis 141
5.9.5. Protein purification 142
5.9.5.1. Purification of OGT 142
5.9.5.2. Purification of p53 143
5.9.5.3. Purification of Alpha-synuclein 143
5.9.6. SDS-PAGE and Western-Blot analysis 144
5.9.7. Glycoprotein Staining 145
5.9.8. In vitro O-GlcNacylation 146
5.9.9. O-GlcNAc enzymatic labelling 147
5.9.10. Spin labelling 149
5.9.10.1. *In vitro* CuAAC spin labelling reaction 149
5.9.10.2. *In vitro* strain-promoted inverse-electron-demand Diels–Alder cycloaddition (DA_{inv}) spin labelling reaction 150
5.9.11. In cell fluorophore labelling of PTM proteins 151
5.10. Spectroscopic methods 152
5.10.1. Bio photometry 152
5.10.2.1. CW EPR spectroscopy 152
5.10.2.2. DEER experiment 153
5.11. Spin labelling efficiency identification (TLC/MS) 154
5.12. Data analysis and spectra simulation 154

5.13. Mass spectrometry analysis 155
5.13.1. In-gel digestion procedure of Coomassie stained proteins 155
5.13.2. Mass spectrometry measurements 155
5.14. DEER measurements for exploring the binding site of glycomacromolecules on the structure of human norovirus capsid protein. 156
List of abbreviations 159
Appendix 162
Reference list 167

CHAPTER I

Introduction



"Science is simply the word to describe a method of organizing our curiosity."
by Tim Minchin

1. Introduction

Among carbohydrates, lipids and nucleic acids, proteins constitute one of the four major classes of bio-macromolecules. It is well established that the structural features and dynamic behaviour of proteins are directly related to their functions, their interactions and activation/deactivation cycles in the native environment. Besides understanding the cellular processes, determination of the protein structure and dynamics is often of paramount importance to unravel the underlying conditions for formation and progressing of pathological processes. Here, we are interested in such features of the proteins as their ability to participate in cellular communication, or to contribute to different metabolic processes and pathways in living systems. At the same time proteins take part almost in all cellular tasks: they might catalyse chemical reactions that are crucial for post-translational regulation processes and assist the folding and activation of other biomolecules.

The whole variety of cellular functions such as transcription, translation, epigenetic modifications and signal transduction is regulated by reversible O-GlcNAc cycling. Furthermore, O-GlcNAcylation affects apoptosis, the cell cycle, and response to the stress, it is relevant for substrate recognition, protein folding, as well as protein fold protection, it plays role in immune system (development, proliferation, and activation of T and B cells) and signaling (*e.g.,* calcium, insulin, and mitochondrial signaling). Dysregulation of O-GlcNAc is supposedly linked with several pathologies including diabetes, cancer, and neurodegenerative disorders.

At the end of 2022, the topic of protein glycosylation has been distinguished after the chemists Barry Sharpless, Morten Meldal and Carolyn Bertozzi have been awarded the Nobel Prize ''for the development of click chemistry and biorthogonal chemistry''. In particular, Carrolyn R. Bertozzi used biorthogonal chemistry inside living organisms to illuminate the cells by adding chemical reporter to the glycans.

It is thus a great honour to write a thesis about bio-orthogonal click chemistry reaction performed on the glycosylated proteins, which summarizes 5 years of research work in the group of Malte Drescher in collaboration with the group of Valentin Wittman at the Chemistry Department of the University of Konstanz. In our project the O-linked glycosylation modification as an essential post-translational modification was studied using bio-orthogonal click chemistry reaction and continuous wave (CW) and pulse EPR spectroscopy as spectroscopic tool. The bio-orthogonal reaction gives possibility for protein tagging with different spectroscopic markers and their detection in a living cell environment such that it does not interact or interfere with the native biochemistry of the

cellular system. Thus, different intracellular processes can be investigated without disturbing the normal chemistry of the cells.

In this thesis, like in a number of other works, the design of research heavily relies on the developments of Barry Sharpless, who pioneered a new kind of chemistry (so-called click chemistry) that was later used by Carolyn Bertozzi to link smaller molecules with the carbohydrate frame of a glycoprotein, to DNAs and RNAs as well as to different cell membrane constituents (such as glycolipids). So far, the identification of glycans in cells was performed mainly via fluorescence and microscopy techniques utilizing fluorescence labels. In the current thesis a significant effort was invested in establishing bio-orthogonal spin labelling for detection of O-linked glycosylation on the protein structures by EPR spectroscopy. In particular, this required revisiting and revising the existing labelling approaches to achieve higher labelling efficiency and higher overall yields, necessary for EPR experiments.

Special features of EPR spectroscopy that provide unique opportunities for biological structure determination are:

1. selectivity, as it detects only unpaired electrons;
2. long-range distance distributions, above 10 nm in advantageous cases, can be measured by pulse EPR;
3. broad applicability in any phase and over a wide range of temperatures and local concentrations.

EPR spectroscopy can be used to study proteins that cannot be crystallized for X-ray crystallography studies, proteins that are too large for study by NMR in solution, and proteins in a much more natural environment than is possible for X-ray crystallography, including the membranes and even in whole cells and tissues, environments that can confound other physical methods. In addition to being selective, EPR is also applicable to a wide range of biological molecules.

EPR spectroscopy has also certain limitations as compared to the NMR spectroscopy. The use of EPR spectroscopy requires site directed or stochastic spin labelling, whereas NMR is applicable to the native state of biomolecules, perhaps after isotope labelling. However, EPR spectroscopy provides much longer-range distance constraints as compared to NMR, and also EPR spectroscopy can be applied to unstructured or partially unstructured biomolecules, while application of NMR needs a possibility of peak assignment, which is usually not possible for flexible parts of

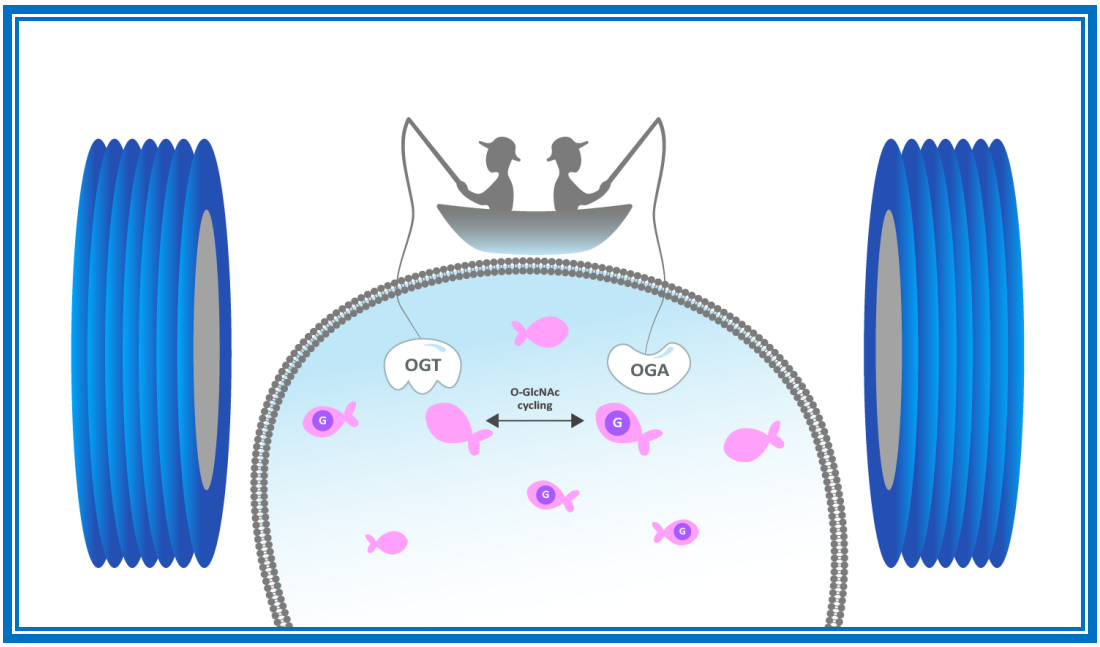
biomolecules.[1] On the other hand, NMR is superior in the number of constraints obtained from a single measurement. Overall, these two spectroscopies are highly complementary as structural tools, which is confirmed by a number of joint NMR/EPR spectroscopy publications.[2]

Also, methods based on fluorescent labelling and spin labelling are highly complementary for determination of biomolecules structure and dynamics: fluorescent methods are very efficient in determining dynamics but experience certain difficulties in distance determination, while EPR spectroscopy can only probe local dynamics in limited time range, but it is very accurate in determining distance distributions.

The thesis is structured as follows: After the introductory Chapter 1, Chapter 2 gives the general introduction to the topic of the thesis that is based on the state of the to date published literature about post-translational modifications (PTMs), chemical reporter strategies, respective bio-orthogonal ligation reactions, and the principles of EPR spectroscopy in application to studies of the PTMs of proteins. After clarifying the theoretical basis and the relevance of the spectroscopic tools for the study of PTMs, the aims of the project, results and their discussion are summarized in the Chapter 3. Chapter 4 contains materials and methods description.

CHAPTER II

State of knowledge



"Science and literature are not two things, but two sides of one thing."
by Thomas Huxley

2. State of knowledge relevant for protein glycosylation, metabolic glycoengineering, and EPR studies

2.1. Post-translational modification of proteins

Post-translational modifications (PTMs) of proteins are natural physiological biochemical events, essential in mammalian cells. PTMs include a variety of processes[3] such as covalent or noncovalent modifications, usually enzymatic in nature, that occur in the Golgi Apparatus and in the Endoplasmic Reticulum (ER). PTMs result in attachment of different functional groups ranging from small methyl groups to huge oligosaccharides to protein's structure.[4,5,6] Upon PTM, structural changes can also occur at the main peptide chain level, as for the formation of disulfide bridges, and in proteolytic cleavage of a protein at a peptide bond. Ubiquitination, lipidation, SUMOylation, phosphorylation, acetylation, methylation, glycosylation, disulfide bond formation, hydroxylation are the main classes of PTMs. PTMs are often catalysed by enzymes, and often, PTMs are reversible modifications of biomolecules which occur in response to changing conditions. This is specifically valid for glycosylation or phosphorylation,[7] which can be added and removed by cellular enzymes. Of course, PTMs can be also irreversible like myristoylation[8] or oxidative PTM, which is common for heart disease.[9] Taken the variable types of functional groups and their modification sites (some PTMs occur on a single type of amino acids while some PTMs occur on multiple types of amino acids), overall, more than 400 different types of PTMs are known. A large number of proteins associated with versatile diseases are inclined to post-translational modifications. PTMs directly affect the function(s) of a protein or may act in a concerted manner. Thus, it is mandatory to elucidate the effects that PTMs may have on the functions of a protein, if we want to understand those functions.

A limited number of suitable experimental tools are available for the study of the biomolecules structure after PTMs. Difficulties of determination of PTMs in proteins are related to the specificity of such protein modifications. Variable chemistry was applied to understand nucleotide-derived PTMs such as the reversible addition of phosphate groups, one or multiple ADP-ribose moieties or AMP, as well as protein-related phosphorylation, ubiquitination and SUMOylation and other types of PTMs. The majority of PTM detection methods have been developed in combination with enrichment strategies to provide the best opportunity to identify, validate, and study the function of regulatory PTMs for a protein of interest.[3,4] However, there is a limited number of proteins

where site specific PTM's identification have been developed. Usually, PTMs are analysed by downstream methods like western blot or mass spectrometry to determine if a protein of interest (POI) is post-translationally modified.[5]

Glycosylation is one of the most crucial types of PTMs to be investigated in detail. The complexity, diverse structure and modification forms create a whole variety of different kinds of glycosylation modifications of proteins.[2] Thus, not surprisingly, glycobiology is one of the rapidly growing fields in the natural sciences. It is studying the structure, biology, biosynthesis, and evolution of glycans (other names: carbohydrates, oligosaccharides, sugar chains) that are widely distributed in nature and of the proteins that recognize them with broad relevance to many areas of basic research, biotechnology, and biomedecine.[10] The field of glycobiology includes the chemistry of carbohydrates, the enzymology of glycan formation and degradation, the recognition of glycans by specific proteins, role of glycans in complex biological systems, and their analysis by different techniques. Glycosylation as a post-translational modification will be considered in more detail regarding its variety, mechanisms and functions in the next sections.

2.1.1. Biological importance of glycans

Here, glycans will be referred to as specifically the carbohydrate portion of a glycoconjugate, mainly a glycoprotein. Many biochemical processes of cells depend on carbohydrate metabolism because it can rapidly provide ATP to support various life activities. In addition, carbohydrate metabolism also has the functions of regulating fat metabolism, producing dietary fibres, facilitating detoxification, and enhancing intestinal function. Also, carbohydrates have an anabolic function. Abnormal carbohydrate metabolism often causes a variety of diseases, such as obesity, diabetes, downstream cardiovascular and metabolic diseases, and others.[11]

The biological roles of glycans are particularly prominent in the assembly of complex multicellular organs and organisms, which require interactions between cells and the surrounding matrix.[12] Protein glycosylation is acknowledged as one of the major post-translational modifications, with significant effects on protein folding, conformation, distribution, stability and activity. Glycosylation encompasses a diverse selection of sugar-moiety additions to proteins that ranges from simple monosaccharide modifications of nuclear transcription factors to highly complex branched polysaccharide changes of cell surface receptors. Carbohydrates in the form of asparagine-linked

(N-linked) or serine/threonine-linked (O-linked) oligosaccharides are major structural components of many cell surfaces and secreted proteins.[13,14]

The glycome, the complete set of glycans produced by a cell, is a dynamic indicator of the cell's physiology. Changes in the glycome reflect the changes in the cell's developmental stages and transformation state of the cell.[15]

Glycobiologists have known for decades that the structures of glycans, which decorate all eukaryotic cell surfaces, change with the onset of cancer (the first demonstration that cancer glycans differ from glycans on healthy cells was done by E. Meezan & P. W. Robbins in 1969)[16], and inflammation[17,18,19]. The glycosylation pattern of a cell is therefore a code for cellular physiology. Meanwhile, an understanding of this code at both molecular and functional levels is starting to emerge. Although glycobiologists are still at the early stages of deciphering this code, their results have already inspired novel methods to detect glycans characteristic of disease conditions, to prevent disease glycan formation and to destroy cells that display them.

Glycans and glycan-binding proteins modulate numerous physiological functions. They are involved in reproductive and embryology processes. For example, in multiple steps in fertilization of male and female reproduction.[3] Also, genetic modifications that eliminate initial steps of major glycan synthetic pathways and of some monosaccharide biosynthetic pathways generally result in embryonic lethality, with one exception being deletion of individual genes needed for initiation of mucin O-GlcNAc pathway, likely because of functional redundancy among the 20 different polypeptides: O-GlcNAc transferases. It has been heavily documented that aberrant glycosylation patterns, including both the under- and overexpression of naturally occurring glycans, as well as neoexpression of glycans normally restricted to embryonic tissues, are a hallmark of the tumor phenotype.[20, 21] Also, following birth, mammals produce milk containing an array of glycoproteins and some species-specific milk oligosaccharides. Glycans affect functions of all classes of blood cells. For example: N- and O-glycans of certain glycoproteins affect differentiation, adhesion, and survival of leukocytes. Cardiac muscle integrity and optimal cardiovascular function depend on various glycans. The influence of glycans has been proven on the action of the endocrine system: O-GlcNAc on the proteins of the nucleus and cytoplasm modulate insulin action, and aberrant O-GlcNAcylation is linked to hyperglycemia; N-glycans may also play role in type II diabetes. Glycans also make their functional changing contributions in the various mammalian systems: pulmonary physiology, gastroenterology, hepatology, nephrology, neurobiology, oral and skin biology.[3]

The ability to visualize and monitor these changes in cells and in tissue samples would advance our understanding of the detailed roles of glycans in life and diseases and provide new diagnostic tools for their treatment.

2.1.1.1. Architecture and functions of glycans

The central paradigm driving research in molecular biology has been that biological information flows from DNA to RNA to protein. The genome, comprising the DNA of an organism, includes all the genes that produce the glycome, which comprises all of the glycans made by an organism. Within an organism, almost every cell that contains a nucleus and mitochondria has an identical genome. Cells typically differ in the portion of the genome and, therefore the glycome, they express. Thus, the glycan complement of a cell depends on which genes are actively transcribed and which transcripts are translated and stably expressed. Transcription, splicing, translation, and post-translational processing may vary depending on the state of differentiation and the physiological environment of a cell. Under different environmental conditions the glycan repertoire of a cell represents a subset of all the glycans that an organism is capable of making.

Biological functions of glycans can be divided into broad categories:

- structural contributions can be described by extracellular scaffolds (cell walls and extracellular matrices, protein folding and functions),

- energy metabolism described by carbohydrates as carbon sources for storage and manipulation of animal behaviour: pollination and seed dispersal,

- information carriers, e.g. molecular patterns recognized by glycan binding proteins.

At the same time, glycan binding proteins are also divided into two groups: intrinsic glycan binding proteins, which recognize glycans from the same organism and extrinsic glycan binding proteins, which recognize glycans from another organism. In this work intrinsic glycan binding proteins will be discussed. Such proteins typically mediate cell-cell interactions or recognize extracellular molecules, but they can also recognize glycans on the glycocalyx of the same cell.

Glycans serve protective, stabilizing, organizational, and barrier functions. The glycocalyx represents a substantial physical barrier, the external location of glycans on most glycoproteins can provide a general shield, protecting the underlying polypeptide from recognition by proteases (mucins), blocking antibody binding (viral glycoproteins), and even protecting entire tissue surface

from microbial attachment (as in the case of mucins). It is important to mention that the same glycosylation change can have different effects in various cell types. The effect may depend on the structure of the glycan, on the binding protein and on the biological context. Given all of the above consideration, it is difficult to predict the functions that a given glycan or a given glycoconjugate might mediate.[22]

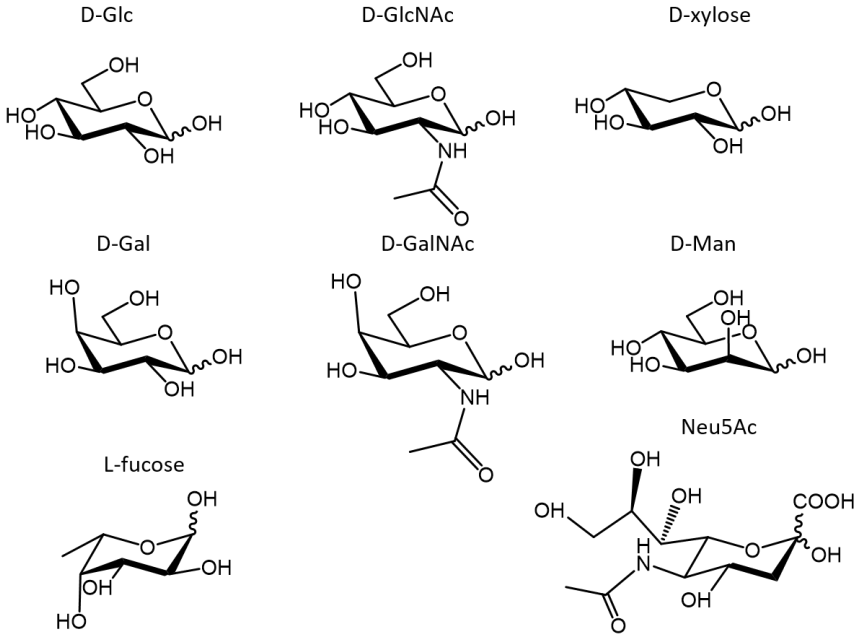


Figure 1. Chemical structures of monosaccharides precursors derived from glucose and present eukaryotic glycans.

Mammalian cell glycans made within the ER/Golgi contain eight precursor sugars: D-glucose (D-Glc), D-mannose (D-Man), D-galactose (D-Gal), N-acetyl-D-glucosamine(D-GlcNAc), N-acetyl-D-galactosamine (D-GalNAc), N-acetyl-neuraminic (sialic) acid (Neu5Ac), L-fucose (Fuc), D-xylose (D-Xyl) (see Figure 1).[23] All these precursors can be derived exclusively from glucose. Some occur in the diet or are indirectly salvaged from glycans degraded within lysosomes.[3] The relative contributions of *de novo*, salvage and dietary pathways are quite variable but mostly unknown.[24, 25] How the preferences translate into physiology/pathology probably depends on the relative biosynthetic loads of different tissues. For instance, during embryonic development chondrocytes carry a heavy glycosylation burden,[26] and in postnatal growth, hepatocytes make most plasma glycoproteins.

Glycobiology studies the structure, biosynthesis and biology of glycans. The *glycomics* is a subset of glycobiology that studies glycome - one of the most complex entities of sugars in nature - on the same complexity level with genomics and proteomics. In other words, glycomics is the systematic

study of all glycans, or carbohydrate chains that are covalently linked to lipids or proteins of a given cell type or organism.[27]

2.1.1.2. Cellular organization of glycosylation

Most well-characterized pathways for the biosynthesis of major classes of eukaryotic glycans, for both secretory proteins and membrane proteins, are passing through an ER and Golgi compartments. George Palade indicated that the most cell-surface and secreted proteins in eukaryotic cells are co-translationally translocated into the ER where they are folded, modified, and subjected to quality control mechanisms. In 1974 George Palade together with Albert Claude and Christian de Duve were awarded the Nobel Prize in physiology or medicine field related to the glycobiology "*for the discoveries concerning the structural and functional organization of the cell*".[28] Proteins then make their way via an intermediate compartment (IC) through multiple stacks of the Golgi apparatus, finally being distributed to various destinations from the *trans*-Golgi network (TGN).[29] Secretory pathway proteins include N-glycosylated, O-glycosylated, modified with glycosylphosphatidylinositol (GPI) anchors, and some called proteoglycans are modified with attached glycosaminoglycan chains.[30] Each of these modification pathways has an individual enzyme. N-linked glycans and GPI anchors are preassembled before being transferred to proteins and then further modified in the ER-Golgi pathway.[31] The stepwise assembly of O-linked glycans and flycosaminoglycans, as well as the glycosylation of lipids, involve reactions in both the ER and Golgi (see Figure 2).

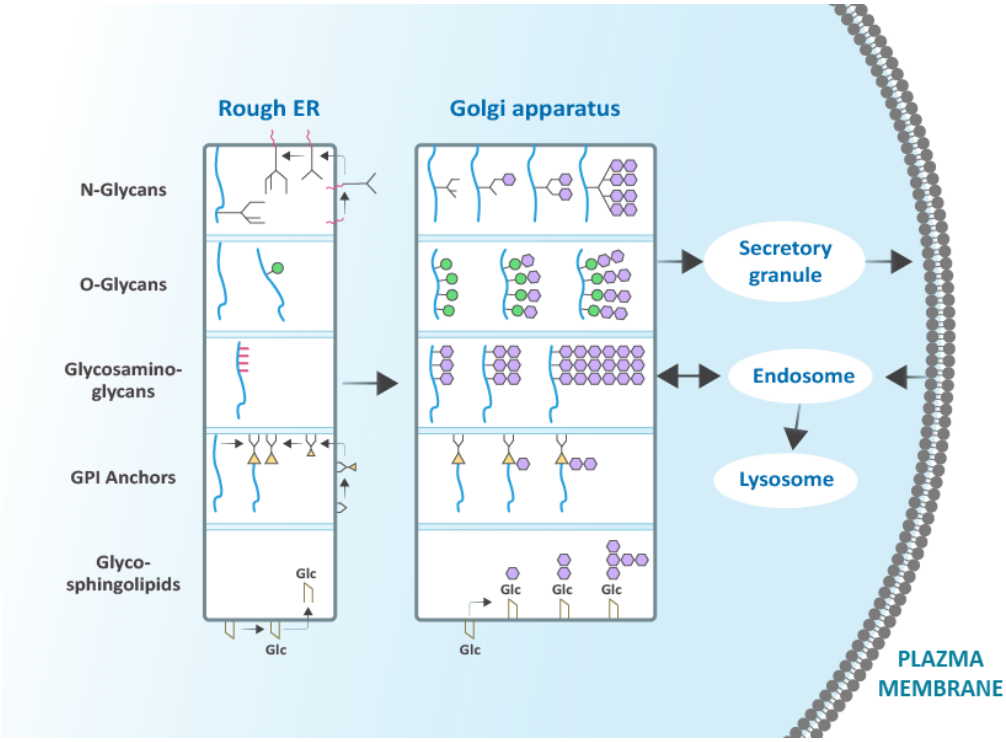


Figure 2. This illustration outlines initiation and maturation of the major types of glycoconjugates in the endoplasmic reticulum (ER) Goldgi plasma membrane pathway in animal cells. Figure is based on.[29]

2.1.1.3. Major classes of glycoproteins: N-glycans and O-glycans

Glycans are classified by their linkage to the lipids, other glycans or protein moieties.[27] Glycoproteins consist of individual glycans and glycan chains linked to nitrogen and oxygen atoms of amino acid residues and are thus termed *N*-glycans and *O*-glycans, respectively. *N*-glycans consist of *N*-acetylglucosamine (GlcNAc) attached by a β1-glycosidic linkage to the nitrogen atom of the amino group of Asn (N) at the consensus glycosylation motif Asn-X-Ser/Thr (in which X denotes any amino acid except for Pro). These branched and highly heterogeneous *N*-glycan structures consist of a core glycan containing two GlcNAc residues and three mannose (Man) residues. Perhaps the most diverse form of protein glycosylation is *O*-glycosylation, in which glycans attach to the oxygen atom of the hydroxyl groups of Ser (S) or Thr (T) residues. *O*-glycans can be further subclassified on the basis of the initial sugar attached to the protein and the additional sugar structures added to the initial glycan (see Figure 3). For example, mucin-type *O*-glycosylation denotes that the initial glycan is *N*-acetylgalactosamine (GalNAc); mucin-type glycans can be further classified on the basis of the glycans attached to the initial GalNAc.[6] Other types of *O*-glycans, such as *O*-linked fucose (Fuc) and *O*-linked Man, often occur in specific proteins or protein domains, such as epidermal

growth factor (EGF) repeats, thrombospondin type I repeats (TSR) or dystroglycan. *N*-glycans and *O*-glycans are often capped with negatively charged sialic acid. *O*-GlcNAc is a unique type of *O*-glycosylation that is synthesized by *O*-GlcNAc transferase; it occurs in the cytosol and nucleus.

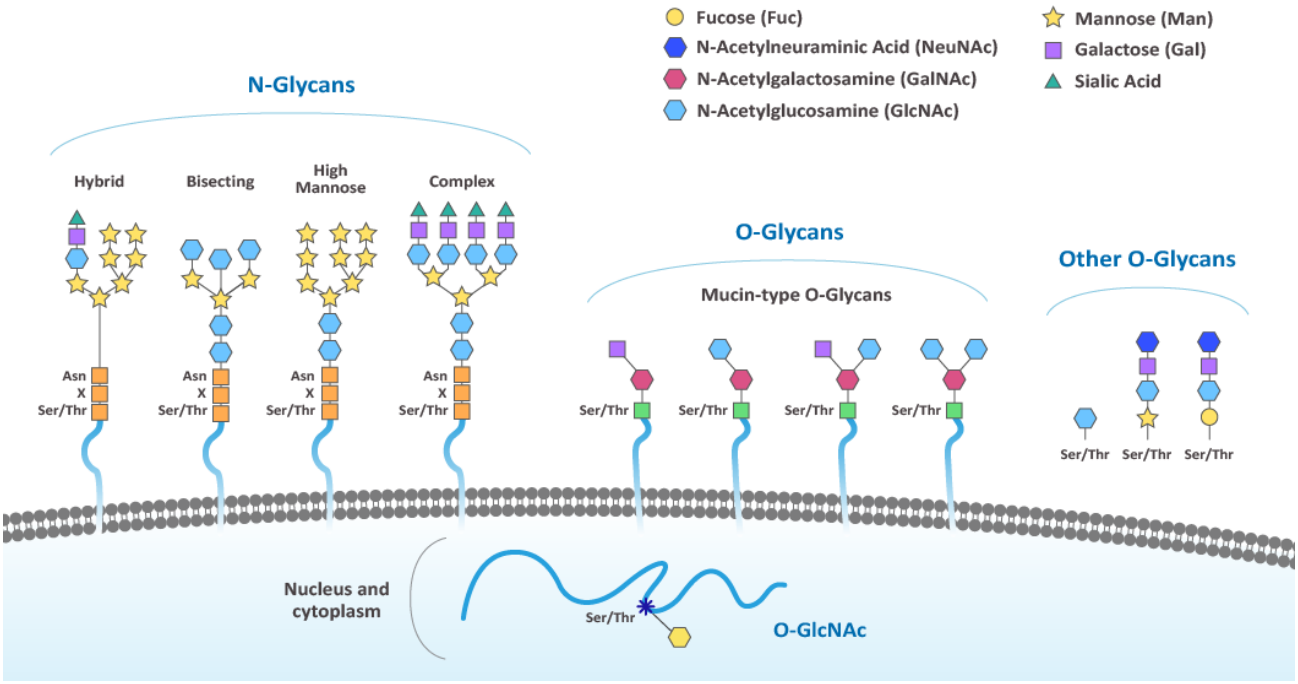


Figure 3. N-linked and O-linked glycosylation types. This drawing is based on.[29]

The core structures of major glycan classes tend to be conserved across many species. However, there can be considerable diversity in outer-chain glycosylation, even among relatively similar species. Such interspecies variation in glycan structure indicates that some glycan sequences do not have fundamental and universal role in all tissues and cell types in which they are expressed.

2.1.2. Biosynthesis of O-linked glycans

Glucose is the central monosaccharide in carbohydrate metabolism, and it can be converted directly or indirectly into all other sugars. Glucose is first converted to glucose-6-P hexokinase. In the glycolytic pathway, glucose-6-P is converted to fructose-6-P by phosphoglucose isomerase or into glucose-1-P by phosphoglucomutase (see Figure 4).[29]

Enzymatic synthesis of uridine diphosphate N-acetylglucosamine (UDP-GlcNAc) begins with the formation of glucosamine-6-P from fructose-6-P by transamination using glutamite as the $-NH_2$ donor by the enzyme glutamine-fructose-6-phosphate transaminase (GFAT). Glucosamine-6-P is then N-acetylated via an acetyl-CoA-mediated reaction to form N-acetylglucosamine-6-P and then

isomerized to N-acetylglucosamine-1-P via a 1,6-bis-phosphate intermediate. Similar to the other activation reactions, N-acetylglucosamine-1-P then reacts with UTP to form UDP-GlcNAc and pyrophosphate. In general, UDP-GalNAc can arise from two pathways. First one is the result of reaction between N-acetylglucosamine-1-P with UTP. N-acetylglucosamine-1-P is formed by the special kinase that is distinct from galactose-1-kinase. UDP-GalNAc can also be formed by epimerization of UDP-GlcNAc using the same NAD-dependent epimerase that converts UDP-Glc to UDP-Gal.

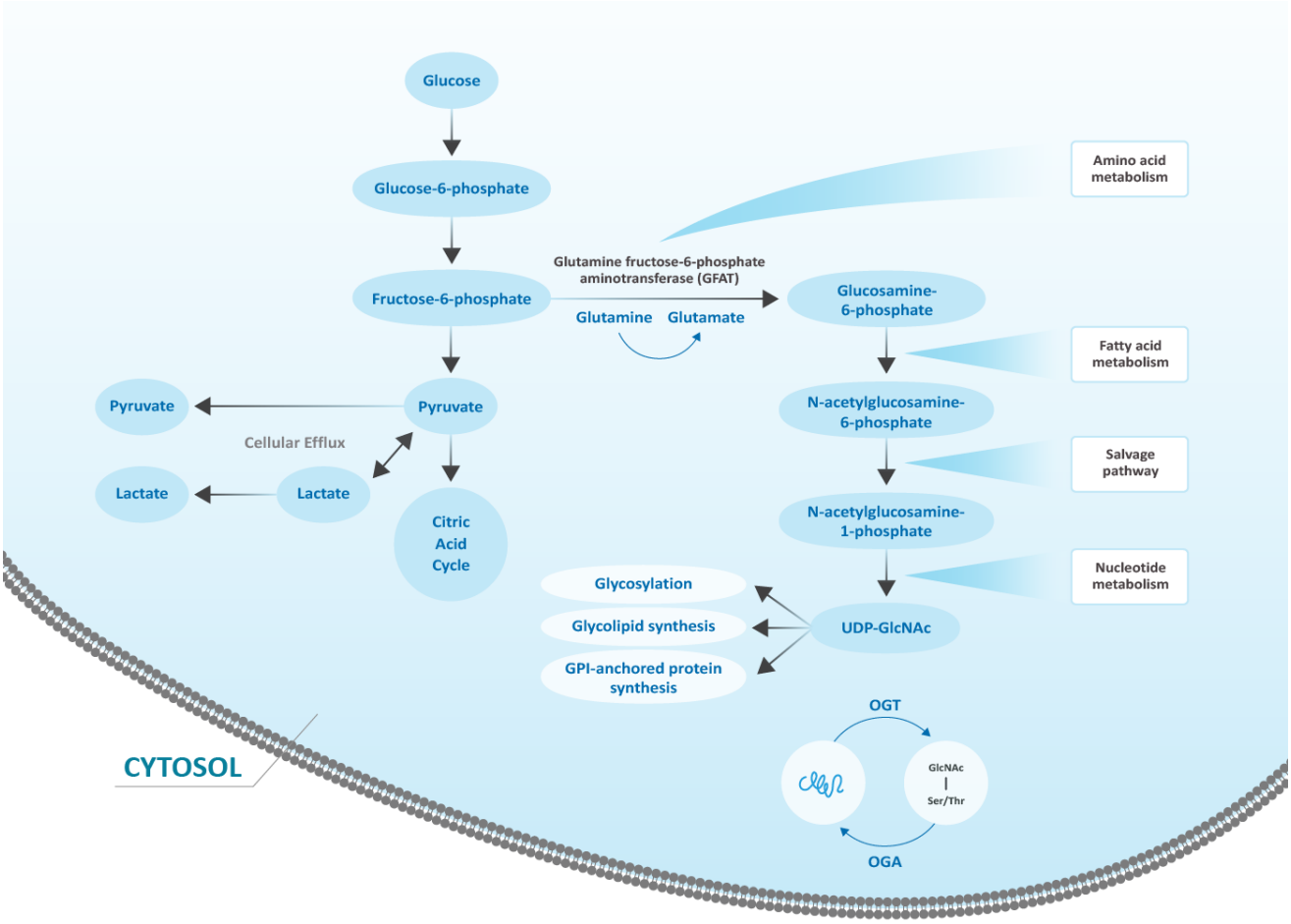


Figure 4. Schematic representation of O-linked glycans' biosynthesis. Figure is based on the reference.[32]

Alternatively, GlcNAc can be directly phosphorylated to form N-acetylglucosamine-6-P by a kinase that also synthesized N-acetylmannosamine-6-P from N-acetylmannosamine. Phospho-N-acetylglucosamine mutase then converts N-acetylglucosamine-6-P to N-acetylglucosamine-1-P. This route may account for the efficient salvage of GlcNAc from lysosomal degradation of glycans. Glucosamine can also be salvaged following sequential phosphorylation and acetylation.

2.1.2.1. O-linked GlcNAcylation of proteins

O-linked GlcNAcylation was discovered in 1984, and since then has been found on serine or threonine residues of thousands of proteins in the nuclear, cytosolic, and mitochondrial compartments of different mammalian cells. In the following sections the main principles of the protein glycosylation are briefly described. Glycosyltransferase and glycosidase are responsible for the assembly, processing, and turnover of glycans. The biosynthesis of glycans is primarily determined by glycosyltransferases that assemble monosaccharide moieties into linear and branched glycan chains (see Figure 4). In many cases, they catalyse a group-transfer reaction in which the monosaccharide moiety of a simple nucleotide sugar donor (electrophile) substrate (for example: UDP-Gal, GDP-Fuc, or CMP-Sia) is transferred to the acceptor (nucleophile) substrate (see Figure 5 as an example of GalNAc/GlcNAc salvage pathway for the protein O-GlcNAcylation).

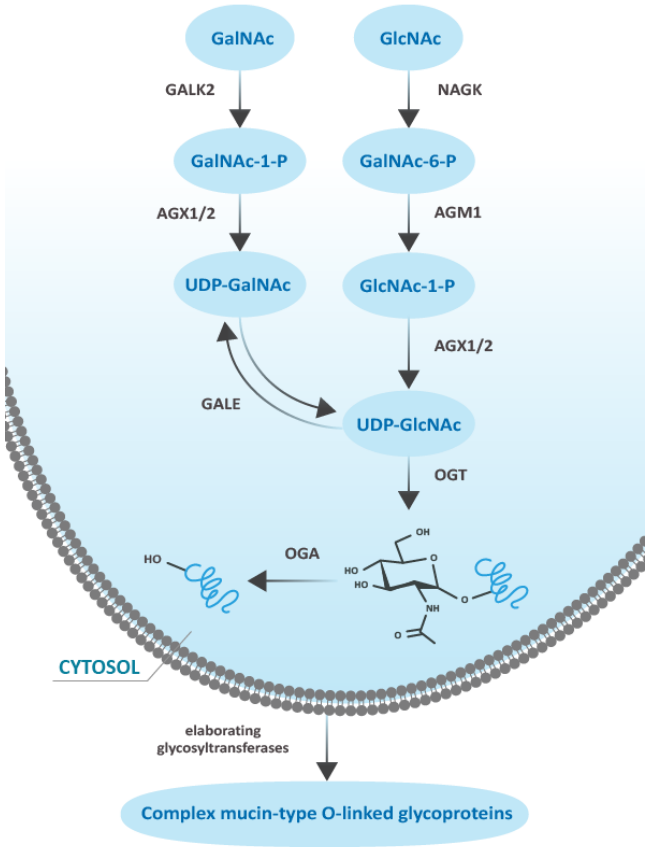


Figure 5. The GalNac (left side of image) and GlcNAc (right side of image) salvage and O-GlcNAc signaling pathway. Current image drawing is based on the reference: Boyce, M. *et al. PNAS, 2011*.[33]

2.1.2.1.1. Metabolic O-linked glycosylation

The glycosyltransferases that transfer sugar directly to the polypeptide chain of a protein or glycoprotein recognize their acceptor substrate in a number of different ways. The polypeptide GalNAc transferases, responsible for the initiation of mucin-type O-glycans, act after the protein has been folded and transported to the Golgi.[10,29] The polypeptide O-GalNAc transferases do not recognize a specific sequence motif, and, in general, they transfer N-acetyl galactosamine to the side chain hydroxyl group of serine and threonine residues found in relatively unstructured regions of the folded proteins. Some polypeptide O-GalNAc transferases possess a lectin domain that serves to direct the glycosyltransferase to regions of the polypeptide that already possess O-glycan chains. In this way, regions of polypeptide that have a high degree of O-glycan substitution, typical of mucin structure, can be synthesized. Mucins are the class of glycoproteins carrying the greatest number of O-GlcNAc glycans. In addition to the O-GalNAc linkage formed by the polypeptide O-GalNAc transferases, a number of the glycosyltransferases can glycosylate the side chain hydroxyl groups of serine and threonine to generate O-GlcNAc, O-Fuc, O-Glc, O-Man, and O-Xyl linkages. Specificity for a particular serine and threonine residue is achieved in different ways. Enzymatic addition of the single N-acetylglucosamine (GlcNAc) monosaccharide to specific protein in the cytosol, nucleus and mitochondria.[6,7] O-GlcNAcylation is catalyzed by the O-linked N-acetylglucosaminyl transferase (OGT) and removed by the enzyme O-GlcNAcase (OGA) (see Figure 5).[34,35,36] In case of O-GlcNAc transferase (OGT) any obvious consensus sequence is associated with acceptor substrate binding specificity. Some amino acid-consensus sequences or glycosylation motifs used in the formation of glycopeptide bonds are listed in Table 1.

O-linked glycopeptide bond	Consensus sequence or peptide motif
GlcNAc-α-Thr	Thr-rich domain near Pro residues
GlcNAc-α-Ser/Thr	Repeat domains rich in Ser, Thr, Pro, Gly, Ala with no specific sequence
GlcNAc-β-Ser/Thr	Ser/Thr-rich domains near Pro, Val, Ala, Gly EGF modules (Cys-X-X-G-X-Ser/Thr-G-X-X-Cys)
Glc-β-Ser	EGF modules (Cys-X-Ser-X-Pro/Ala-Cys)
Glc/GlcNAc-Thr	Rho:Thr-37; Ras, Rac; Cdc42:Thr-35
Gal-Thr	Gly-X-Thr (X: Ala, Arg, Pro, Hyp, Ser)
Gal-β-Hyl	Collagen repeats (X-Hyl-Gly)
GlcNAc-α-1-P-Ser	Ser-rich domains (e.g., Ala-Ser-Ser-Ala)

Table 1. Glycosylation motifs for the formation of glycopeptide bonds

The family of polypeptide N-acetyl-α-galactosaminyltransferases (ppGalNAcTs) is initiating the biosynthesis of O-linked glycoproteins and modifying them in the secretory pathway. The consensus sequence definition for the ppGalNAcTs makes the prediction of mucin-type O-linked glycosylation difficult based just on primary sequence.[37]

2.1.2.1.2. Metabolic glycoengineering

Glycoengineering is a method that allows to improve the properties of proteins by changing their glycosylation.[38] Metabolic glycoengineering (MGE) is a method which modifies glycoprotein structure by treating cells with derivatives of monosaccharides which were beforehand synthetically obtained. MGE is also referred to as metabolic oligosaccharide engineering (MOE) in case glycoprotein modification is performed by oligosaccharide.[39] MGE generates new possibilities for the study of biological systems. The non-natural sugar compounds participate in a targeted biosynthetic glycosylation pathway after what they can be exploited as a "tag" for bio-orthogonal and chemoselective ligation reactions (*e.g.* fluorescent labelling, spin labelling) (see Figure 6). In the same moment, using MGE/MOE is a daunting task, because of the large variety of different structures of the glycosylation cores. Despite the difficulties, an evolving repertoire of sugar analogs and the increasing number of metabolic pathways that can be targeted are continuously observed.

MOE was pioneered on the sialic acid biosynthetic pathway.[40] In its first decade, MOE was primarily utilized to modify the cell surface without directly or explicitly altering the biology of the host cell[41] or by directing drugs to the modified glycans. With time MOE demonstrated its abilities to modulate biological responses directly,[42,43]and to control cell-substrate binding[44] and gene expression.[45] Another fairly recent development of MOE is the extension of this technology beyond the sialic acid pathway to include pathways that process GlcNAc,[46] GalNAc[47] and fucose.[48]

For the MOE the cell line selection is an important task. The metabolic incorporation of monosaccharide analogs can vary significantly depending on the glycosylation pathway targeted, the monosaccharide analog used, and even the species involved. Still, by now MOE technology can be used essentially for any mammalian cell type such as for human cancer cell lines as well for healthy cells.[49]

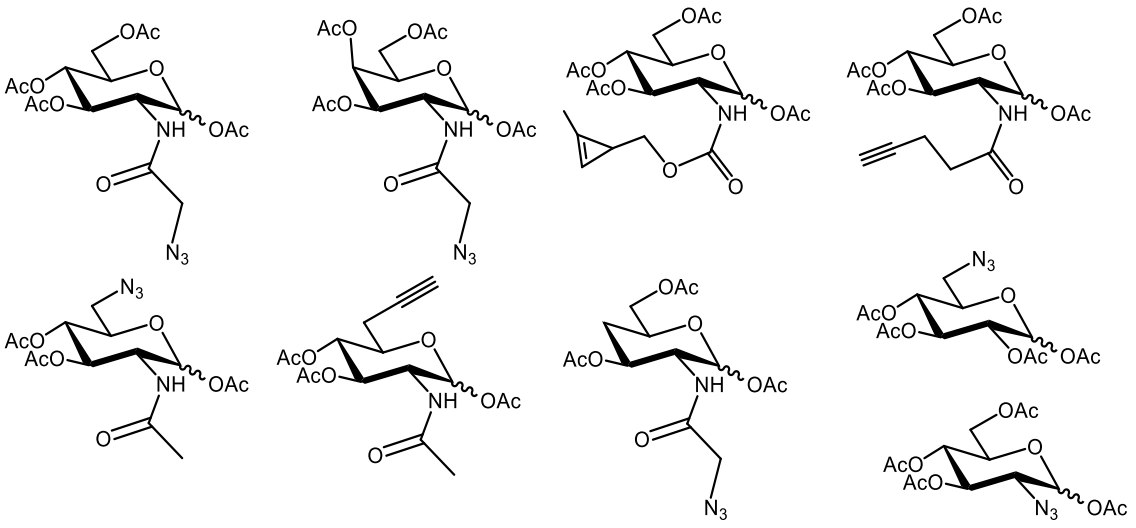


Figure 6. Chemical structures of sugar reporters to target protein O-GlcNAcylation.

2.1.2.1.3. Chemo-enzymatic O-linked glycosylation *in vitro*

The native concentration of any protein in mammalian cells is rarely high, and, at some point, it is difficult to perform glycosylation detection and quantification with the low *in cellula* protein concentrations in the range of nM or few µM.[50] A strategy here would be to perform MGE/MOE *in vitro* and then either to also study the protein properties *in vitro* or to inject the modified protein in cells and perform the *in cellula* experiments. The strategy which is giving the possibility to study O-GlcNAcylation of the proteins *in vitro* is enzymatic glycosylation. O-GlcNAcylation cycling is controlled by two enzymes: glycosyltransferase OGT (O-GlcNAc transferase) and the glycoside hydrolase OGA (*O*-GlcNAcase). Some publications demonstrated chemo-enzymatic methods for probing unmodified O-GlcNAc sites in the biological systems.[51,52] To imitate *in vitro* conditions for O-linked glycosylation, OGT enzyme or its analog is needed. Quite often the proteins have rather complicated structure with open and closed glycosylated sites. Such sites are dynamically regulated.[51] Few approaches were represented for assessing O-GlcNAc level by using OGT enzyme in combination with uridine 5'-disphopho-N-acetylazidoglucosamine (UDP-GlcNAz).[51] A labelling method for mucin-type O-linked glycoproteins was presented by using N-acetyl-alpha-galactosaminyltransferases (ppGalNAcTs) with 5'-disphopho-N-acetylazidogalactosamine (UDP-GalNAz) that modifies proteins in the secretory pathway (see Figure 7).[37]

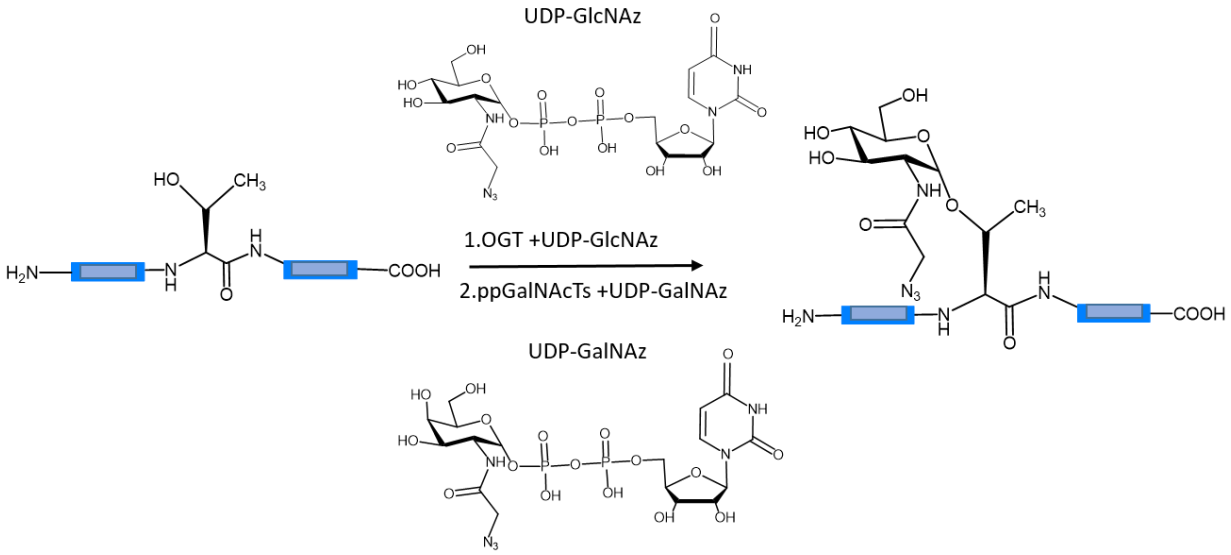


Figure 7. Reaction that demonstrate the initiation of mucin-type O-linked glycan biosynthesis by the 1) OGT with UDP-GlcNAz and 2) ppGalNAcTs with UDP-GalNAz. Figure is based on the reference.[51]

Second strategy of chemo-enzymatic labelling which allows to perform O-GlcNAcylation of closed sites was presented by using of β-1,3-N-acetylgalactosaminyltransferase (B3GALNT2) in combination with UDP-GalNAz to detect already existing glycosylated sites of the protein (see Figure 8).[53]

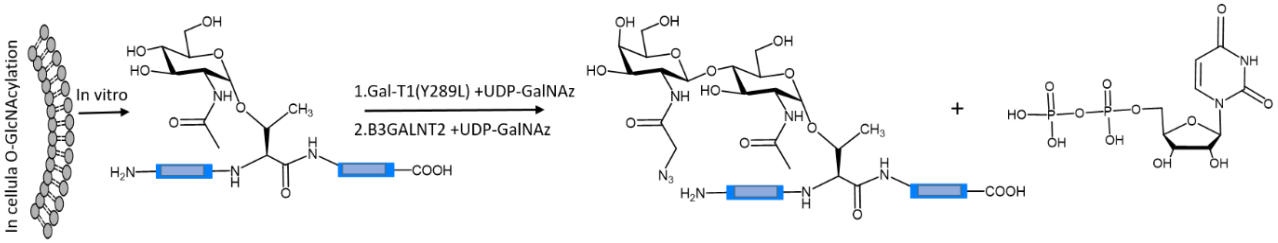


Figure 8. Enzymatic reaction in vitro of O-linked glycan biosynthesis over existing O-GlcNAcylation site. Figure is based on the reference.[54]

Utilizing glycosyltransferases (GTs) in the chemo-enzymatic synthesis of oligosaccharides has been in the past limited by different restriction factors such as: availability of the enzymes, availability of the activated monosaccharide building blocks, inhibition of glycosylation pathway by the released nucleotide phosphate and the enzyme substrate specificity.[54] To overcome the inhibition effects of by-products and consumption of expensive sugar-nucleotides, elegant sugar-nucleotide regeneration systems have been explored but have shown limited usability due to rather complex compositions.[55]

2.1.2.2. O-linked GlcNAcylation on the cell membranes

Glycosylation processes also affect the lipids and proteins in the cell membrane. Layer of glycans over the cell membrane forms glycocalyx which controls many cellular biological functions, in particular cellular communication and interactions.

Like for the soluble glycoproteins, the biosynthesis of glycocalyx components is a complex, multi-step process which occurs at different locations in the cell.[56] In the section 'Biosynthesis of O-linked glycans' it was already mentioned that biosynthesis of glycoproteins and glycolipids is performed in ER and the Golgi apparatus.[57,58] After the protein folding, newly synthesized protein is prone for the glycosylation process in ER. Most commonly, N-type glycans are transferred onto peptide chain and only a single O-glycosylation sugar is attached in the ER. Maturation step in the Golgi apparatus is associated with adding further sugars onto peptide chain. At the end of biosynthesis, the glycosylated products are packed into secretory granules, relocated to the cell membrane, and secreted.[56] As the result, the cell-surface carbohydrates can be endowed with novel structural and chemical features.

Multiple factors can affect the amount of sugar analogs incorporated and displayed on the cell surface, including the cell type, cell density, incubation time, specific sugar analog, and metabolic status of the cell.[59]

2.1.3. Interplay between O-GlcNAcylation and phosphorylation PTMs

Multifunctional proteins are involved in diverse and often unrelated functions and may perform different tasks in different biological environments.[60] It is usual for proteins to perform multiple functions in the context of their intra- and extra-cellular interactions.[61]

O-GlcNAc sites resemble phosphorylation sites, it means that phosphorylation and O-linked glycosylation PTMs may occur by steric competition for occupancy at either the same or a proximal amino acid site.[62,63] Both PTMs are reversible. In case of phosphorylation PTM, the attachment and removal of O-phosphate is controlled by kinases and phosphatates.[64,65] The protein phosphorylation is in general regulated by about 518 specific kinases and about 200 phosphatases,[66,67] while only two enzymes regulate the addition and removal of *O*-GlcNAc to/from thousands of protein targets. In this case phosphorylation of sugar residues by ATP-dependent kinase occurs at C-2 of O-xylose in proteoglycans and at the position of O-mannose in α-dystroglycan after the mannose has been further glycosylated.[3]

Reversible phosphorylations induce changes in the secondary and tertiary structures and control the modular interactions of proteins.[68] In addition, phosphorylation and alternatively glycosylation control the functional behavior of a growing list of proteins.[69,70] For instance, promotion of proteolysis may result from phosphorylation of serine/threonine or tyrosine residues.[71] Alternatively, resistance to proteolysis may result from the presence of carboxyl or sulphate groups in glycoproteins .[72] It leads to the competitions between different PTMs at the given Ser- or Thr-site.

Also known is the process which is called phosphoglycosylation or glycophosphorylation. In this process a sugar phosphoryl is transferred from a nucleotide sugar donor directly to a serine residue on a protein.[29]

2.2. O-GlcNAc online resources and computational modelling

The protein structural features directly influence their functions in different biological environments. As was often mentioned above, many factors are generating varying modifications of glycosylation forms and sometimes even phosphoglycosylated or even phosphorylated forms of post-translational modification. Nowadays many experimental results are describing O-GlcNAcome ability of ~5000 human proteins in approximately 1700 published articles. To be constructive and productive in the future, glycobiology meta-analysis research more and more requires the reliable databases (which will combine and include all kinds of glycoprotein characteristic parameters known to science). The scientific group from Medical college of Wisconsin created “The O-GlcNAc Database” based on the analysis of the amino acid sequence surrounding 7002 *O*-GlcNAc sites identified from different species which give opportunity to get O-GlcNAcylation information of different proteins.[73]

There is an analogue data base called “O-GlcNAc Home”, which has a O-GlcNAcAtlas functionality based on the O-GlcNAcylation results of the last 33 years of study and consists of two datasets. Dataset-I describes the group of proteins with 14530 unambiguous O-GlcNAcylated sites and includes 5517 unique proteins. Dataset-II contains 2424 unique proteins with 10054 ambiguous O-GlcNAcylated sites.[74] For each protein, comprehensive information, including species, modified sites, site mapping methods and literature reference, is provided.[74] O-GlcNAc Home is a useful resource that allows to research O-GlcNAc sites of the proteins quickly and reliably. All these information databases provide comprehensive understanding of O-GlcNAcylation PTMs which was accumulated over for the long period of research.

The well-structured and comprehensive information of the databases opened an opportunity to perform computational analyses and modelling. One of the programs which was used during preparation of this thesis is GlycoSHIELD program, developed at the Max Planck Institute for Biophysics in Frankfurt.[75] The program provides the computer simulations that may be used for the glycan structure modelling and dynamics study.

Another important source GlyGen represent a combination of the possibilities that were demonstrated by previous programs and give a brilliant opportunity to get glycosylation characteristics including the type of glycosylation for any protein models from UniProtKB.[76] The features of that program are based on the glycoinformatics databases and tools that combine information from related disciplines, including genetics, proteomics, pathology, etc.[76]

2.3. EPR spectroscopy

2.3.1. Theoretical background of EPR spectroscopy

One of the key applications of Electron Paramagnetic Resonance (EPR) spectroscopy is to study the structure and dynamics of paramagnetically tagged biomolecules. In particular, EPR is allowing to determine structural ensembles of intrinsically disordered proteins,[77] as well as structures of soluble and membrane proteins. Most common spin labels for bio-EPR applications are based on the nitroxide radicals.[78,79] The basic concepts of EPR spectroscopy are similar to the more broadly used analytical method of Nuclear Magnetic Resonance (NMR). Both methods deal with the interaction of magnetic moments with electromagnetic radiation in static magnetic field. The difference is only that in EPR spectroscopy the interaction of unpaired electron spins are studied, while in NMR the interactions of nuclear spins are relevant. EPR spectroscopy is more sensitive than NMR spectroscopy due to the larger gyromagnetic ratio of electron spins as compared to the gyromagnetic ratio of protons: due to the higher gyromagnetic ratio, in EPR one has higher frequency of resonant electromagnetic radiation in a given static magnetic field and thus larger Boltzmann polarization of electron spins as compared to nuclei polarization in NMR. Longitudinal relaxation times of spin labels at typical pulse EPR experiment temperatures (usually, around 50 K for nitroxide based spin labels) are of the order of few milliseconds. Transverse relaxation times of nitroxide spin labels can vary between several hundreds of nanoseconds and several microseconds, depending on the degree of deuteration of the sample and on the presence of relaxation enhancers, such as *e.g.* paramagnetic metal ions. EPR spectroscopy can be generally divided into two main branches: continuous-wave (CW) and pulse EPR spectroscopy. In this chapter the essential details of spin label dynamics measurements by CW EPR and spin-spin distance distributions measurements with the pulse EPR technique double electron-electron resonance (DEER) are overviewed. These two methods are mainly considered in this thesis. Also, in this thesis nitroxide spin labels will be the main spin label type used. In the following sections a brief overview of EPR basics and key properties of the listed spin labels are given.

2.3.2. Spin interactions and spin Hamiltonian

A ‘spin’ S is a discrete degree of freedom that transforms like an angular momentum under rotations. It is a shorthand for a quantum-mechanical degree of freedom with a discrete set of basis

states $|S_z\rangle$, labeled by the quantum number Sz $= -$S, . . . , +S (the "z basis"). Furthermore, the basis states $|S_z\rangle$ transform like angular momenta under spin-space rotations.[80,2,81]

The interactions of a paramagnetic center in a static external magnetic field B_0 are described by the static spin Hamiltonian:

$$\mathcal{H}_0 = \mathcal{H}_{EZ} + \mathcal{H}_{ZFS} + \mathcal{H}_{EE} + \mathcal{H}_{HF} + \mathcal{H}_{NZ} + \mathcal{H}_{NQ} + \mathcal{H}_{NN},$$

where $\mathcal{H}_{EZ}$ is the electron Zeeman interaction between the electron spin and the external static magnetic field B_0 expressed in an arbitrary coordinate system often. $\mathcal{H}_{EZ}$ is the dominant spin Hamiltonian term:

$$\mathcal{H}_{EZ} = \beta_e \boldsymbol{B_0 g S}/\hbar$$

β_e is the Bohr magneton,

$\boldsymbol{B_0}$ is the transposed magnetic induction vector B_0,

$\boldsymbol{g}$ is the so-called g-tensor, which is connecting the mechanical and the magnetic moment of the electron. If $\boldsymbol{g}$ can be substituted by a scalar value g, it is often used in a form of product $\beta_e g/\hbar = \gamma$, also called gyromagnetic ratio,

S is the electron spin vector operator,

$\hbar$ is Planck 's constant

$\mathcal{H}_{EE}$ term describes the electron-electron dipolar interaction. The electron-electron dipolar interaction is included in the $\mathcal{H}_{ZFS}$ or $\mathcal{H}_{EE}$ dependent on the strength of this interaction. Usually, if electron-electron dipolar interaction can be excited in pulse EPR experiments, this is written as $\mathcal{H}_{EE}$ term. For a weak electron-electron coupling between spins 'A' and 'B' only it's secular part (with respect to the electron Zeeman interaction) has to be considered:

$$\mathcal{H}_{EE} = \hbar\omega_{dd}(3cos^2\theta_{AB} - 1)$$

with

$$\omega_{dd} = \frac{\mu_0 \beta_e^2 g_A g_B}{4\pi\hbar} \cdot \frac{1}{R_{AB}^3}.$$

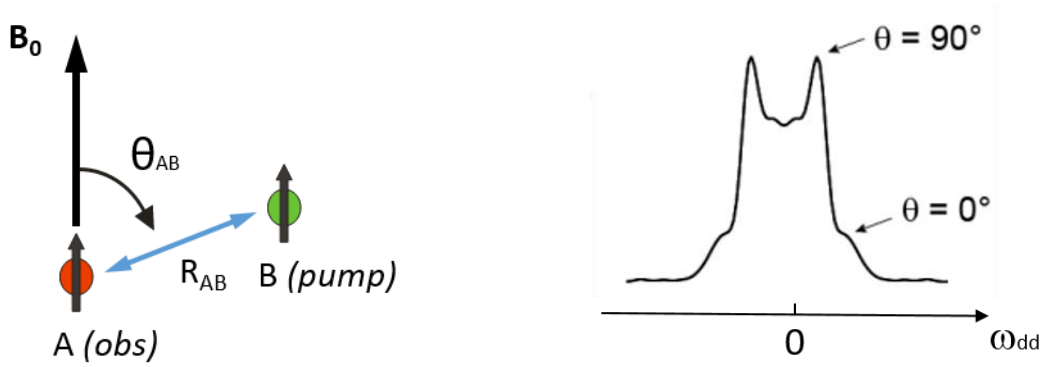


Figure 9. Graphical representation of the dipolar coupling between two spins 1 and 2 with interspin vector R_{AB} that draw an angle with the external magnetic field B_0 (left image) and Pake-pattern (calculated dipolar powder spectrum) for randomly oriented molecules.

In the Figure 9, R_{AB} is the spin-spin vector's length, θ_{AB} is the angle between the spin-spin vector and the static magnetic field, g_A and g_B – the g-values of A-spin and B-spin, and μ_B is the Bohr magneton.

The $\mathcal{H}_{HF}$ term contains all the hyperfine couplings between the electron spin and the surrounding nuclear spins.

$$\mathcal{H}_{HF} = \boldsymbol{SAI}$$

A is the hyperfine tensor,

I is the nuclear spin vector operator

$\mathcal{H}_{NZ}$ is the nuclear Zeeman interaction which describes the coupling of a nuclear spin I to the external field B_0.

$$\mathcal{H}_{NZ} = -g_n \beta_n \boldsymbol{B_0 I}/\hbar$$

I is the spin nuclear number,

g_n is the nuclear g-factor, analogous to the described above electron g-factor,

$\mathcal{H}_{NQ}$ is the nuclear quadrupole interactions term for all nuclei with nuclear spin quantum numbers I>1/2, which is related to non-spherical charge distributions in the corresponding nuclei.

$$\mathcal{H}_{NQ} = \boldsymbol{IPI},$$

$\mathbf{P}$- is the nuclear quadrupole tensor

$\mathcal{H}_{NN}$ is the nuclear spin-spin interactions term. For example, the dipole-dipole interaction between two nuclear spins I_i and I_k can be described as:

$$\mathcal{H}_{NN} = \boldsymbol{I}_i \boldsymbol{d}^{(i,k)} \boldsymbol{I}_k$$

$\boldsymbol{d}^{(i,k)}$ - is the nuclear dipole-coupling tensors.

For nitroxide-based spin labels, nuclear Zeeman, nuclear quadrupole and nuclear spin-spin interactions are not particularly important for the CW EPR and pulse dipolar spectroscopy measurements. Thus, for our purposes, it will be sufficient to restrict the spin Hamiltonian to the electron Zeeman, hyperfine and electron-electron interaction terms:

$$\mathcal{H}_0 = \mathcal{H}_{EZ} + \mathcal{H}_{HF} + \mathcal{H}_{EE}.$$

2.3.3. Basic principles of magnetic resonance

2.3.3.1. Magnetic moment of the electron spin, Zeeman effect and Larmor theorem

An EPR experiment is based on the quantum evolution of the electron magnetic moment $\vec{\mu}$ associated with electron's spin operator $\vec{S}$. For spin S=1/2 it is known that the quantum state of the electron's spin can always be presented in a vector form. Typically, to describe an electron spin in an external magnetic field, the direction of the applied external magnetic field is z-axis. A single unpaired electron has a spin quantum number S=1/2. It becomes quantized with two orientations in its eigenstates: 'up' $|\alpha\rangle$ ($m_s = 1/2$) and 'down' $|\beta\rangle$ ($m_s = -1/2$), with the sense of quantization along the direction of the magnetic field vector $\overrightarrow{B_0}$. The two energy levels of an electron in a magnetic field are differently populated and split by the energy difference ΔE. The population difference between the spin-up and spin-down states is determined by the sample temperature and can be computed from the Boltzmann distribution law.

The Zeeman interaction can also be defined using the magnetic moment operator, as in the following equation:

$$\left|\hat{H}_z = -\hat{\mu}_z B_0 = g_e \mu_B \hbar^{-1} B_0 \hat{S}_z\right|$$

Here, B_0 is the magnitude of the static magnetic field, and $\hat{\mu}_z$ is the z-component of the electron magnetic moment vector operator. This operator is proportional to $\hat{S}_z$ and for the electron with a negative charge –e it is described by:

$$\hat{\mu}_z = -g_e \mu_B \hbar^{-1} \hat{S}_z,$$

where: g_e is g-factor of free electron which is approximately equal 2.002,

$\hbar$ is the reduced Planck's constant,

$\mu_B = e\hbar/2m_e$ is the Bohr magneton.

The Zeeman effect describes the energy difference between spin-up and spin-down states in the applied static external magnetic field. The Zeeman effect determines the relation between the applied field and the resonance energy in EPR. For paramagnetic centers with g-value close to 2, the resonance frequency of 9.5 GHz approximately corresponds to the static magnetic field of 0.35 Tesla, while the EPR frequency of 35 GHz approximately corresponds to the static magnetic field of 1.25 Tesla. Because for an unpaired electron its magnetic moment is coupled to its spin, which is analogous to a mechanical moment, the dynamics of the electron magnetic moment in a static magnetic field follows the Larmor theorem: if the magnetic moment is tilted from the equilibrium orientation parallel to the static magnetic field, then it starts to process around the direction of the magnetic field. The essence of the EPR phenomenon is that if a microwave irradiation at the Larmor frequency is applied to an electron spin in magnetic field, then its precession around the static field direction is initiated, which is accompanied by absorption of the microwave energy.

2.3.3.3. Relaxation, saturation and EPR line shapes

Transverse relaxation time T_2 or phase memory time T_m describes the time scale of the loss of coherence in an ensemble of electron spins. Longitudinal relaxation time T_1 is called spin-lattice relaxation, and it determines how quickly the electron spin system can return to the equilibrium state. Transverse relaxation times are usually measured in a variable-time Hahn echo experiment, while the longitudinal relaxation times are measured either in the Inversion Recovery experiment or in the Saturation Recovery experiment. In pulse EPR, the maximum possible time length of a pulse experiment is determined by the transverse relaxation properties of the paramagnetic species, while the repetition rate is determined by the longitudinal relaxation time. In CW EPR

experiment, for a homogeneous line shape, the peak-to-peak linewidth ΔB_{pp} of a first-derivative EPR spectrum is given by

$$\Delta B_{pp} = \frac{2}{\sqrt{3}\gamma T_2}.$$

Here, γ is the electron magnetogyric ratio.

Relaxation processes are highly dependent on molecular and electronic structure and on the temperature of EPR measurements. The on-resonance amplitude (I) of the detected EPR line can be expressed as

$$I = \frac{1}{1 + \gamma^2 B_1^2 T_1 T_2}$$

where:

B_1 - the microwave magnetic field applied at the sample,

T_1 – the electron spin-lattice relaxation time,

T_2 – the electron spin-spin relaxation time.

This equation demonstrate that the saturation factor s is decreasing with the increase of the microwave power (~B_1^2), while it is increasing for shorter electron spin-lattice relaxation time T_1 and for shorter electron spin-spin relaxation time T_2.

This equation demonstrate that the saturation factor s is decreasing with the increase of the microwave power ($\sim B_1^2$), while it is increasing for shorter electron spin-lattice relaxation time T_1 and for shorter electron spin-spin relaxation time T_2.

The line shape might be inhomogeneously broadened in case the sample has unresolved hyperfine splittings or e.g. a distribution of g-values/a g-tensor anisotropy. In that case the isotropic line shape must be convoluted with the underlying resonance positions distribution, thus forming a complex spectral shape. The homogeneous EPR linewidth can also be dependent on the position in the homogeneously broadened EPR spectrum. When the weak unresolved inhomogeneous contributions dominate, then the EPR line shape is often resembling the Gaussian curve shape, meaning that the EPR spectrum is the sum of a large number of narrower individual homogeneously broadened lines that are each shifted if frequency with respect to others.

Saturation

The saturation factor $S=(\gamma B_1)^2 T_1 T_2$ needs to be considered for the line broadening at high microwave radiation power levels. For the large microwave power regime $(\gamma B_1)^2 > 1/T_1 T_2$, the peak-to-peak intensity increases by a factor 1 + S. If a weak signal needs to be detected with maximum signal-to-noise ratio it is advantageous to increase power beyond the linear regime, but not necessarily to the maximum available power level. For very strong irradiation B_1 the term 1 can be neglected in the denominator of equation below for the magnetization vector components. The on-resonance amplitude of the absorption line is then given by:

$$M_y(\Omega = 0) = M_0/\omega_1 T_1\,, \qquad \langle \omega_1^2 T_1 T_2 \gg 1 \rangle$$

i.e., it is inversely proportional to ω_1. In this regime, the amplitude decreases with increasing power of the microwave irradiation.[82]

2.3.4. Continuous wave EPR spectroscopy and rotational dynamics detection

Various structure and dynamics information can be resolved from the data analysis of continuous wave (CW) EPR, leading to enhanced understanding of protein biophysics.[83] Starting with detecting, quantifying, and understanding paramagnetic species in natural and synthetic materials, more interesting and sophisticated systems/problems have been found to be accessible only to EPR.

CW EPR offers the local dynamics of the labelled protein and report the local polarity, crowding and changes in these upon alteration in the environment. The crucial advantage is the possibility to isolate the labelled site of protein under study without interference of the surrounding biomolecules. CW EPR spectroscopy methodology development in combination with spin labelling (especially when site-directed spin labelling is possible) give a wide spectrum of application in biological problems.[84]

Typical information obtained from CW EPR can also include the depth of a residue being buried inside of the membrane bilayers and the helical structure of the transmembrane domain based on the water and/or oxygen accessibility probed by CW EPR.[85,86]

2.3.4.1. Determination of spin Hamiltonian parameters from CW EPR spectra

Isotropic spin Hamiltonian parameters (or average values from the anisotropic parameters) can be determined from the ambient-temperature CW EPR spectrum in solution. For the rapidly,

isotropically tumbling radical (most conveniently this is measured at X-band), the spin Hamiltonian contains only the isotropic parts of the Zeeman and hyperfine interactions:

$$\hat{H} = g_{iso}\mu_B B_0 \hat{S}_z + \sum_i a_{iso,i} \hat{S}_z I_{z,i}$$

2.3.4.2. Spin label dynamics determination in cell and *in vitro*

EPR spectra also contain information about spin label tumbling in the corresponding environment.[87] In CW EPR spectroscopy on solutions, generally, four motional regimes can be defined, based on gradually decreasing relation between the inverted rotational tumbling time and the spectral anisotropy in frequency units: (i) in the isotropic limit (very fast tumbling), the anisotropy is averaged to zero, and symmetric homogeneous lines are recorded; (ii) in the fast-motion regime (fast tumbling), the system is close to the isotropic limit but some residual effects of the anisotropy are observed, such as differences in the line widths and peak-to-peak amplitudes for the lines originating from the hyperfine sub-spectra with different total anisotropy; (iii) in the slow motion regime (slow tumbling) the line shapes of individual hyperfine components start to deviate from the homogeneous Lorentzian shapes and contain information both on the tumbling time and underlying spectral anisotropy; (iv) in the rigid limit (very slow tumbling), no averaging of the anisotropy occurs anymore and the EPR line shape is essentially the same as in the case of a frozen glassy sample. The Figure 10 illustrates the characteristic spectral shapes of a nitroxide radical in the mentioned four rotational tumbling regimes.[83]

Rotational mobility of a molecule is quantified by the rotational correlation time τ_r, which is the time that is required to rotate through one radian.[88] The relation of τ_r to the temperature T and the viscosity η is given by:

$$\tau_r = \frac{4\pi\eta r^3}{3k_B T}$$

where r is the hydrodynamic radius according to Stokes-Einstein theory, *T* is the temperature, and k_B is the Boltzmann constant.

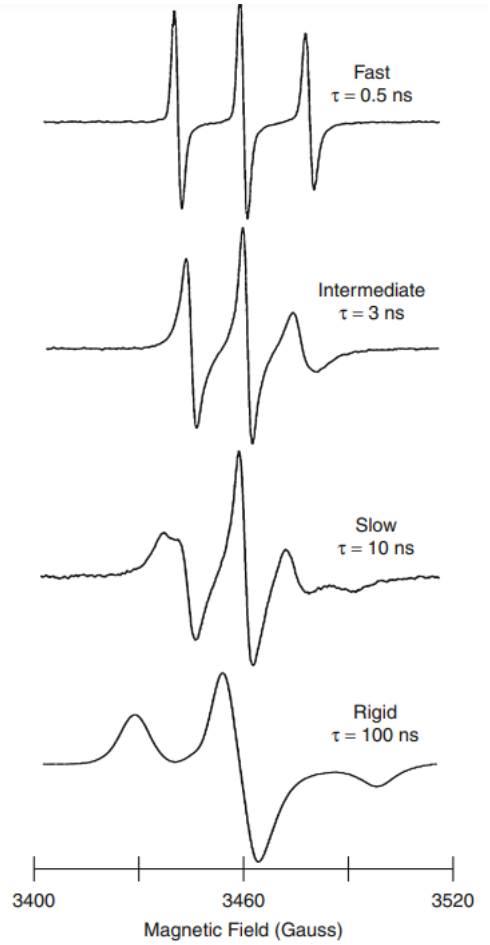


Figure 10. The characteristic spectral shapes of a nitroxide radical in the mentioned four rotational tumbling regimes: free label, fast regime, intermediate regime and slow regime.[89]

The rotational mobility depends on the size and geometry of the molecule and on its environment. If the paramagnetic species is associated with a large molecule, e. g. a protein, the overall dynamics is influenced by the linkage, adjacent secondary structure elements, and the size of the entire conjugate. A free tumbling nitroxide, such as unbound free nitroxide radical, reveals a very fast motion EPR spectrum with three sharp lines of nearly equal height.[90]

The spectrum of nitroxide spin labels is characterized by a g-tensor (Zeeman interaction) and a hyperfine tensor describing the interaction of the unpaired electron with the ^{14}N nucleus in its vicinity.[2] In the case of proteins labelled with nitroxides, the CW-EPR spectrum can be recorded in solution in a wide variety of temperatures, including room temperature (RT). In the following sections of this thesis the advantage of the possibility to obtain from CW EPR spectra information on the dynamics[91] of the spin labels is utilized. Information on the environment (viscosity, polarity, solvent accessibility) of the spin labels can also be obtained in CW EPR.[92] Due to the sensitivity for the dynamic events in the time range of ps-ns, nitroxide-based on site-directed spin labelling (SDSL)-EPR spectroscopy can be used to monitor structural dynamics changes map interaction sites, and monitor folding events[93]. On the other side, pulsed dipolar electron paramagnetic resonance (PD-EPR) experiments can be carried out to characterize protein conformational ensembles, changes in their equilibrium and representation, protein oligomerization and interactions.[94]

2.3.4.3. Spin counting and spin labelling efficiency determination

The number of spins in a protein can be determined in a spin counting experiment. In this experiment, the concentration of biomolecules (*e.g.* protein molecules) is determined from the UV/Vis absorption, and for the same sample the spin concentration is determined from the double integration of the CW EPR spectrum and comparison to the CW EPR spectrum intensity from a reference solution with known spin concentration. By comparing the biomolecule concentration and the spin concentration, one can thus determine the spin labelling efficiency.[95] This is a valuable test which was used in the current work in order to determine the levels of the O-linked glycosylation of the protein (detected via spin labelling).[96] Due to the strong differences in the rotational tumbling regimes between the free spin labels and the attached spin labels, it is usually not difficult to discriminate between the free and the attached spin label component. Sample purification is repeated, in case free spin labels are detected in the CW EPR spectroscopy experiment during the spin counting. The CW EPR spectroscopy measurements for spin counting can be thus considered as an additional sample quality control step.

2.3.5. Rapid-scan spectroscopy

Rapid scan is an emerging technique in EPR spectroscopy. This method has an improved sensitivity as compared to the conventional CW EPR spectroscopy and is characterized by a better time resolution. Regarding its technical realisation, rapid scan method lies in between CW and pulsed EPR spectroscopy and combines the advantages of both techniques.[97] For the rapid scan measurements the microwave power is constant, the magnetic field is centered at resonance. The modulation coils have an additional setup to rapidly scan the spectrum thousands of times per second. Provided that the scan width is large enough, the whole resonance is traversed twice: once with increasing and once with decreasing field. Unlike conventional CW, both the in-phase and out-of-phase component of the reflected microwave are recorded directly in a quadrature detection regime as the voltage output of a crystal detector. The ability to detect a periodic signal that can be averaged to attenuate components that are not coherent with the scan substantially improves the signal-to-noise as it has the same effect of a very narrow comb filter.[98] Moreover, acquisition of the absorption and dispersion allows to apply the Kramers-Kronig relations thereby reducing the required measurement time by half.[99] Since the time on resonance is short compared to relaxation times, higher microwave powers can be applied without saturating the spin system. This is quite

similar to pulsed EPR spectroscopy where the resonance condition is also satisfied only for a short period of time.

2.3.6. Double electron-electron resonance (DEER) spectroscopy

2.3.6.1. Distance distribution determination

A specific advantage of pulse EPR spectroscopy is that the strong broadening due to the g-tensor and hyperfine anisotropy can be removed from the echo signal by refocusing techniques, and thus weak interactions that are otherwise hidden in the inhomogeneously broadened line shapes can be detected.

In the current work the pulsed electron-electron double resonance (abbreviated as PELDOR or DEER, see Figure 11) for the structural investigation of spin labelled proteins with and without post-translational glycosylation is used. DEER technique is most commonly applied to the doubly spin labelled bio-molecules (e.g. with two nitroxide-based spin labels, or with spin labels and/or naturally occurring paramagnetic centers).[2,100] Four microwave pulses are applied at two different frequencies, often referred to as the spin A frequency (detection frequency) and the spin B frequency (pump frequency). The three microwave pulses (the 1st, the 2nd and the 4th, as it is depicted in Figure 11) are applied at the detection frequency (ν_{det}), corresponding to the 'spin A'. The 3rd microwave pulse is generated at the pump frequency (ν_{pump}), the 'B spin'-frequency. The delay τ_1 separates the first two detection pulses.

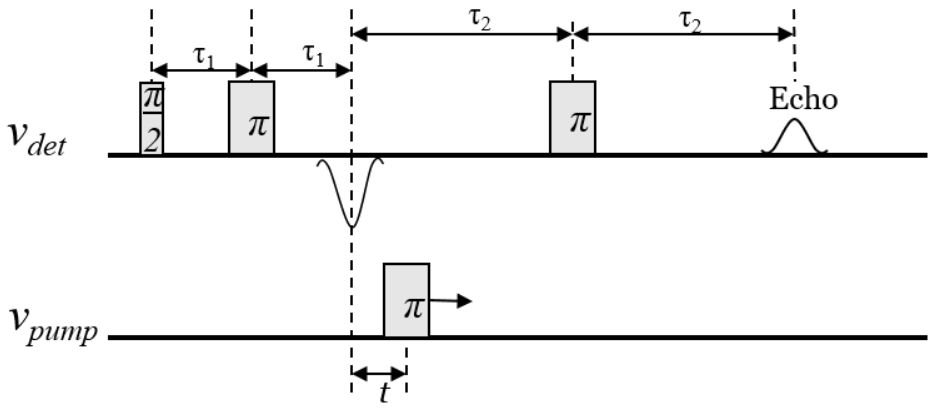


Figure 11. 4-pulse DEER pulse sequence at the observer frequency.

After the first π-pulse Hahn echo is appearing on the A spins. Next, another refocusing on the A spins is achieved by applying a second π-pulse after a time τ_2. During this refocusing, the pump pulse is applied, which is off-resonance with respect to the detected spins, but which inverts a fraction of other spins (B-spins), and thus generates oscillations of the spin echo signal with the electron-electron dipolar coupling frequency.

Due to dipole-dipole coupling between A and B spins, the local magnetic field at the A spins changes so that the precession frequencies of the A spins are shifted. The echo intensity of the A spins recorded as function of the time t yields the DEER time trace, where the echo intensity oscillates with $\cos(\omega_{AB}t)$.

2.3.6.2. DEER data processing

Extraction of distance information from DEER data is possible because interelectron dipolar interaction frequency is proportional to the inverse cube of the spin-spin distance. To obtain reliable data, at least 1.5 full oscillation periods have to be recorded in DEER experiment. The experimental DEER time-domain signal can be written as a product of a 'form factor' arising from the dipolar interactions[101] between the small number of spins (typically two) within a labelled molecule, *F*(*t*), and a background signal, *B*(*t*), arising from a large number of weak intermolecular dipolar interactions:

$$D(t) = F(t) \cdot B(t)$$

Properly accounting for the intermolecular background signal is necessary to determine the desired intramolecular distance distributions that are reflected in *F*(*t*). The intermolecular background signal in the case of a homogeneous distribution of spin labelled molecules in the sample, and the sizes of molecules much smaller than the mean intermolecular distance, can be written as:

$$B(t) = B_0 \cdot exp(-\beta t).$$

With $\beta = \left(2\pi\mu_0 \mathrm{g}^2 \mu_B^2 / 9\sqrt{3}\hbar\right) \cdot N_A c\lambda$. Here, μ_0 is the magnetic permeability of vacuum, g is the electron g-value, μ_B is the Bohr magneton, N_A is the Avogadro number, c is the molar concentration, and λ is the B-spins flip probability of the pump pulse. If spin concentration is not accurately known, or the spin distribution is not homogeneous, then usually the intermolecular

background is fitted and the DEER time trace is divided by the background fit to obtain the form factor trace.

The calculation of the distance distribution $P(R)$ from an experimental time-domain DEER signal $F(t)$ is an ill-posed mathematical problem. Therefore, small variations in $F(t)$ can lead to large variations in the $P(R)$ obtained. A common way to address this issue in the analysis of DEER data is to impose some degree of smoothness on $P(R)$ either by adding an adjustable smoothness factor to the fit criteria via Tikhonov regularization (TR)[102,103,104,105] or by assuming some smooth functional form, such as a sum of Gaussian components, to model $P(R)$.[106,107] In the discrete matrix calculations, conversion of the form factor data to the distance distribution can be written as:

$$\boldsymbol{F} = \boldsymbol{KP}.$$

The corresponding discrete Tikhonov regularisation, as used in the DeerAnalysis software is given by

$$G(\alpha) = \|\boldsymbol{F} - \boldsymbol{KP}\|^2 + \alpha\|\boldsymbol{LP}\|^2 = \rho + \alpha\eta.$$

Here, $\boldsymbol{L}$ is the second derivative operator acting on the distance distribution $\boldsymbol{P}$, and α is called 'regularization parameter.'

Application of the TR method requires that an estimate of the background factor be made *a priori* by fitting the latter portion of the time-domain signal $D(t)$. This estimated background factor enables the determination of a background-corrected signal as an estimate of $F(t)$ that is then analysed to give a distance distribution. Error estimations are typically made *a posteriori* by assessing the effects of the background correction and of the experimental noise at an arbitrary statistical-significance level.

Different types of spin-labels can be used to perform DEER experiments on systems having two spin-labels or a combination of spin labels and native paramagnetic centres. By these experiments, it is possible to extract distance distributions between the paramagnetic centres $P(r)$ in the nanometer range (typically, 1.5-8.0 nm), by measuring the dipolar couplings.[100] It is also possible to study the conformational distribution of a protein or protein domain and eventual changes caused by the protein interaction with a partner biomacromolecule or a ligand. Moreover, by *e.g.* labelling different components of a protein complex, it is possible to characterize the oligomerization state and the respective constant(s).[74, 93, 108] Because of the fast relaxation of typical spin labels and native paramagnetic sites, DEER experiments are generally carried out at cryogenic temperatures

(10-60 K) in frozen glassy solutions. The main drawback of metal-based and Trityl spin-labels is the insensitivity of their CW-EPR spectra to structural dynamic changes of the labelled region; therefore, they can only be used for distance measurements between paramagnetic centres by PD-EPR. In contrast, nitroxide spin labels can be successfully used both for CW EPR and pulse EPR experiments.

2.3.6.3. Some examples of DEER applications

DEER in combination with SDSL, using two spin labels per biomolecule provides a means to measure distances. This technique has been also used to determine protein structures,[109] or to monitor structural changes.[110] Main accent in Malte Drescher's group at University of Konstanz was made on the structural study of intrinsically disordered proteins.[111,112,113] DEER applications were reported for protein structures in different circumstances, such as aggregation or fibrillation.[114] The results of DEER measurements can demonstrate the kinetic structural changes in the protein model under the influence of Cu(II) or metal ion-protein interactions.[115]

The application of EPR spectroscopy to determine the binding of a carbohydrate multivalent GlcNAc derivative to plant lectine wheat germ agglutinin (WGA) was described for the first time in our group.[116] This method allowed to determine the binding mode of the multivalent interaction. Employing a combination of CW EPR and pulsed EPR spectroscopy Patrick Braun and Bettina Nägele presented evidence for the chelatine binding affinity of divalent ligand to GlcNAc binding sites of the WGA dimer.[117]

EPR technique is broadly used also to study different protein modifications in biological systems. For example, an EPR application to examine free radicals in melanin polymers and the human melanoma malignum cells was reported.[118]

It was explored by EPR spectroscopy how spin-labelled fatty acid molecules are binding to human serum albumin (HSA) in human serum.[119] This is a suitable method for the identification of patients with pancreatic adenocarcinoma through detection of its changes of fatty acid binding to albumin.

In previously published EPR studies, mobility parameter values of spin labeled glycocalyxes of different mammalian cell lines were reported.[120] Such mobility measurements report about cells capability to absorb (get chemically attached) sugar on the cell surface through O-linked glycosylation. Another publication about fluorescence microscopy study of the glycocalyx reported similar results.[121]

In our project EPR is considered as a potential tool for studying the O-GlcNAc modification of the protein structure and discovering related modification of the protein functions. For the first time EPR technique was used in the group of Gail E. Fanucci and Zhongwu Guo to investigate the heterogeneous environment of cell surface glycans in select cancer and normal cells. For that goal combination of metabolic glycan engineering (MGE) coupled with nitroxide spin-labelling (SL) was used on the cell surface glycans.[120] Here, this research field will be extended by aiming at spin labelling glycosylated proteins that act in the cytoplasm *i.e.* inside cells.

2.4. Site directed spin labelling

Most of the biomolecules are intrinsically diamagnetic. Some of them might have biologically natural radicals which are involved in metabolism, signalling, and stress response such as oxygen radicals: $HO^{\bullet}$, $NO^{\bullet}$, $O_2^{\bullet}$. The presence of some paramagnetic transition metal ions is possible in protein active sites such as: Cu(II), Mn(II, III, IV), Fe(III), Ni(I, III), Mo(V). Paramagnetic species can originate from the amino acid site chains which are involved in electron transfer: ($Trp^{\bullet}$, $Tyr^{\bullet}$) or semiquinone and flavin cofactors.[13] The absence of too many native paramagnetic species in biomolecules helps avoiding background signal and opens many possibilities to get an EPR signal from the target biomolecule. However, this also means that in order to make protein structure accessible for the study by EPR spectroscopic methods, the protein has to be spin labelled *in cellula* or *in vitro*, and, if possible, this should be done in a site directed way. For the first time a combination of the site-directed spin labelling (SDSL) and EPR spectroscopy was applied in the study of protein structural dynamics.[122,123]

SDSL technique is based on selective labelling of a certain biomolecule site, and thus it requires a chemically specific label attachment reaction to be possible. So far, the main efforts were invested in strategies of SDSL of the proteins, with spin labels targeting specific amino acid sites. The longest employed and still one of the most favoured targets in protein SDSL is the cysteine residue, because of its unique nucleophilicity, low abundance in the protein sequences and handy redox behaviour (Figure 12).[124] The most popular cysteine labelling strategies involve labels containing a thiosulfonate ester, maleimide, or iodoacetamide, and are attached to cysteines via electrophilic substitution or Michael addition.

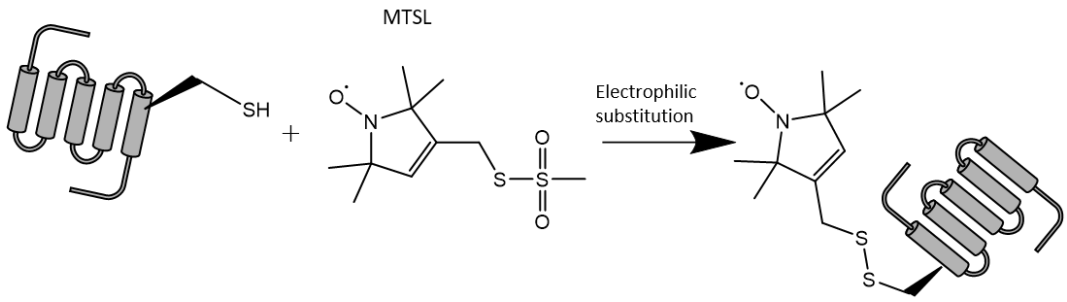


Figure 12. Reaction of the protein's cysteine site labelling by MTSL spin label.

To date, there is a plethora of available spin-label designs. Their varieties might be classified in two main groups by the bioconjugation reactive specificity to the object of study and their paramagnetic properties in EPR evaluation content. SDSL-EPR provides insights into the structural dynamics and conformational ensembles of various macromolecules, including proteins, lipids, nucleic acids and their interactions.[84] The suitable design of spin label and bioconjugation reaction of spin labels to the object of study, in our case protein, is the key task. During the last decade, spin labelling development has made further immense progress, due to improving of the possibility for *in cellula* bioconjugation and increasing the environment stability of spin labels.

Different types of spin labels for SDSL are described in literature: nitroxyl radicals (Methanethiosulfonate spin label (MTSL), 3-Maleimido-PROXYL (M-Proxyl), IA-Proxyl); metal-based labels (Gd(III), Fe(III), Cu(II)…); triarylmethyl radicals (Trityl or TAM) and photo-indusible (photo accesible) spin labels.[125,126] For the cases, when the targeting of cysteines is not possible or not convenient, spin-labels specific for Tyrosines[127] or non-canonical amino acids (ncAA).[128]with corresponding reaction site for the spin labelling reaction have been designed (Figure 13).[129] In case of non-canonical amino acids in use, the SDSL specificity is achieved by using the specific reactions, not found in cells: currently, the most popular approach is based on variants of click reaction (Figure 13). In our work the nitroxide-based spin labels were used.

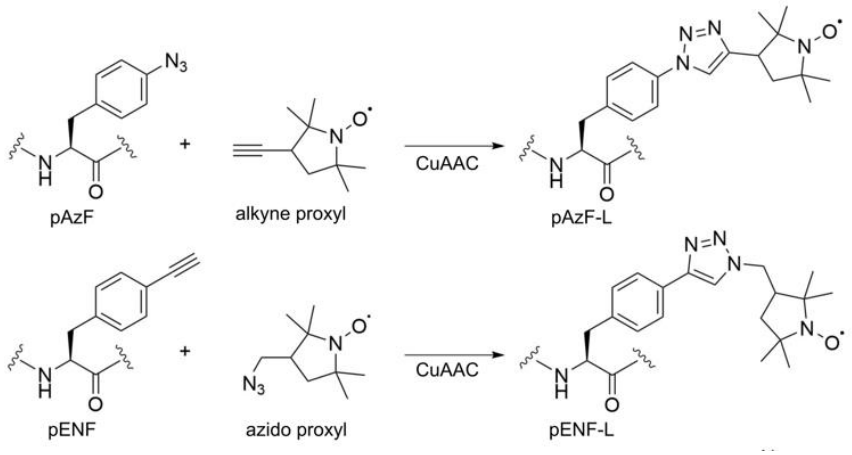


Figure 13. Spin labelling of non-canonical amino acids (ncAA) with azido- or alkine- reaction sites for site-directed spin labelling (SDSL) via CuAAC.[129]

Globally, in the present thesis project, the aim was to exploit spin labelling for *in vitro* and *in cellula* EPR applications. As paramagnetic centers, nitroxides are very convenient, exhibit no background signal in cells, and offer the possibility to extract auxiliary information from low-temperature and room-temperature CW EPR as well as from additional pulse EPR experiments. Pioneering work on *in vitro*, and *in cellula* EPR has mostly exploited labelling strategies based on nitroxide radicals. Up to now, numerous biomolecular structures have been refined and modified to customize spin label properties for specific applications.

However, nitroxides in general suffer from limited lifetime in the *in-cellula* or *in-vitro* reducing environments.[130] To address this problem, the paramagnetic moieties of nitroxide radicals have been further stabilized by substituting shielding methyl groups with alkyl groups and by embedding them in more stable pyrrolidine systems.[131]

The typical variants of the SDSL-EPR experiment design generally make use of one or two residues available for spin labelling: Singly labelled proteins can be used for continuous-wave EPR (CW-EPR) experiments to determine the local dynamics of the spin-label.[132] Doubly labelled proteins are used for DEER[101] or analogous experiments to obtain structural constraints.

2.4.1. The stability of the radicals in biological systems

The currently most commonly used nitroxide spin-labels are low-spin (S = ½) paramagnetic centres, and are stable radicals in which the unpaired electron is delocalized on the N-O group inserted into a heterocyclic five-member ring (pyrrolidine, see Figure 14). The target bimolecular site and its chemical properties determine the conjugation strategy as well as stability and rigidity of the

resulting bond. While labelling strategy and linker stability have to be considered for experimental approach, linker rigidity (see Figure 14) has a high impact on spectroscopic characteristics, decides about the label's usability to sense environmental changes, and defines accuracy in label-to-label distance determination.

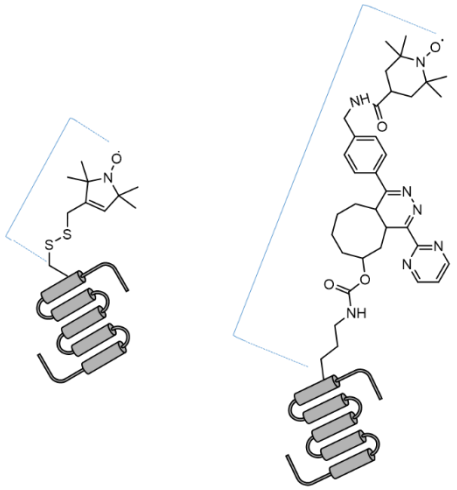


Figure 14. Chemical structures of nitroxides with a different linkers and stable radicals with five – and six- heterocyclic rings.

The stability requirements for the *in cellula* EPR experiments are stricter than for the *in vitro* EPR, due to the often quite strong radical's reduction in cells. Some known cell-stable spin labels are not well water soluble, such as many trityl radicals, thus impeding in-cell labelling. The stability of the nitroxide radicals is highly dependent of the structure of the radical and the nature of cellular environment. Nitroxide radicals are among the most stable radicals in biological environments in cases when the access to the oxygen from the air persists, so that the radicals can be recovered by oxidation. However, in the oxygen-less environment, the reduction of nitroxides limits many EPR experiments, especially at room temperature. In the same time, *in cellula* EPR applications often demand longer incubation times at physiological temperatures, as for instance in an application where slow folding kinetics were involved.[130]

Shock-freezing of the sample can prevent complete spin label loss due to reduction, so that the pulse EPR experiments, such as DEER can be performed *in cellula* at cryogenic temperatures. However, in turn, such freezing precludes *in cellular* ambient temperature CW EPR applications. Alternative spin label compounds might need to be considered for the *in cellula* studies by EPR spectroscopy. Still, the sensitivity of nitroxide's reduction in biological environments is not yet fully systematically studied and it is thus difficult to accurately predict its range of applicability in the *in cellula* EPR.

2.4.2. Spin-labelling approaches for glycoproteins & chemical reporters on the sugars

2.4.2.1. Cu(I)-catalyzed azide-alkyne 1,3-dipolar cycloaddition (CuAAC)

Regarding to the different chemical reporters on the sugar moieties variety of spin labelling reactions are possible.

The term “Click chemistry” was introduced in 2001 year by Karl Barry Sharpless. Azide-alkyne cycloaddition reaction was retrieved due to the interest in the materials, pharmaceutical, and other industrial fields in terms of capacity to generate different varieties of compounds within the research aims In 2002, the Meldal[133] and Sharpless[134] laboratories individually brought forward the ability of copper(I) salts in accelerating this cycloaddition at room temperature or with moderate heating, furnishing exclusively the 1,4-regioisomer with minimal workup and purification. Cu(I)-catalyzed azide-alkyne 1,3-dipolar cycloaddition (CuAAC), well known as the “click chemistry reaction”. 1,3-dipolar cycloaddition reaction of organic azides and alkynes, producing covalently linked molecules via a 1,2,3-triazole.[135]

Click chemistry azide-alkyne cycloaddition reaction is thermodynamically-favored reaction and fulfilling the most important criteria of one product yield production. The necessary components of the reaction are commercially available and monosubstituted alkynes and azides can be easily synthesized. In our case alkyne proxyl with nitroxide radical is reacting with azido group of the sugars that is used for the MGE. The reaction is conducted under aqueous conditions, at room temperature (see Figure 15 and Figure 16).

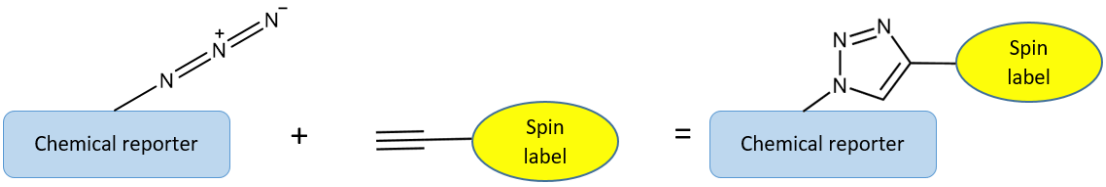


Figure 15. Representation of the basic principle of spin labelling of a chemical reporter via the copper-catalyzed azide-alkyne cycloaddition (CuAAC) reaction.

The active catalyst Cu(I) is generated from Cu(II) salts, in our case it is $CuSO_4$, using sodium ascorbate as the reducing agent. At the same time, sodium ascorbate is an efficient reduction agent of the nitroxide radical. Thus, it turned necessary to use an optimized reaction protocol with

reduced levels of sodium ascorbate, in order to keep sufficient amounts of intact paramagnetically active spin labels. [129,136] An extensive search for a replacement reaction is undergoing.

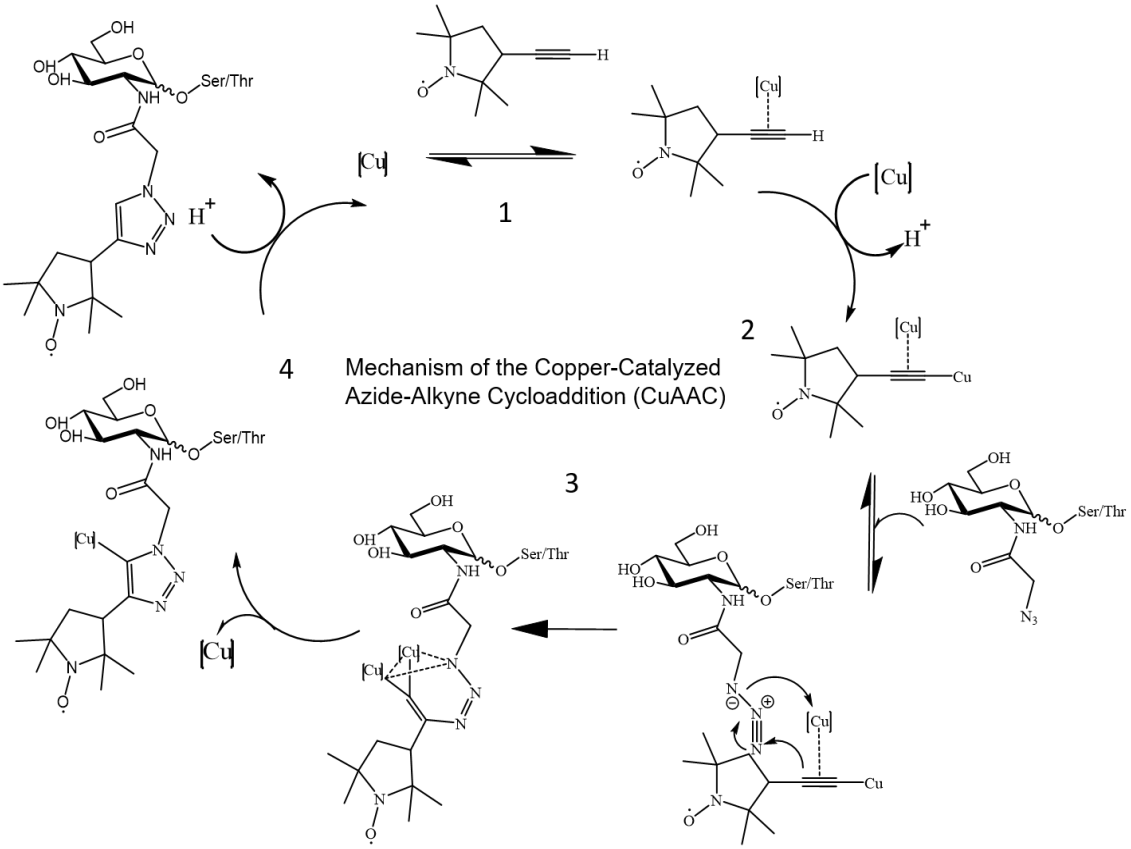


Figure 16. The mechanism of the copper-catalyzed azide-alkyne cycloaddition (CuAAC) for the specific case of spin labelling.[137]

2.4.2.2. The Diels–Alder reaction with inverse electron demand

It was shown that the Diels–Alder reaction with inverse electron demand (DA_{inv}) of 1,2,4,5-tetrazines[138] with strained dienophiles, such as trans-cyclooctenes fulfils the requirements of a bioorthogonal ligation (see Figure 17).

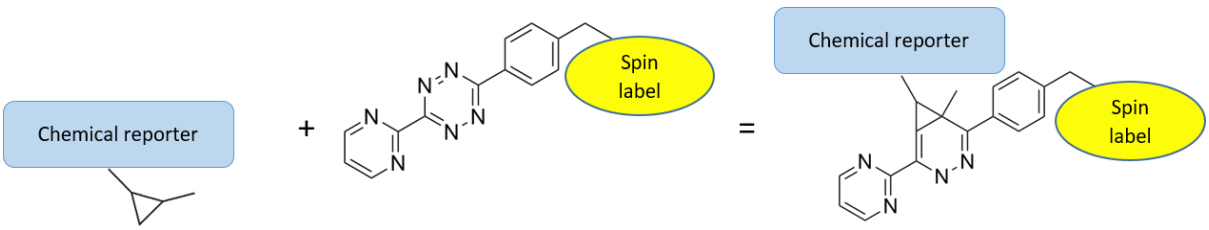


Figure 17. Representation of the basic principle of spin labelling of a chemical reporter via Diels–Alder with inverse electron demand (DAinv) reaction.

Usually, the cyclic alkenes or kinetically stable tetrazines[130, 139] are expected to be too large for being efficiently metabolized by biosynthetic pathway. Application of the DA_{inv} between terminal alkenes and 1,2,4,5-tetrazines in the prep- aration of carbohydrate microarrays was also reported.[140] The fact that terminal alkenes are hardly found in biological systems and are completely absent in proteins makes them a promising reporter group.[140] The schematic orbital representation of the DA_{inv} with normal and inverse electron demand can be seen below in the Figure 18.

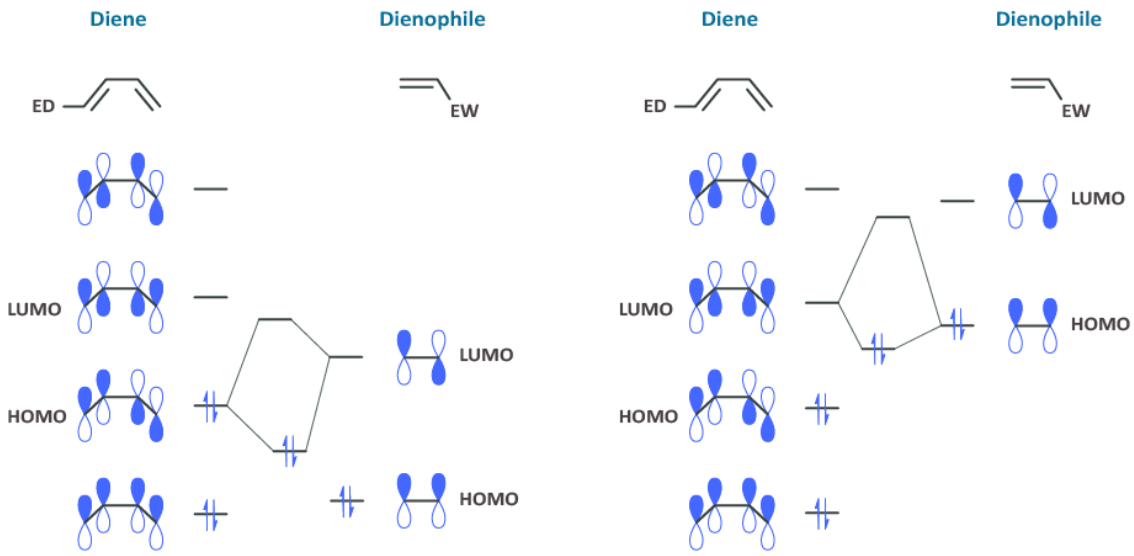


Figure 18. Orbital schemes for Diels-Alder reactions with (left image) normal and (right image) inverse electron demand. ED = electron donating substituent, EW = electron withdrawing substituent. Based on [141, 142]

The Diels-Alder type reaction - tetrazine-trans-cyclooctene ligation constitutes a non-toxic biomolecule labelling method of unparalleled speed that is ideally suited for *in vivo* / *in cellula* labelling at low concentrations. A tetrazine functionalized molecule A reacts with a trans-cyclooctene (TCO)-functionalized molecule B thereby forming a stable conjugate via a dihydropyrazine moiety.

2.5. Fluorescent labelling spectroscopy for understanding the biological processes in glycoproteins

As already discussed, the usage of chemical reporters for the detection of protein glycosylation is better known as metabolic glycoengineering (MGE) or metabolic oligosaccharide engineering (MOE) (in particular, such approaches are being developed and applied by the groups of Prof. Dr. Wittmann and Prof. Dr. Zumbusch at the University of Konstanz).[143] Mostly microscopic tools like fluorescence-lifetime imaging microscopy (FLIM) in combination with Förster resonance energy transfer (FRET), have been reported for the visualization of fluorescently labeled carbohydrates in whole glycome of PTM proteins.[143,144,145,121,146]

FLIM-FRET is a technique to determine POI localisation and specific interactions with the complementarily labelled target/partner molecule. The advantage of FLIM technique is that instead of light intensity, the contrast in the micro-image is given by the changes of the fluorescence lifetime, i.e. the picture obtained is less prone to random artefacts due to the fluorescence intensity fluctuations, and is much more accurately reporting the local environment conditions changes for the fluorophores.

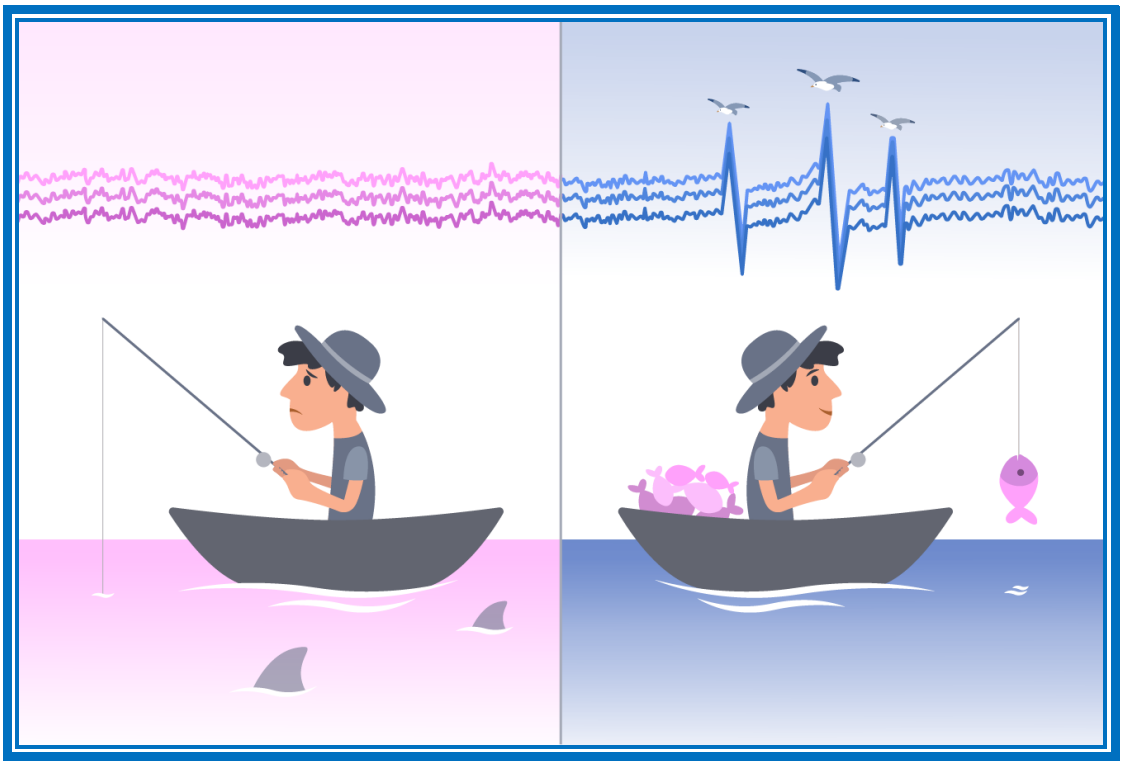
These advantages come at a price that not all the powerful features of the FRET techniques are available in the FLIM-FRET. First, being an in-cell technique, FLIM-FRET is no longer easy to use as a single-molecule method – rather local properties of small ensembles of fluorescent labels are probed. Second, due to the sensitivity of the fluorescence time to a number of environment conditions, such as pH, temperature, solvent, and local viscosity, the efficiency and accuracy of distance determination with FRET is getting significantly compromised.

Under these circumstances it appears advantageous to exploit the DEER spectroscopy for obtaining distance constraints, since DEER is also on average better equipped than FRET in the aspect of accurate distance and distance distribution determination.[147,148] However, modifications of the MGE/MOE approach developed for fluorescent labelling are required before this can be used in SDSL. Most importantly, in EPR and especially in the pulse EPR one requires significantly larger amounts of labelled molecules as well as better labelling efficiencies per target site. As will be described in detail in the results section of this thesis, these MGE/MOE modification challenges turned out to be quite difficult to address. As a result, in this thesis the SDSL optimization for glycoproteins with only a selected amount of proof of principle EPR results reported will be mainly described. Nevertheless, the usability of DEER spectroscopy in structural studies is nowadays well

established and does not necessarily need further proofs. And, thus, one can state beforehand that establishing the appropriate quality SDSL routines can definitely add a useful and powerful new tool to the toolkit of glycosylation studies.

CHAPTER III

Designing strategy for the biological sample preparation



"Data is great, but strategy is better."
Steven Sinofsky

3. Designing strategy for the biological sample preparation

Before the experimental results are described, it is necessary to explain in detail the evaluation of different approaches used in the EPR-related protocols, as well as the criteria for the protein model selection. The present chapter thus serves as a preliminary analysis for the experimental results chapter.

Altogether, this chapter and the following chapter "Main results and discussion" summarize the results of developing an O-GlcNAc-specific spin labelling approach, applicable for structural studies using EPR spectroscopy. The structure and location of glycosylated sites is a largely unexplored topic in EPR spectroscopy. At the same time, EPR spectroscopy is a promising method to study of O-GlcNAc modified proteins. The following topics were addressed in particular: development of the biological sample preparation of O-GlcNAcylated protein for the study *in vitro*; finding the suitable protein models for the study of the O-GlcNAcylation modification by EPR spectroscopy; proteins investigation for in cellular O-GlcNAcylation via MGE. After this, EPR approach optimization/method establishment of glycosylation identification was performed, followed by the evaluation of theprotein modification level, inhibitors influence, as well as identification of protein-specific glycosylation site by DEER spectroscopy.

3.1. The general strategy of O-GlcNAcylated protein structure study by EPR

In general, one can say that the MGE in combination with fluorescence mainly address the problems of intermolecular interactions, partitioning of glycoproteins in different cell compartments, local glycoprotein concentrations and similar. Here, an attempt is made to extend this methodology to target also structural questions at molecular or atomic level (*i.e*. at nanometer or sub-nanometer length scale). Due to some critical differences between EPR spectroscopy and fluorescent spectroscopy (required labelling efficiencies, glycoprotein concentrations), changes and optimizations are required at many steps of the general sample preparation protocol.

An example of the MGE sample preparation can be summarized in the following steps:

1. Expression of the target protein in mammalian cells (HEK, HeLa, …);
2. In cell labelling, by adding sugar with the specific biorthogonal reactive site;
3. In cell studies (e.g. different types of single-molecule fluorescence experiments).

In EPR, so far the spin labelling was mostly targeting unmodified cysteines or unnatural amino acids.[129,128] An example coarse-grained overview of SDSL on a target protein can be described as follows:

1. Protein expression is typically performed in prokaryotic *E.coli* cells

2. The protein of interest is purified and spin-labelled *in vitro*

3. EPR experiments are performed either *in vitro* or *in vivo*. In the latter case, spin labelled protein is introduced into the target cells, which may be different from *E.coli* (*e.g*. by electroporation, osmotic shock, microinjection).

The goal of this thesis is to combine the EPR-related points with the specific restrictions of the MGE-related protocols. From the comparison with fluorescent studies, one can immediately recognize some key points:

1. For EPR, the concentration of labelled protein lies, in the micromolar range, whereas fluorescence measurements can be performed using sub-picomolar amounts of labelled molecules, the latter better suiting the typical physiological protein concentration.

2. Related to the first point, protein expression yields need to be enhanced. The mammalian cell-based expression provides substantially lower yields as compared to the expression in the *E.coli* expression system. However, *E.coli* expression system is not appropriate for glycosylation, and thus cannot be used for our purposes.

3. Due to the need of performing the spin labelling and some EPR experiments *in vitro***,** a suitable protein purification protocol needs to be developed for some EPR experiments *in vitro.*

4. A protocol for spin labelling of the glycan groups of a glycoprotein needs to be established. Here one can partially rely on the previous fluorescence studies and on the work of attaching spin labels to glycocalyx.[121,143]

The overall workflow thus can be described as depicted in Figure 19, and consists of the following steps.

Expression of the target protein in mammalian cells;

Performing MGE with added modified sugar (chemical reporter), which contains reactive site for the *in vitro* and in the future perspective *in cellula* bio-orthogonal labelling;

Isolating and purifying the modified protein of interest from the cell environment;

Spin labelling of the O-linked glycosylated protein, followed by an additional purification to remove the unattached spin labels;

EPR measurements *in vitro* to detect and quantify the O-GlcNAc modification of the protein of interest.

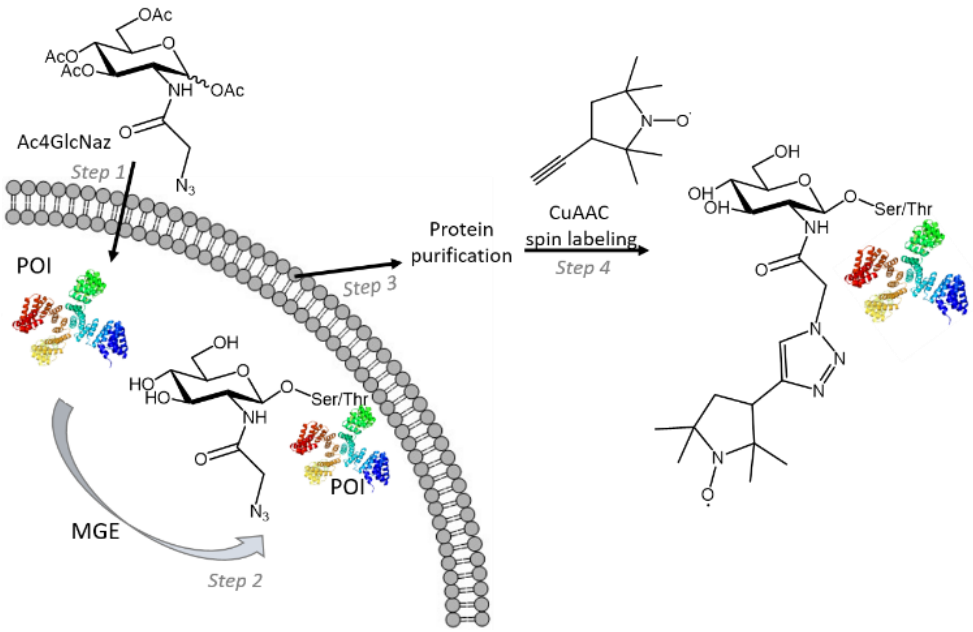


Figure 19. Schematic representation of the general experimental strategy. The target protein is expressed in HEK293T cells. $Ac_4GlcNAz$, after penetration to the cells, is deacylated and converted to GlcNAz for further metabolic pathway. After MGE and protein purification, the attached azido sugar is selectively labelled via Copper click chemistry with Alkyne Proxyl spin label for EPR investigation.

3.2. Optimisation of the biological sample preparation

3.2.1. Suitable mammalian cell line and modified sugar selection

Glycosylation processes are going differently in different types of cells. Mammalian proteins which require various kinds of PTMs are ideally expressed in mammalian cells. Our primary question was: Which mammalian cell line is better for the protein expression accompanied by the O-linked glycosylation and MGE?

Accurate literature study revealed the relation of the question under study with the glycosylation level on the glycocalyx. Glycocalyx is a dense layer of glycans on the cell membrane surface. Changes in the glycocalyx composition quite often correlate with glycosylation processes in cells. Since our strategic methodology assumes the penetration of the sugar moieties through the cell membranes, the amount of the sugar delivered into the cell membrane is within main interest. In that way it is important to avoid the high level of modified sugar losses due to its attachment to the

glycocalyx on the cell surfaces. For the first time cell surface glycans imaging with biorthogonal chemical reporters was done by C. R. Bertrozzi.[149] Recently the study of Jeremias E.G.A. Dold in the Valentin Wittmann's group at the University of Konstanz has demonstrated the effect of MGE on cellular sialic acid levels.[121] That study reported remarkable differences between the cell interior and the cell surface. The similar work was performed by the groups of Gail E. Fanucci and Zhongwu Guo, using for the first time EPR spectroscopy as an analytical method in glycocalyx study.[37] They exploited the similar strategy of azides incorpioration into cell surface glycans followed by a click chemistry reaction with dibenzocy- clooctyne-N-(3-oxo-propylamine) nitroxide spin label. Both publications, the EPR and the fluorescence microscopy study, demonstrated that on the surface of HEK 293T cells the attachment of Ac4GlcNAz sugar is not happening, while this is an abundant process on other cell lines like Ls174T, HepG2 or HeLa.[120]

The selection of the type of O-linked glycosylation is determined by the need to target proteins in cytoplasm and cellular nucleus (and not the ones on the cell membrane). To this respect, O-GlcNAcylation has been identified to specifically add sugar mostly on the cytoplasmic or nuclear proteins that remain inside the cells: the mentioned work of Dold and Wittmann reported MGE investigation in the presence of Ac_4GlcNAz and demonstrated quite low level of Ac_4GlcNAz attached to the cell membranes (as compared to other sugars).[121] These results determined our choice to use the HEK 293T cell line as the protein expression system in our spin labelling study.

The enzymatic activity of OGT on adding sugars to Ser/Thr sites includes both reactions with uridine diphosphate (UDP)-GlcNAc and UDP-GalNAc, thus giving a possibility for the incorporation of GalNAc or GlcNAc sugars, native in mammalian cells. In that way GlcNAz is a substrate for the GlcNAc Salvage Pathway in cells, as it was described earlier in the second chapter (see Figure 5). The enzymatic activity of OGT on adding sugars to Ser/Thr sites can tolerate uridine diphosphate (UDP)-GlcNAz which gives a possibility to accept Ac_4GlcNAz in the glycosylation reaction. In relation to these facts, Ac_4GlcNAz sugar was used for MGE in our experiment.

3.2.2. Protein expression methods

The principle of the protein expression from the DNA plasmid follows the central dogma in molecular biology: the genetic information is encoded in DNA, which is then transcribed to RNA and translated to protein sequence. Most DNA constructs utilize viral promoters (CMV, RSV, SV40) for mammalian expression. To convert DNA into working protein, the translation process involving RNA

is involved. During transcription process, the genetic information in DNA is transferred to a messenger RNA (mRNA). After that, mature mRNA molecules have to leave the nucleus and move to the cytoplasm where in cytoplasm the ribosomes are located. In the next step, the translation of mRNA started with the formation of a complex structure on the mRNA. And after elongation phase in the translation process, peptide bonds between the first and second neighbour amino acids are formed by a peptidyl transferase activity. The schematic representation of the DNA translation to polypeptide (protein) is demonstrated in Figure 20 (left).

As discussed above, the complex process of protein translation and post-translational glycosylation requires the protein expression to be conducted in mammalian cells: the transient expression in mammalian cells for PTMs and appropriate protein folding along with the glycosylation followed by the desired downstream MGE is necessary. The selected protein expression method should also be efficient enough for the high-quality protein production and for its good quantitative yield.

In the presented project, for the major part of the experiments, calcium-phosphate mediated DNA transfer into HEK293T cells and expression with Lipofectamine transfection reagent have been chosen. HEK293T cells were transiently transfected by calcium-phosphate co-precipitation using the required DNA plasmid and additional components for transfection: MQ water, 2x HBS buffer and $CaCl_2$ solution (see Materials and Methods for detailed protocol).

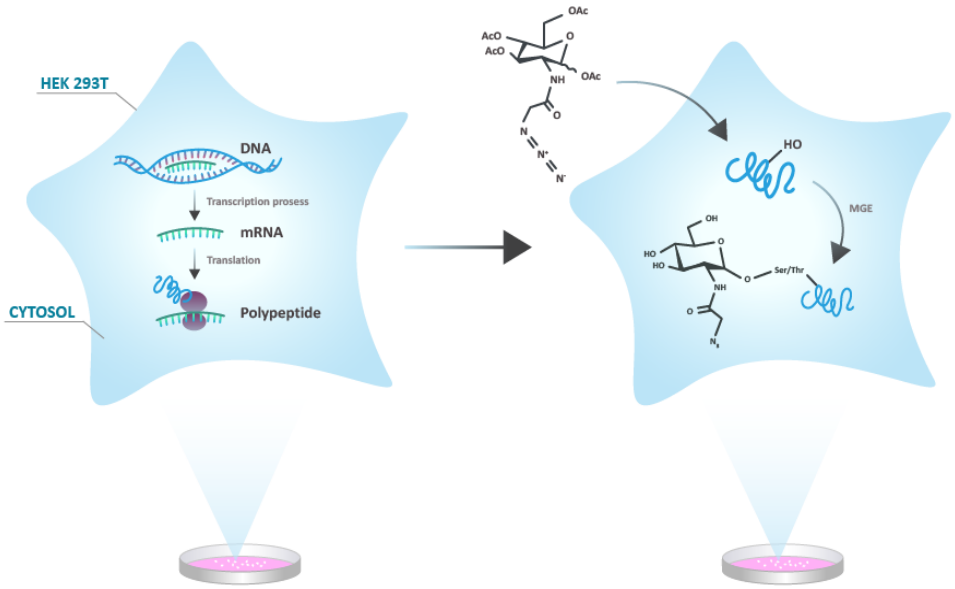
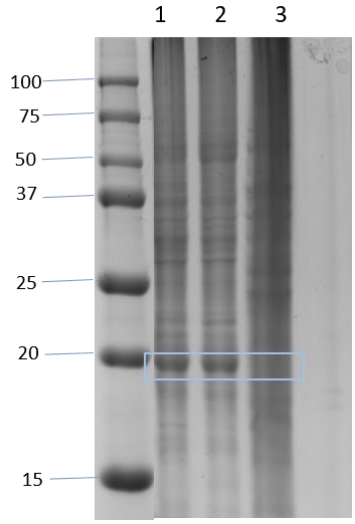


Figure 20. Schematic representation of the *in cellula* experimental strategy. The target DNA plasmid delivered in HEK293T cells to express the protein of interest (left). On the second cellular experimental step Ac_4GlcNAz is deacylated and converted to GlcNAz for further salvage pathway that lead to the protein O-linked glycosylation (right).

While for the most experiments this was the basic expression method, for delicate plasmids Lipofectamine transfection methods was used. In case of the Lipofectamine-mediated transfection, commercially available Lipofectamine 2000 Transfection Reagent together with Opti-MEM medium were used according to the protocol (described in Materials and Methods).

If transfection is successful, both methods produce high protein yields (see Figure 21).



1. Lysate of HEK293T cells after expression with Alpha-crystallin DNA plasmid using calcium-phosphate transfection methods;
2. Lysate of HEK293T cells after expression with Alpha-crystallin DNA plasmid using lipofectamine mediated DNA transfection methods;
3. Lysate of HEK293T cells before expression.

Figure 21. SDS-PAGE analysis of lysated HEK293T cells before and after expression with Alpha-crystallin DNA plasmid using calcium-phosphate and lipofectamine mediated DNA transfection methods.

The work with small amounts of the transfection cells gave the possibility to use Lipofectamine method, while for large amounts of the cells to be transfected the calcium-phosphate co-precipitation method was needed due to the experimental costs - Lipofectamine 2000 Transfection Reagent is quite expensive.

3.2.3. Metabolic glycan engineering

MGE is a unique strategy which allows delivering sugar moieties with required chemical reporter into the cells with expressed protein of interest. For the first time chemical reporters (see Chapter II Figure 6) were developed to target O-linked protein glycosylation. An appropriate chemical reporter has to be accepted and processed *in cellula* by the enzymes, which drive the metabolic process of glycosylation, and then attached at the protein glycosylation sites. To achieve this goal, some crucial parameters like permeability into the cells and diffusion in the cytoplasm had to be prescribed (covered). Regarding the fact that most chemical reporters barely get through the cellular membranes due to their polarity, the acetyl-group in case of the Ac_4GalNAz or Ac_4GlcNAz

sugar provides peracetylated form which is making the chemical reporter more hydrophobic and, as a result, better cell permeable (see Figure 22). After integration into the cell, the acetyl-group is removed by cellular esterases.[150]

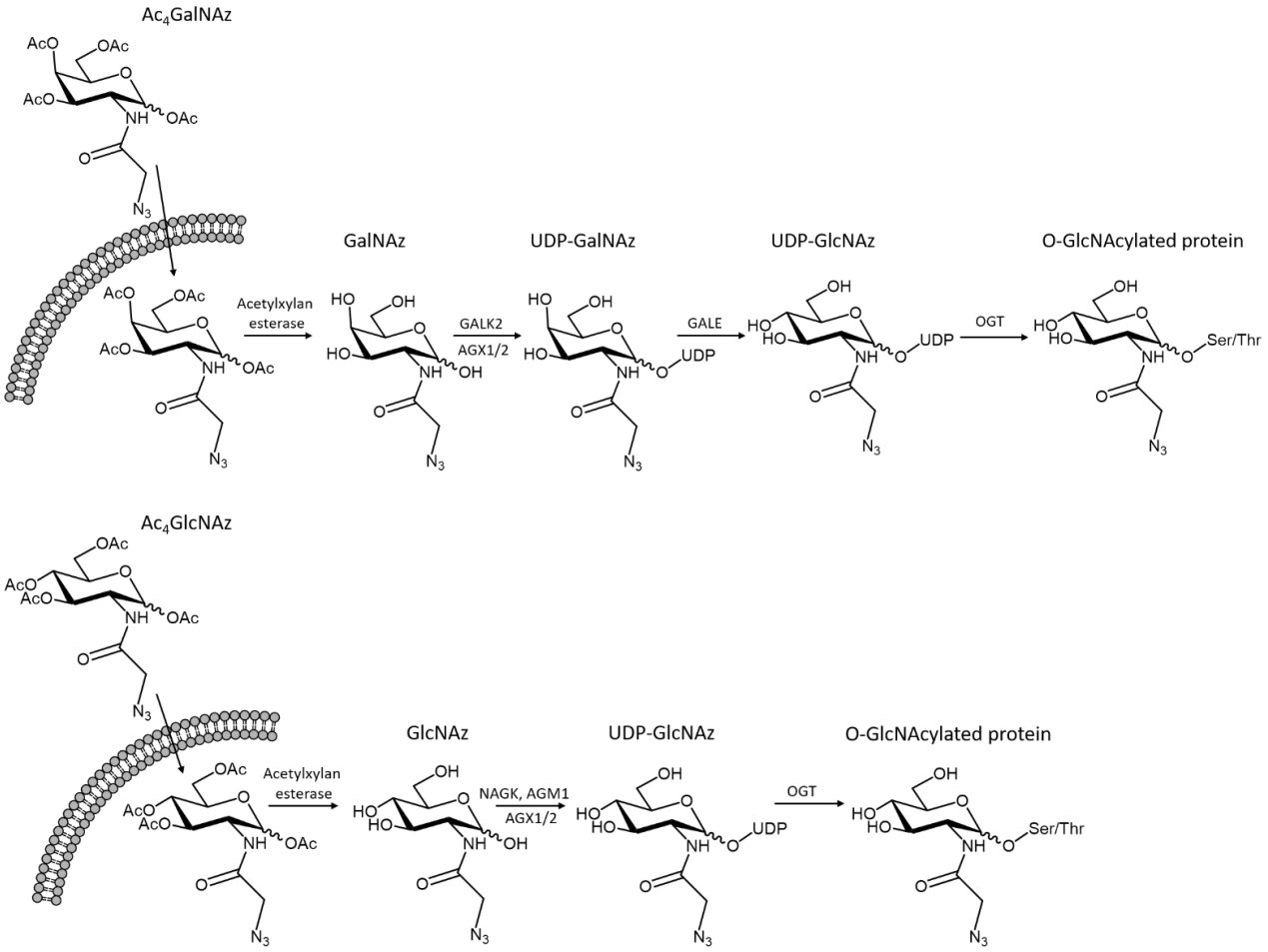


Figure 22. Schematic representation of the MGE strategy. Ac_4GalNAz and Ac_4GlcNAz accepted by metabolic pass as unnatural GlcNAc analog. In case of adding Ac_4GalNAz isomerization produce UDP-GlcNAz by the epimerase UDP-glucose 4-epimerase (GALE). In case of Ac_4GlcNAz UDP-GlcNAz formed without isomerization step in its own salvage passway.

For the O-GlcNAz protein modification UDP-GlcNAz form is required. A good acceptance by all enzymes (GlcNAc kinase, GlcNAc phosphomutase, and UDP-GlcNAc or UDP-GalNAc) is needed to form UDP-GlcNAz. A greatly increased nucleo-cytoplasmic protein glycosylation with O-linked GlcNAz (so called O-GlcNAzylation) have been proven.

3.2.4. Protein Purification Tag

Regardless of the primary goal to establish the in-cell EPR methodology for glycosylated proteins, in this work it was necessary to conduct both spin labelling and EPR measurements *in vitro*, in the interest of the method validation and quantification. For that purpose, the O-GlcNAcylated protein had to be purified. Nowadays different methods of purification are known and probably the most broadly used and efficient one is the tagging purification by targeting columns. To use this protein purification strategy, cloning gene of interest has to contain a protein tag before it will be transfected into cells for the protein expression. In this case the resulting protein allows to be captured by the purification matrix. A wide variety of protein tags are available, in our work His-tag, HA-HA-tag, GFP-tag were tested (see Figure 23).

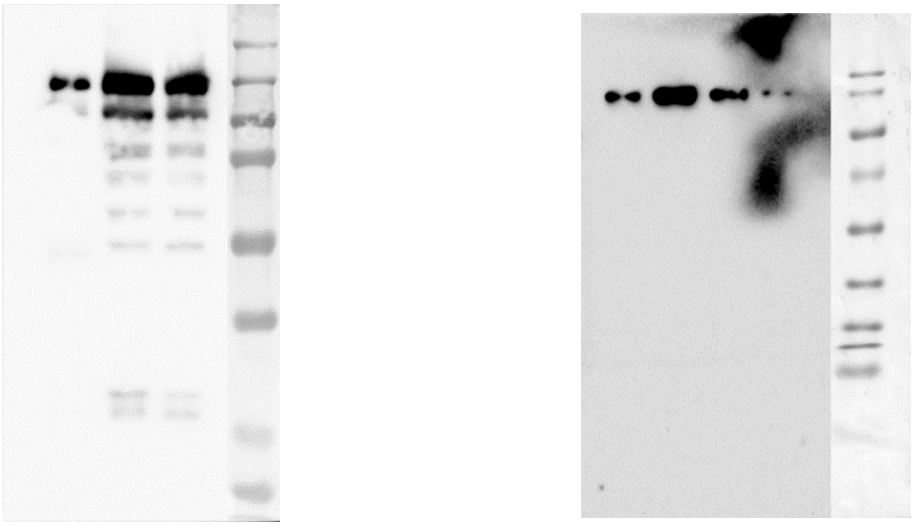


Figure 23. Western blot images as a quality control of OGT protein purification with HA-HA-tag (left image)and GFP-tag (right image) using anti OGT antibody.

GFP-tag has demonstrated a very good performance in the purification procedure and could be also used for microscopy control experiments. However, due to its high price, the GFP-tag was not feasible to use when one had to scale the protein production up in order to produce the sample amounts sufficient for EPR experiments. Therefore, while GFP-tag was regularly used in the preliminary small-scale tests, the final protein production and purification was relying on the His-tag and Ni-NTA column purification technique.

3.2.5. Lysation and protein purification

For the cell lysate extraction one can use a handful of methods including viral, enzymic, chemical, osmotic or acoustic (sonication) methods. Chemical lysis is the most commonly used method, which was exploited to deal with small amounts of cells (up to 2-3 10 cm Petry dishes). The chemical lysis required the following reagents: MQ water, EDTA, NaF, NaCl, 5% Triton X-100, β-glycerophosphate, $NaVO_3$, Tris. Sonication was used as a substitute for the chemical lysis (see Figure 24), as the most suitable method for large cell culture volumes (after starting to work with 30 cm Petry dishes to produce EPR samples).

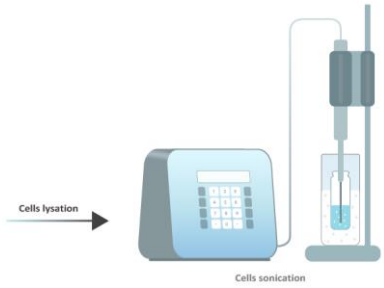


Figure 24. Sonication lysation system

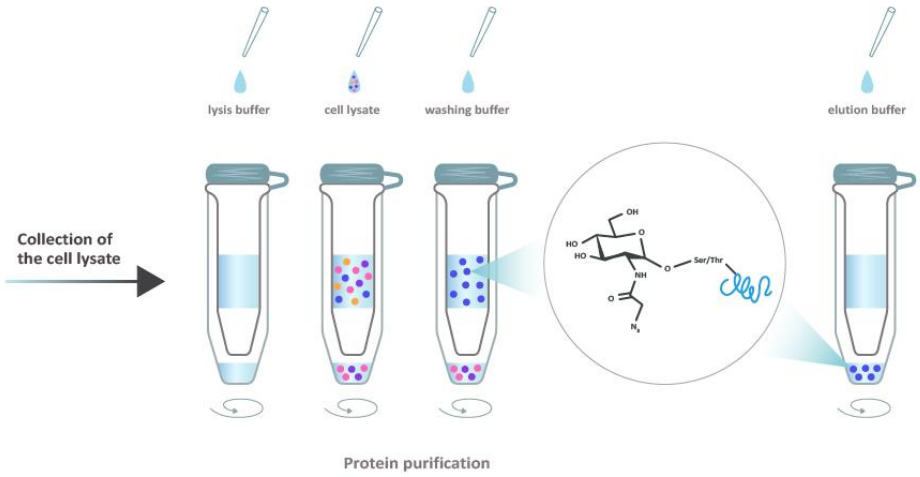


Figure 25. Schematic demonstration of the protein purification process on the column.

After the collection of cell lysate by centrifugation, protein purification was performed by using Ni-NTA resin or GFP-agarose beads for the proteins with His-tag or GFP-tag respectively (see Figure 25).

3.2.6. Spin labelling by Click chemistry azide-alkyne cycloaddition

Azide-alkyne cycloaddition reaction was retrieved due to the interest in the materials, pharmaceutical, and other industrial fields in terms of capacity to generate different varieties of compounds within the research aims. Sharpless and Morten Meldal, independently of each other,

developed an elegant and efficient chemical reaction: the copper catalysed azide-alkyne cycloaddition and have got the Nobel prize in Chemistry for that development in 2022. Click chemistry azide-alkyne cycloaddition reaction is thermodynamically-favored reaction and fulfilling the most important criteria of one-product-yield production. The necessary components of the reaction are commercially available and monosubstituted alkynes and azides can be easily synthesized. In our case alkyne proxyl, a nitroxide radical, is reacting with azido group of the sugars that was used for the MGE. The reaction is conducted under aqueous conditions at room temperature. Figure 15 and Figure 16 summarizes the mechanism of the copper-catalyzed azide-alkyne cycloaddition (CuAAC) for the specific case of spin labelling.[137] The active catalyst Cu(I) is generated from Cu(II) salts, in our case it is $CuSO_4$, using sodium ascorbate as the reducing agent. At the same time, sodium ascorbate is an efficient reduction agent of the nitroxide radical. Thus, an optimized reaction protocol with reduced levels of sodium ascorbate had to be used, in order to keep sufficient amounts of intact paramagnetically active spin labels. [129 ,136,] In brief, the spin labelling reaction between chemical reporter and spin label consists of the following steps:

Step 1. The short incubation of proxyl radical that is containg alkine group in reaction mixture of $CuSO_4$, sodium ascorbate, tris((1-hydroxy-propyl-1H-1,2,3-ztiazol-4-yl)methyl)amine (THPTA) ligand. This results in the Cu(I) ion complex with the alkine moiety (π-bound copper), thus making it activated.

Step 2. The following reaction forms the copper σ-bond at the monosubstitute position of the alkine proxyl.

Step 3. The π-bound copper coordinates the azide group of the sugar moieties, after that the six-membered copper matallacycle is formed. In this moment the σ-bound copper atom acts as a stabilizing donor ligand.

Step 4. The final element of the Copper-Catalyzed Azide-Alkyne Cycloadition (CuAAC) mechanism cycle is the photonolysis delivery of the triazole product by closing the catalytic cycle from the triazolyl-copper derivative ring.

3.2.7. Isolation of O-GlcNAcylated protein of interest after CuAAC reaction

After spin labelling reaction our aim is to prepare a biological sample that should contain just the O-GlcNAcylated protein with the alkyne proxyl label attached. To conduct the final purification of the MGE modified and spin labelled protein of interest a variety of different approaches to wash out

the redundant components of the Cu-click reaction and the excess of the free spin label were tested: molecular weight cut-off (MWCO) cellulose ester membrane, dialysis membrane made of polypropylene, ultrafiltration membranes produced of polypropylene – a material with low binding tendency. During abundant washing repeats that were necessary to avoid the unattached free alkine proxyl radical in the final biological sample, also some part of the MGE modified and spin labelled protein might get lost due to the weak interaction of the sugar moieties with the surface of the filtration membranes. Room temperature CW EPR spectra were used to determine the yield after protein purification and isolation. Thus, after the washing step, along with the removal of the EPR signal from the free spin label, also a part of the signal of the MGE modified and spin labelled protein might be missing. Indeed, during the first experiments, in which MWCO cellulose ester membrane was used, the assumption that some amount of the O-GlcNAcylated protein might get stuck on the cellulose-based membrane was tested, so that the concentration of protein in the final sample with each additional washing step decreased, which is drastically important for the further EPR measurements. This assumption was confirmed by the biophotometric measurements of the protein concentration before and after MWCO membrane, which demonstrated reduction of the protein concentration after several washing steps. After that observation, in the following experimental work the isolation systems made of polypropylene material with low binding propensity were only used. The schematic representation of the isolation system principles for the spin labelled MGE modified protein is illustrated in the Figure 8.

3.3. Selection of a protein model for EPR measurements

Considering the conditions of O-linked glycosylation PTM of protein and features of spin labelling and related EPR methodology, in particular in relation to eventual structural studies, the selected protein model has to fulfil some restrictions. The main criteria of the protein model selection can be formulated as follows:

- it has to be a protein belonging to the glycoprotein group - a member of the class of glycoconjugates;
- it should have highly distinctive O-linked glycosylation sites;
- protein model needs to be easy for purification, and it should be a cytoplasmic protein (not a membrane protein);

- protein with multiple glycosylation sites can be advantageous for primary detection and optimization of the EPR signal intensity;
- protein with only one or two glycosylation sites can be advantageous for optimizing the site-specific DEER measurements.

One can immediately recognize from the last two points that at different stages of the EPR methodology optimization different protein models might be necessary (e.g. multiple glycosylation sites for the proof-of-principle O-GlcNAcylation detection or 1-2 glycosylation sites to demonstrate the detection of the exact structural constraint). For that the revision of the literature dedicated to the study of O-GlcNAcylation modification of the proteins by various analytical tools, mostly different Mass-Spectroscopy methods (MS-MS, LC MS, TOF MS, ESI MS) was made.

By using glycoprotein data bases many publications were analysed and in an easier way different glycoproteins that might fit our protein model requirements for EPR study were considered.

Regarding the novelty and unpredictability of the new features appearing in the EPR spectroscopy of the O-GlcNAcylated proteins, there was an uncertainty with the choice of the protein model that might be good enough for the new approach development. Considering some conflicting criteria listed above, and also that there was an interest in testing the approach on different protein models to prove the more general applicability of the spin labelling approach, several protein models were selected.

3.3.1. Alpha-synuclein

Studies related to Alpha-synuclein (Figure 26) form one of research topics in the group of Prof. Drescher. Alpha-synuclein is a human neuronal protein which is abundant in the brain and found mainly in presynaptic neurons and axon terminals.[151] It regulates synaptic vesicle trafficking and subsequent neurotransmitter release, also it interacts with phospholipids and other proteins.[152] This protein is a common constituent for Lewy's body creation as a result of protein disfunction and aggregation. Lewy's bodies were discovered in neurons of the dementia disease patients.

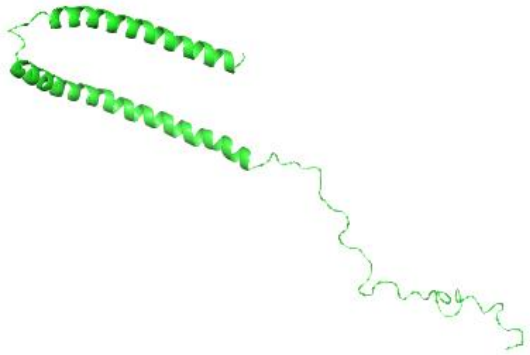
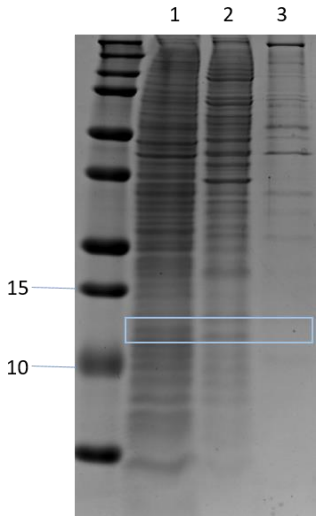


Figure 26. Structure of Alpha-synuclein according to the PDB (pdb code: 1xq8).

Some preliminary work was done to determine that certain O-GlcNAc modifications block the aggregation and toxicity of the Alpha-synuclein.[153] From the given perspective, initially it was planned to use Alpha-synuclein in our project to study its O-linked glycosylation sites by EPR to investigate its biological role and clinically interesting O-GlcNAc modification of Alpha-synuclein in the context of living cells. Based on the O-GlcNAc database search results, Apha-synuclein has the following O-GlcNAc sites: Thr54, Thr64, Thr72, Thr81 and Ser87.[154,155,156,157,158]

In the previously published work the alpha-synuclein with O-GlcNAc modification at Thr72 or Ser87 (the glycosylated site was produced via peptide synthesis and ligation) demonstrated inhibition of aggregates formation.[153] The inhibition of the aggregate formation has also been shown for enzymatically O-GlcNAcylated alpha-synuclein.[158,154] All these studies on O-GlcNAcylated α-synuclein have been performed *in vitro*. The original idea was to detect O-GlcNAcylation sites of Alpha-synuclein *in cellula*. The test experiments, during protein expression in HEK293t and HeLa cells revealed that the expression level of Alpha-synuclein is very poor (see Figure 27) for the future EPR study.



1. Lysate of HEK293T cells before expression (10 µl injected sample);
2. Lysate of HEK293T cells after expression with Alpha-synuclein DNA plasmid (10 µl injected sample);
3. Lysate of HEK293T cells after expression with Alpha-synuclein DNA plasmid (5 µl injected sample);

Figure 27. SDS-PAGE analysis of lysated HEK293T cells before and after expression with Alpha-synuclein DNA plasmid.

Besides differences between HEK or HeLa cells and neuron cells (and thus differences in expression levels), the low apparent expression levels of alpha-synuclein can be also related to its function, in which alpha-synuclein exists mainly as intrinsically disordered monomer form, which is relocated from cytoplasm to the cell membranes and bound there as large α-helix.[159] Such membrane-bound alpha-synuclein might be prone to significant losses under a soluble protein targeting purification protocol.

The observation concerning Alpha-synuclein O-GlcNAcylation in the group of Prof. Dr. Leist also demonstrated poor level of modification, as well as in the group of Prof. Dr. Zumbusch O-GlcNAcylation of Alpha-synuclein with the to date most sensitive detection with anti-O-GlcNAc antibody was not successful.[160]

3.3.2. OGT protein

O-GlcNAc transferase also known as OGT or O-linked N-acetylglucosaminyltransferase is an enzyme that catalyzes the addition of the *O*-GlcNAc post-translational modification.[161] It is the main enzyme in O-GlcNAcylation cycle that attachs the sugar moieties to the Ser or Thr amino acids of the proteins.[162] OGT is generated by alternative splicing of the OGT messenger RNA (Mitochondrial Targeting Sequence) and it exists in three isoforms: two in nucleoplasm – long nucleoplasmic isoform (ncOGT) and short isoform (sOGT) are localized in the nucleus and cytosol; and mitochondrial isoform (mOGT).[163]

Each isoform containing tetra-tricopeptide repeats (TPR) on its N-terminal side (see Figure 28). TPR is involved in protein-protein interactions. In its C-terminal region OGT has catalytic domain that is involved in its glycosyl-transferase catalytic activity.[163]

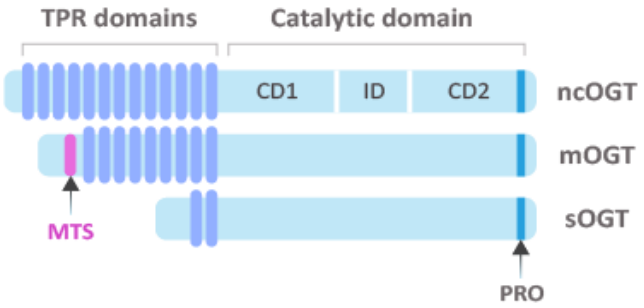


Figure 28. Types of OGT proteins and their domains.

Also in the end of C-terminal phosphoinsositide binding domain is located, dominated PRO domain for Phophosphatidylinositol Phosphate binding domain of OGT that is also involved in the recruitment of OGT at the plasma membrane upon insulin stimulation.[164]

The data from O-GlcNAc database demonstrated good potential of OGT as a protein model (see Figure 29) for the O-GlcNAcylation and spin labelling.

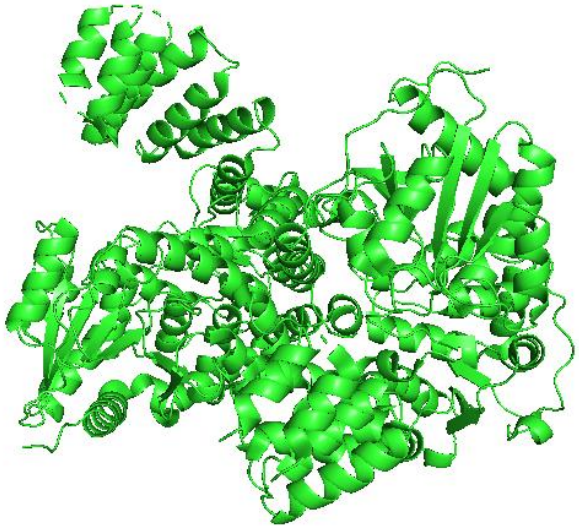


Figure 29. Structure of OGT according to the PDB (pdb code: 2vsn).

In particular, it was a promising model for the proof-of-principle demonstration of the capability to detect EPR signal after MGE and spin labelling, since it contains several glycosylation sites: Ser10, Thr12, Ser18, Thr38, Ser52, Ser56, Thr449. [73, 165,166]

3.3.3. P53 protein

As one of the suitable protein models p53 protein was also tested. It is a globular protein with a distinct domain. The p53 protein is manifesting itself in the role of key tumour suppressor. While p53 has molecular weight of only 43 kDa, due to high number of proline in the sequence it is shown as 53 kDa during SDS-gel analysis. The protein structure consists of five domains (see Figure 30): a trans-activation domain that activates transcription factors; a proline-rich domain; a DNA binding domain that recognize specific DNA sequences; an oligomerization domain (responsible for the tetramerization of the protein); a C-terminus or regulatory domain (recognize damaged DNA, such as misaligned base pairs or single-stranded DNA).[167]



Figure 30. p53 protein sequence of domains: NTD – N-terminal domain; PR – proline-rich domain; DBD – DNA binding domain; OD – oligomerization domain; CTD – C-terminus domain.

The p53 protein (see Figure 31) is well studied as a regulator of transcription in the nucleus, where it exists primary as a tetramer. However, in unstressed cells, p53 held in check by the murine double minute-2 (MDM2) that targeted p53 for transcriptional repression, proteasomal degradation, and cytoplasmic localization.[168]

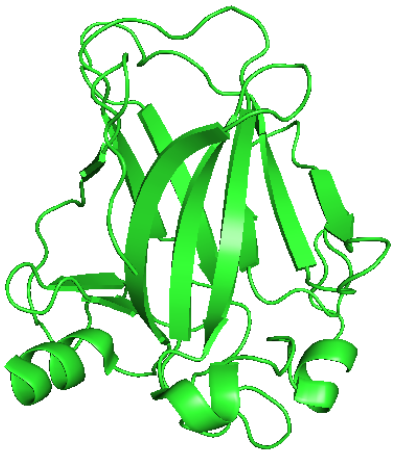


Figure 31. Structure of p53 according to the PDB (pdb code: 7pc6).

Post translational modifications are regulating activation of p53 in response to predisposed oncogenic signals, this process also largely depends on accumulation of p53.[169,170] p53 is known to

be O-GlcNAcylated but the exact number of O-GlcNAcylation sites and the PTM effects on p53 structure and as a result-functions are still not well understood.[171] There is also still a lack of knowledge with respect to the role of p53 post-translational modifications in carcinogenesis and cancer prevention. For instance, understanding of p53 may uncover routes for new strategies in treating and preventing cancer, *e.g*. by restoration of p53 function and selective killing of tumours with mutant TP53.[172]

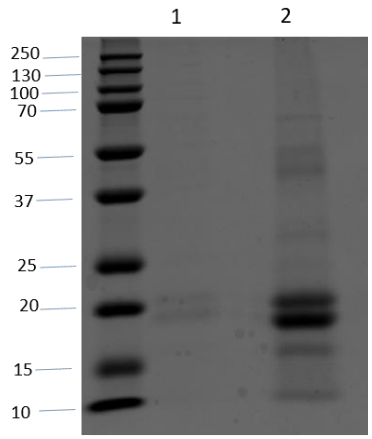
3.3.4. Alpha-crystallin protein

Alpha-crystallin is a member of a small heat shock protein family by sequence comparisons. Like many other heat shock proteins, Alpha-crystallin (see Figure 32) represent chaperon-like properties, including the ability to prevent the precipitation of denaturated proteins and to increase cellular tolerance to stress. It is the major protein of the mammalian lens. Its functions are important for the maintenance of lens transparency and the prevention of cataract.



Figure 32. Structure of Alpha-synuclein according to the PDB (pdb code: 6t1r).

Alpha-crystallins are aggregates assembled from polypeptides of 10 to 25 kDa (see Figure 33) that share common central domain, so called “Alpha-crystallin domain” with variable N-terminal and G-terminal extensions.[173]



1. Alpha-crystallin after elution;
2. Alpha-crystallin after elution concentrated.

Figure 33. SDS-PAGE analysis of purified alpha-crystallin.

It is important to say that the 3D structure of alpha-cystallin remains so far unknown and its structural study possibly requires specific analytical tools since it is not possible to obtain crystals of this protein for X-ray studies. Alpha-crystallin has O-GlcNAcylated site[174] and can be a nice model for demonstrating the possibility of glycosylation modification/spin labelling structural study. Alpha-crystallin is thus a suitable model protein for optimizing the proposed method of MGE linked with spin labelling and DEER measurements, as an application in glycobiology.

3.4. Optimisation of the protocol for the sample preparation for EPR measurements

3.4.1. Regular protocol for the similar biological samples preparation

In glycobiological science immunoprecipitation and fluorescence methods of glycosylation analysis are used. A broad variety of studies was done in this field by combination of chemical reporters for the protein modification and fluorescence lifetime imaging (FLIM, FRET) microscopy. Close to that approach of biological sample preparation, our long term goal is to develop an EPR method that allows to detect and to determine the exact position of O-GlcNAcylated site in the protein structure. For that reason, the experimental strategy that is schematically represented in the Figure 34 was used.

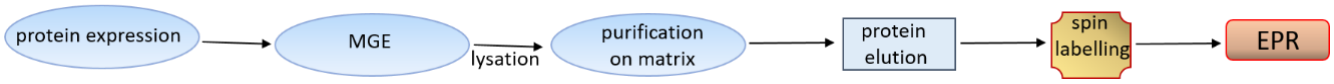


Figure 34. The schematic sequence of the experimental strategy that consists of the part for biological sample preparation and EPR measurement.

For our first strategy OGT-GFP, OGT-HA-HA, and OGT-His DNA plasmids were expressed in HEK 293T cells. After DNA plasmid translation into OGT protein, the cells were treated with Ac_4GlcNAz and Ac_4GalNAz sugars to see which sugar gives better effect on the level of O-GlcNAcylation. After some period of incubation (see Materials and Methods for the detailed protocols) with added chemical reporters, cells were collected and lysated for further protein purification via an affinity tag, encoded in the DNA plasmid at the terminus of the protein (C-terminus for the -GFP and the -His tags, N-terminus for the -HA-HA tag). After this, purified protein was spin labelled in the corresponding working buffer for the protein.

After spin labelling a number of washing steps was required to treat the free spin label excess that might otherwise disturb the EPR signal of spin labelled protein. Unfortunately, multiple repetitions of this strategy didn't reveal any EPR signal related to the spin labelled glycosylated protein. Such negative outcome signalled about the need to consider and revise each step of the sample preparation protocol for EPR measurements.

3.4.2. Step-by-step improvement of the EPR sample preparation protocol

Each single step in the Figure 19 was inspected and optimized, as follows.

1. Expression step improvement

The protein expression level could be conveniently controlled by using the OGT plasmid with GFP tag. Due to its fluorescent properties GFP tag gave us the possibility for immediate visualisation of expression (see Figure 35).

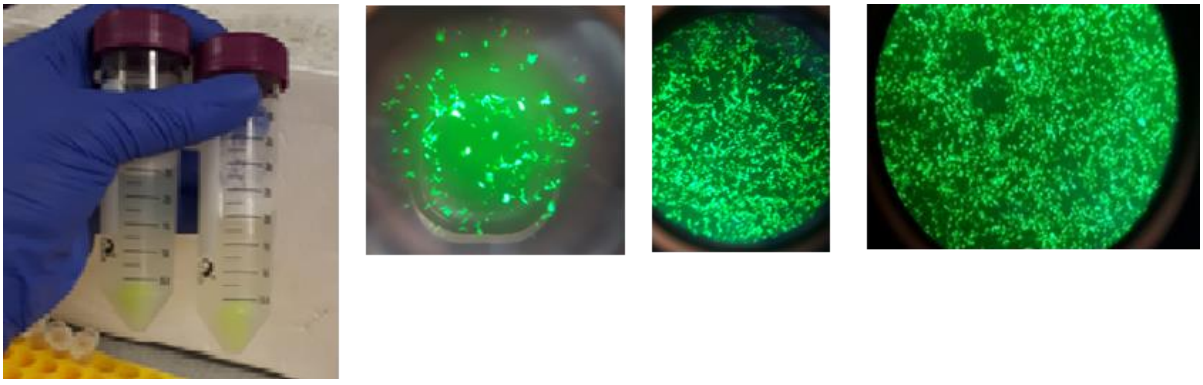


Figure 35. Expressed OGT-GFP protein in HEK293 T cells (left image). Images of microscopy control of protein expression level with GFP-tag in HEK293T cells (three right images).

Transfection using lipofectamin worked well. However, it is expensive to use this transfection method when one has to scale up the amount of the cells for producing the final EPR samples.

Thus, for small-scale protein expression, e.g. for assays or MS characterization, lipofectamine transfection method was often used. In the cases of large scale protein expression, but also for some small scale expressions, the transfection with calcium phosphate was performed, which also demonstrated good efficiency.

Using GFP-tag visualisation, following conditions were optimized: the number of DMEM media exchange steps, the cell culture incubation time, the DNA plasmid concentration used in the transfection. Some representative examples of such optimization are given in Figures 35. The DNA plasmid transfection was performed at the time point when the cell culture reached 80% area coverage of the Petri dish. After transfection and POI expression only a fraction of the cells was revealing GFP fluorescence signal. In the Figure 35, one can see an example of the fluorescence image of the cell culture at the state before the protocol optimization (left microscopy image) and after the protocol optimization (right microscopy image). One can see that after the optimization, transfected cells account for the majority of the cell culture, and are homogeneously distributed over the Petri dish. In particular, it turned out that there is an optimal DNA plasmid concentration level during the transfection, so that if this level is exceeded, then the final yields of the transfected cells and protein of interest decrease. It was also critical for the improved yields to exchange the DMEM medium after the step of transfection (in addition to the medium exchange after 12 hours of transfection, as described in a standard protocol).

2. Purification step.

Different OGT plasmids with encoded affinity tag were tested: GFP-, His-, and HA –HA-tag. For example, HA-tagged recombinant protein can be affinity purified directly from a cell culture lysate or supernatant. The HA-tagged protein binds to the HA-tag specific monoclonal antibody conjugated on an agarose gel. After washing away residual impurities, bound HA-tag proteins can be eluted off the affinity column by high concentration of the HA-tag peptide or by low pH buffer.

It turned out that HA-tagged OGT was prone to substantial degradation (see Figure 36), while for OGT-GFP and OGT-His constructs the degradation levels were clearly reduced (see Figure 37).

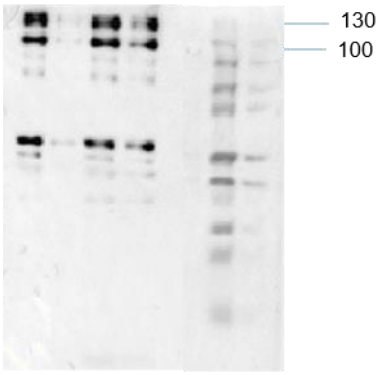


Figure 36. Western Blot image that demonstrates degradation of OGT protein with -HA-HA tag after its purification anti OGT antibody.

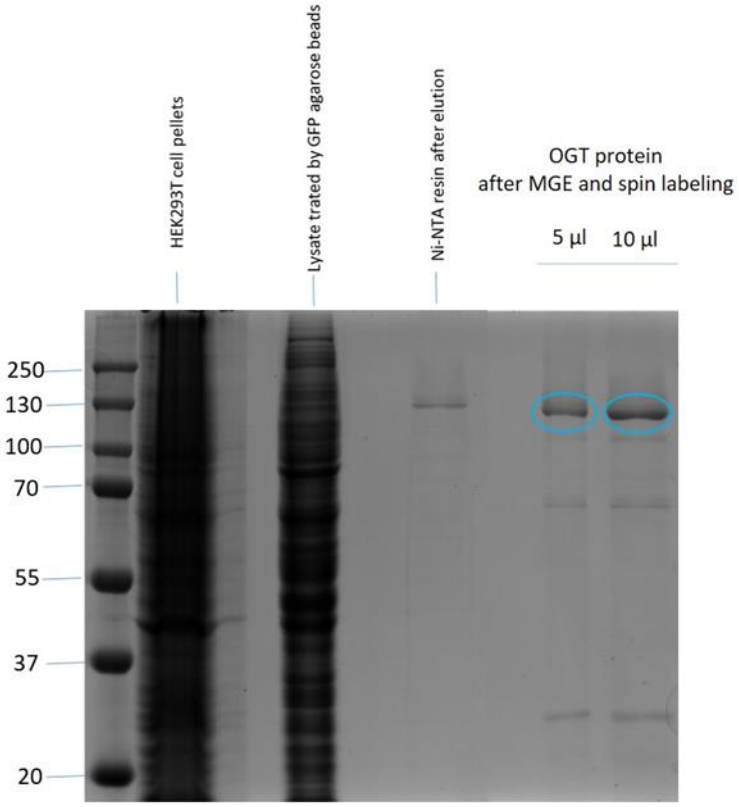


Figure 37. SDS-PAGE as a quality control of the OGT protein purification.

3. Conditions for the CuAAC reaction.

After protein purification, CuAAC reaction was tested with different of sodium ascorbate, which concentrations. Sodium ascorbate is required here to transform Cu(II) to Cu(I). However, at the same time, it can reduce the spin label to a diamagnetic moiety. Thus, one needs to reduce the Na-Asc level, as compared to the standard protocol, so that acceptable levels of intact spin labelled sugars could be obtained. The Na-Asc levels during CuAAC reaction were adjusted according to the spin labelling protocols for non-canonical amino acids spin labelling. [129, 136] At optimized conditions, the overall efficiency of binding reaction between sugar Ac4GlcNAz and Alkyne Proxyl was about

75%, which then includes the actual labelling efficiency and the loss of intact spin labels due to their reduction by Na-Asc. In Figure 38 one can see … (describe TLC data here). The corresponding MS data are shown in the Figure 39, indicating a clear peak of the spin labelled Ac_4GlcNAz.



Figure 38. Thin-layer chromatography (TLC) plates for the reaction between Ac_4GlcNAz and alkyne proxyl spin label.

The CW EPR spectra (Figure 40) also clearly reveal the spin label attachment to the Ac_4GlcNAz. This is best seen from the peak-to-peak amplitude of the high-field line in the EPR spectrum (as compared to the amplitude of the central line). All three EPR spectra of a nitroxide spin label are characterized by the same integral intensity, so that their relative peak-to-peak amplitudes change according to the changes in the line widths. The line width is in turn a measure of the spin label characteristic rotational diffusion time. Stochastic tumbling of a spin label slows down after attachment to the sugar, which is seen in CW EPR as the reduction of the peak-to-peak amplitude of the high field line as compared to the other two lines. Difference in the line broadening due to rotational diffusion appears because of the differences in the summed hyperfine and g-anisotropy between the corresponding three spectral components in the nitroxide EPR spectrum.

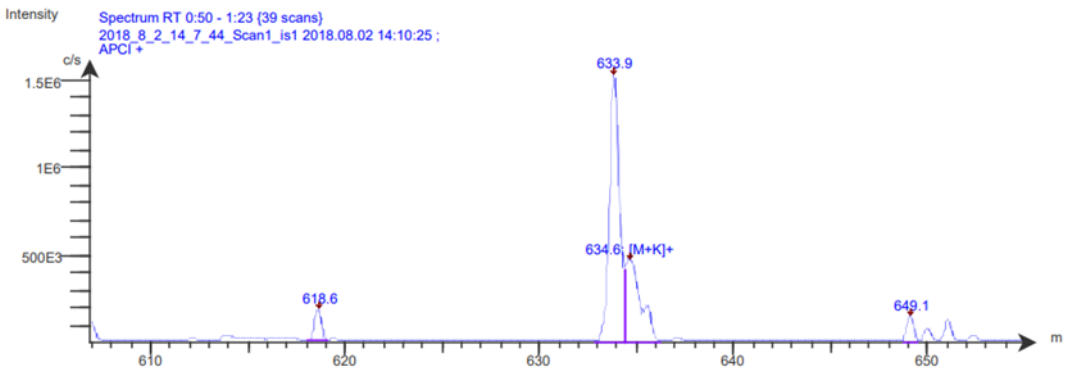


Figure 39. Mass spectrometry measurements on the thin-layer chromatography (TLC) plates (TLC/CMS technique) for the reaction between Ac_4GlcNAz and alkyne proxyl spin label.

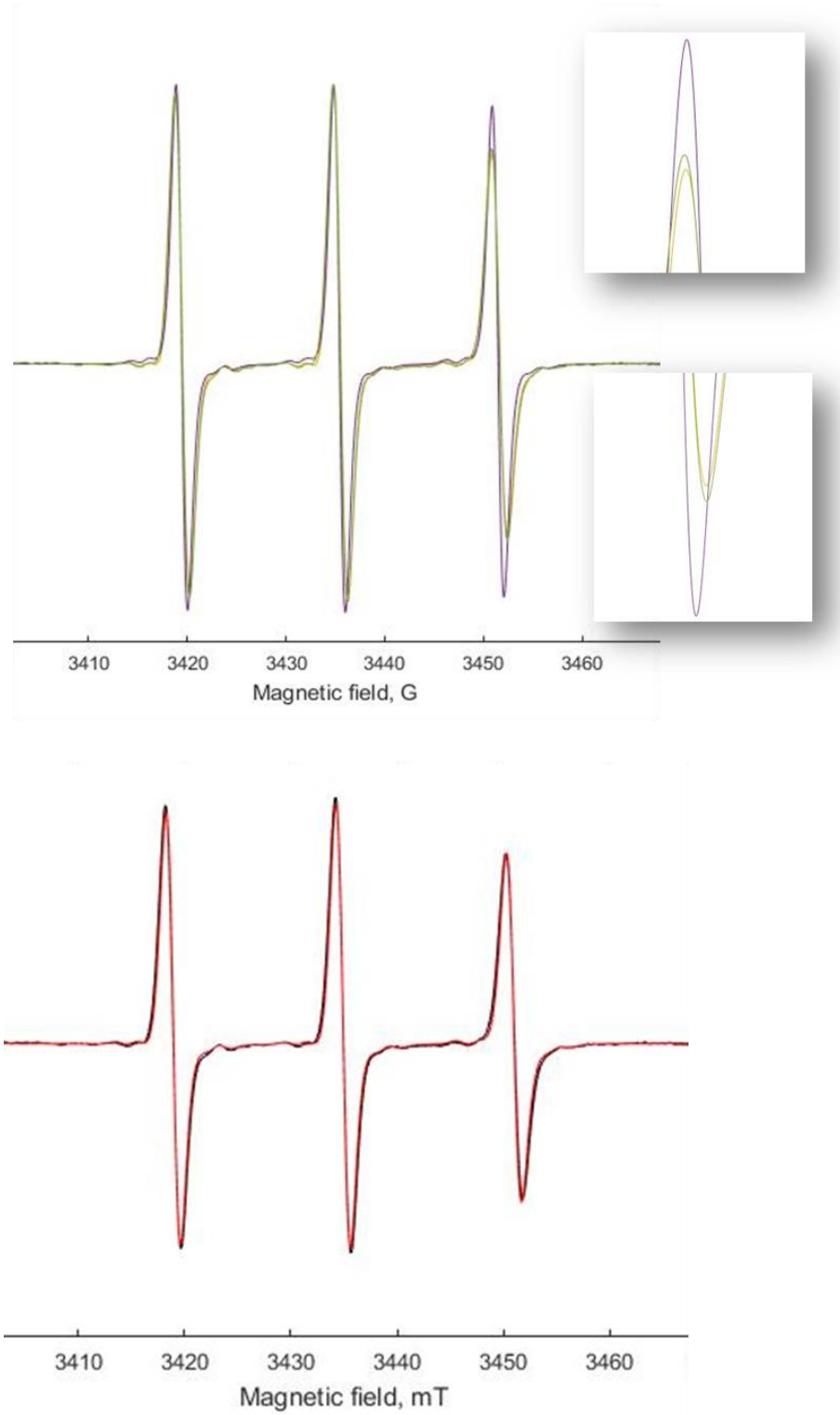
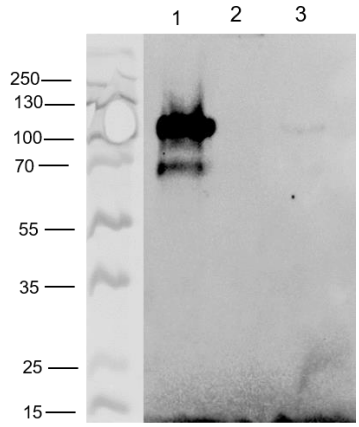


Figure 40. Top panel: Ambient-temperature CW EPR spectra of free alkyne proxyl spin label (purple spectrum), Ac_4GlcNAz spin labelled by alkine proxyl in a CuAAC reaction for 1h, without purification (green spectrum) and after purification on a silica-gel column (yellow spectrum). The insets in the top panel show the top and the bottom parts of the high-field peak for the three spectra. Bottom panel: Ambient-temperature CW EPR spectrum of Ac_4GlcNAz spin labelled by alkine proxyl in a CuAAC reaction for 1h, without purification (black spectrum) and its simulation by a linear combination of independently detected EPR spectra: 75% of the purified spin labelled Ac_4GlcNAz and 25% of the free spin label (red spectrum).

4. Washing step after the CuAAC reaction.

During the sample washing after the CuAAC reaction significant decreasing of the protein of interest concentration was observed. When the amount of HEK293T cells was increased to match the EPR sample requirements and accordingly the concentration of the protein was higher one could observe also substantial protein degradation that prevented from continuing with the further EPR measurements (see Figure 41).



1. GFP agarose beads with OGT-GFP protein modified by Ac4GalNaz (labelled by alkyne proxyl);
2. Sample after elution;
3. Final sample after CuAAC reation and buffer

Figure 41. WB images as a quality control of the OGT proteins purification. A Western blot performed using anti OGT antibody and confirms disappearing of the protein after protein labelling.

5. Suitable number of cells for the EPR sample preparation.

Increasing of the cells amount (from one 10 cm Petry dish to ten Petry dishes, then to three 30 cm Petry dishes, and, finally, to five 30 cm Petry dishes) was the final optimization step to match the spin labelled protein yields with the EPR requirements. For nitroxide spin labels, the sensitivity of CW EPR spectrometer has a detection limit around 1-2 µM spin concentration (in the intermediate stochastic tumbling regime).[2,99] What does this mean? In our case, O-GlcNAcylated protein that with at least 10 µM concentration of attached alkine proxyl to the azido group of protein bonded sugars had to be prepared.

As was mentioned before, O-GlcNAcylation is mostly feasible in mammalian cells. But protein expression in mammalian cells has low productivity and, in this way, the final amount of O-GlcNAc modified and spin labelled proteins, which can then be used for the final EPR measurement, drastically decreases. Regarding the fact that one has no quantitative information about protein O-GlcNAcylation level, it is difficult to evaluate the protein concentration that has to be achieved in the prepared EPR sample.

3.4.3. In vitro O-GlcNAcylation

As was mentioned before, O-GlcNAcylation is a native PTM process in mammalian cells. But protein expression in mammalian cells has low level of production and in this way the final yield of the O-GlcNAc modified and spin labelled protein is low, albeit just enough for the EPR detection.

One possibility to increase the concentration of the protein is to express it in procariotic cells (like *E.coli* – a popular and well established protein expression system used in bio-EPR) and perform O-GlcNacylation *in vitro* enzymatically. Thus, the next step was set out to express the protein in prokaryotic cells (*E.coli*) and, after purification, to establish the similar enzymatic O-GlcNAcylation system *in vitro*. For that goal few previously published protocols were exploited. Compared to other PTMs processes, O-GlcNAczlation doesn't require too complicated biological system to reproduce the glycosylation conditions, since the O-GlcNAcylation cycle is regulated by just two enzymes: OGT and OGA. The main idea of our "*in vitro* O-GlcNAcylation" strategy was to use high concentrations of the glycoprotein and under conditions when just OGT enzyme is present without sugar cleavage enzyme OGA to study what is the highest O-GlcNAcylation level that might be reached. *In vitro* strategy is disturbing the cellular context but the protein O-GlcNAcylation form study is an important step before one can investigate spin labelling *in cellula*.

It was decided to perform the protein expression in *E. coli* cells (this protocol yields much higher concentrations of the protein of interest) and to O-GlcNAcylate the protein *in vitro* according to the protocols (see Methods and Materials chapter).

1. Expression of the protein in *E.coli*.
2. Purification of the protein
3. Attachment of the purified protein to the matrix (see next section for the details of this additional step)
4. In vitro O-GlcNacylation
5. Spin labelling of modified protein and washing from spin labelling excess
6. Elution of O-GlcNac modified protein from substrate

For performing O-GlcNAcylation by Ac4GlcNAz sugar moieties *in vitro* UDP-GlcNAz and enzymatically active form of OGT is needed. The reaction was performed according to the protocol

(see chapter Methods and materials) in the buffer and additionally in presence of $MgCl_2$. The approach was tested on the following three protein models: GFP, alpha-synuclein and TRX.

The prepared sample with alpha-synuclein didn't give us an EPR signal of attached alkine proxyl radical to the O-GlcNacylated alpha-synuclein. Instead of that an EPR signal with characteristic equally split lines, typical for Mn(II) impurities in $MgCl_2$ was obtained (see Figure 42). The presence of magnesium in the reaction buffer is essential for the O-GlcNacylation reaction. Unfortunately, after the washing of free alkine proxyl radical the Mn(II) EPR spectrum is still present, thus setting a substantially higher limit for the nitroxide detection threshold in the final sample.

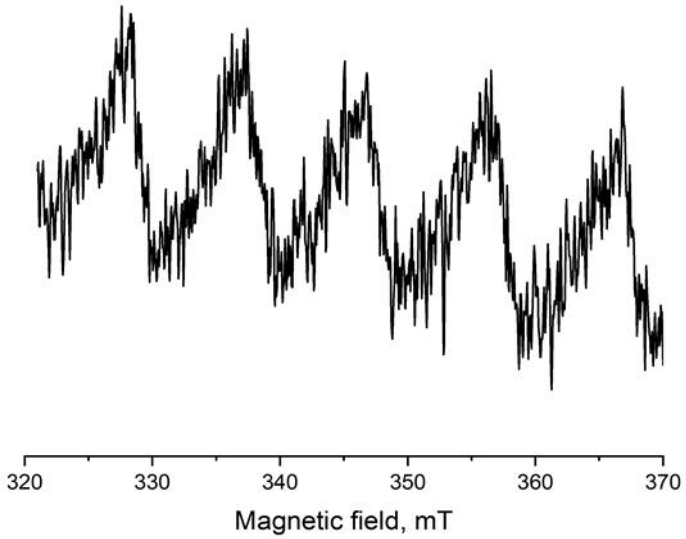


Figure 42. Experimental EPR signal after in vitro O-GlcNAcylation process. The spectrum is typical for Mn^{2+}. Unfortunately, only five lines from the six-line EPR pattern of the Mn^{2+} were recorded.

The same results were observed for the TRX and GFP samples which were also tested as the glycoprotein models for *in vitro* O-GlcNAcylation. To have an independent control that the absence of EPR signal from alkine proxyl is indeed due to the unsuccess of enzymatic O-GlcNAcylation reaction, Peirce Glycoprotein Staining Kit (Thermo Fischer) that is capable of visualizing the O-GlcNAcylated proteins on SDS-gels was used. The gel on the Figure 43 demonstrates not efficient enough modification of both tested proteins as compared to the reference sample Horseradish Peroxidase protein (as positive control).

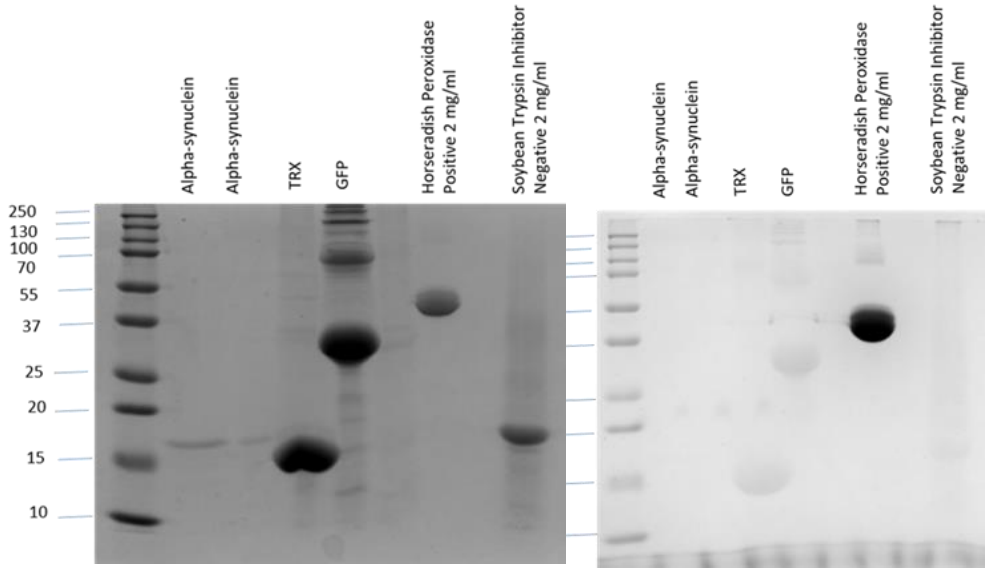


Figure 43. SDS-PAGE after coomassie staining (left image) and after Peirce Glycoprotein Staining Kit use (right image) for *in vitro* O-GlcNacylated Alpha-synuclein, TRX, GFP and positive/negative samples applied to the kit.

3.4.4. Intermediate detection of O-linked glycosylated OGT protein on purification matrix

As described above, our previous attempts to perform EPR measurements on spin labelled O-GlcNAcylated proteins were not successful, mainly because of low yields of the final spin labelled protein, and substantial protein degradation during washing steps. Also *in vitro* O-GlcNAclytion experimens had not led to any success.

Since the major protein losses were related mainly to the elution and following steps, the experimental scheme was updated to the one shown in Figure 44, where spin labelling by alkine proxyl is performing directly after protein attachment by His-tag to the Ni-NTA resin. All washing steps of free radical of alkine proxyl also were performed before modified protein was eluted from purification matrix.

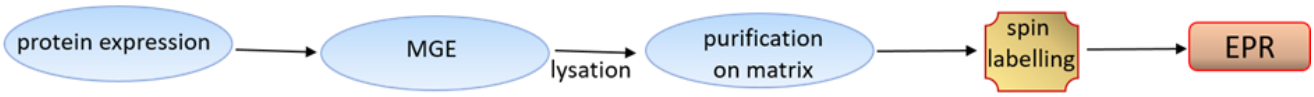


Figure 44. A schematic strategy that was used to observe the first EPR spectrum of prepared sample.

The EPR sample was made of O-GlcNAcylated protein that still be connected to Ni-NTA resin. To a great satisfaction, after this modification of the sample preparation scheme the first positive EPR results were finally obtained (Figure 45, right spectrum).

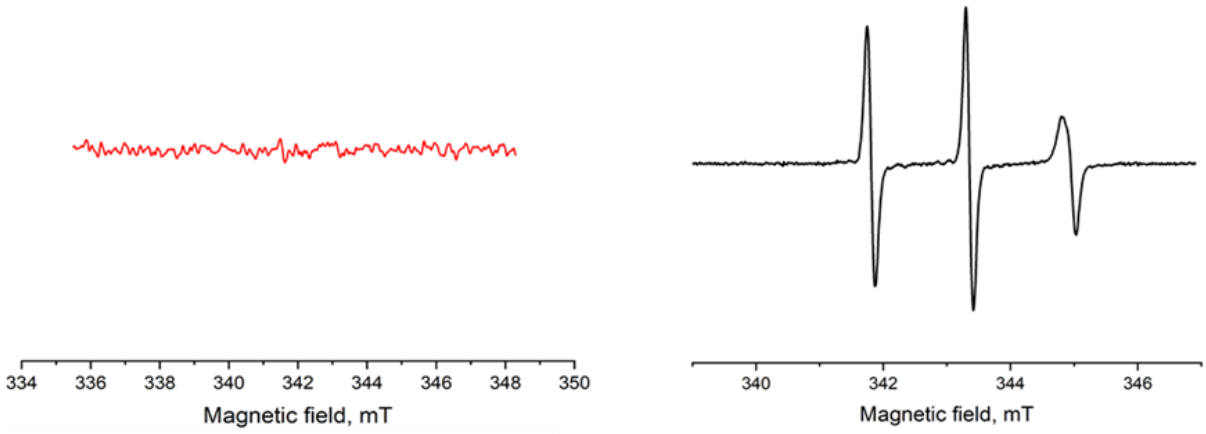


Figure 45. EPR signals of Ni-NTA resin (right spectrum) and of O-GlcNAcylated OGT protein connected to Ni-NTA resin.

Knowing that EPR spectroscopy is sensitive just for the paramagnetic substances, the EPR background signal from the Ni-NTA resin, which was used as a purification matrix for the expressed OGT protein with His tag, was measured. The assumption was that if the resin itself is essentially EPR silent, then one can make a safer detection of the spin labelled protein, prior to its elution. This assumption proved to be correct: the EPR spectra of the free Ni-NTA resin demonstrated no EPR signal (see Figure 45, left spectrum). The shape of EPR spectra detected on the spin labelled O-GlcNAcylated OGT on the Ni-NTA resin, after washing out the constituents of the CuAAC reaction mixture, demonstrated the clear slow motion CW EPR spectrum characteristic for a spin label with restricted rotational freedom due to the protein immobilisation on the resin. The difference between the individual nitroxide line intensities is the clear evidence of the alkyne proxyl attachment to the protein and the absence of substantial amounts of the free spin label.

3.4.5. Final strategy of EPR sample preparation

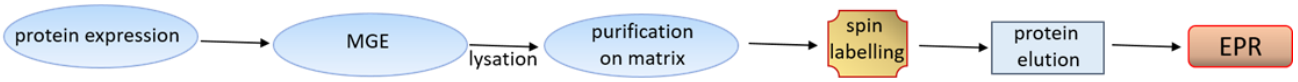


Figure 46. A schematic strategy that was used for the detection of O-GlcNAcylation modification out of purification matrix (Ni-NTA resin).

The final sample preparation scheme was used in this work is represented in the Figure 46. EPR spectra of p53 protein (after MGE in HEK293T cells purification, elution and spin labelling steps) are shown in Figure 47.

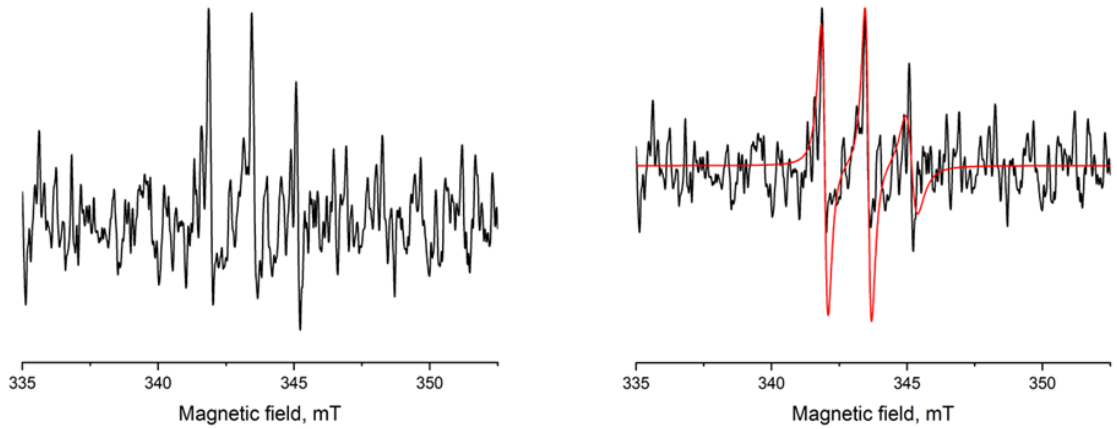


Figure 47. Experimental EPR spectrum obtained from the prepared sample out of 10 units of 10 cm Petry dishes. Signal was weak but simulation (red curve of the right spectrum) confirms nitroxide radical attachment to the p53 protein model.

All the time the weak EPR signal was observed (see Figure 47) after spin labelling reaction improving and decided to increase the amount of the cells (from 10 units of 10 cm Petry dishes to 6 units of 30 cm diameter Petry dishes for the HEK293T cells growing for the p53 protein expression and MGE). Due to these quantitative changes in the sample preparation the EPR signal increased (see Figure 48).

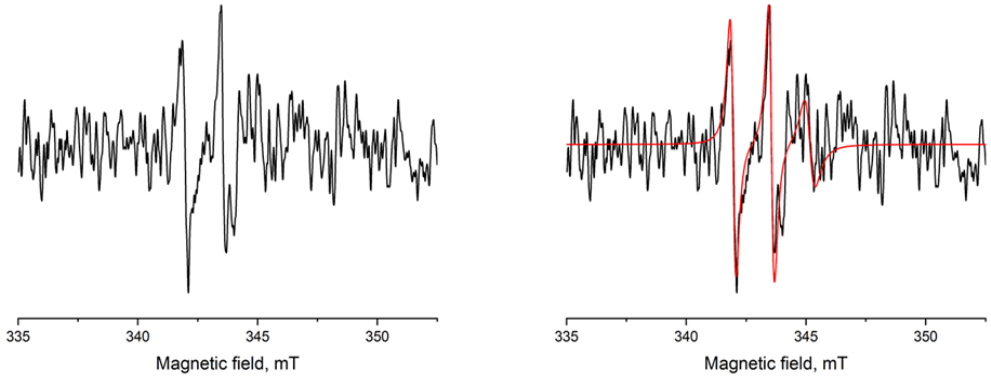


Figure 48. Experimental EPR spectrum obtained from the prepared sample out of 6 units of 30 cm Petry dishes. Signal was amplified due to the increased intensity of the nitroxide lines and simulated (red curve of the right spectrum).

After up-concentrating the sample an EPR signal with a much better signal-to-noise ratio was obtained (see Figure 49). Also negative control, *i.e.* in the absence of OGT protein expression was performed to exclude any contributions from possible impurities that might be on the Ni-NTA beads.

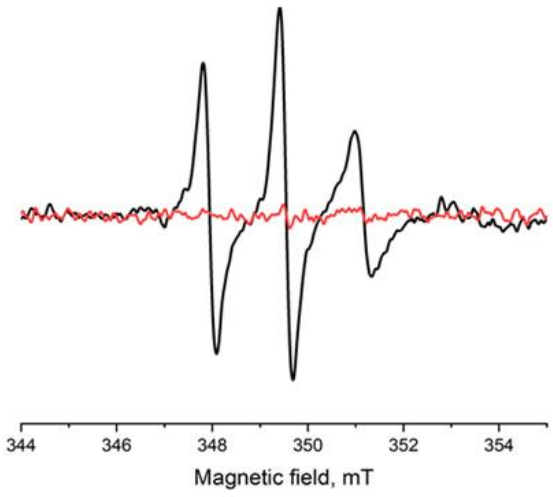


Figure 49. EPR spectra of O-GlcNAz modified OGT protein (black curve) and the sample without OGT protein expression (red curve).

It is essential to understand why did the new experimental scheme work, in contrast to the original one. If one has a look at Figure 34, Figure 44, and Figure 46, some reasonable speculations about the differences in these three strategies can be given. When the spin labelling of O-GlcNAcylated

proteins was performed on Ni-NTA resin before elution, it saved a lot of protein and also further washing from free Akline Proxyl spin labelling excess didn't influence the proteins and thus kept the protein concentration stable. Furthermore, by keeping the protein of interest immobilised on a solid support, the self-degradation processes that were otherwise observed at high protein concentrations could be inhibited. Only after complete washing out of the free spin label, the protein attached to the Ni-NTA resin was eluted and isolation system (MWCO membrane) was used to neutralize the imidazole elution buffer and substitute it by the working buffer of OGT protein. Most likely, these are the key factors, why keeping O-GlcNAcylated protein on the purification beads during CuAAC reaction and washing steps to treat the spin label excess allowed preserving the protein concentration and prevented the protein from degradation.

3.4.6. Spectra of modification by Ac4GalNAz and Ac4GlcNAz

According to the metabolic process (described in chapter State of Knowledge > O-linked GlcNAcylation of the proteins) which demonstrated that GalNAz is converted by endogenous cellular enzymes into UDP-GalNAz, which can be epimerized to UDP-GlcNAz by UDP-galactose-4-epimerase[29] and due to the fact that the modification of proteins with GlcNAz, which is the metabolized form of Ac4GalNAz attached by OGT, O-GlcNAc modification of the proteins was stronger for Ac4GalNAz- than for Ac4GlcNAc-treated samples[160]. The experimental viability statistics for the cells treated by Ac4GalNAz and Ac4GlcNAz was almost the same.[160] Viability level was slightly lower for Ac4GlcNAz comparing to Ac4GalNAz. From the beginning of our project Ac4GlcNAz due to its advantages described in the section "Metabolic glycan engineering" of the current chapter was used. Knowing the statistics of Doll work[160] it was decided to perform experiment with two different sugars. Both sugars have been shown to solely end up in protein O-GlcNAcylation. From the EPR spectra one could see that samples prepared at the same conditions but with different sugar's treatment show slightly different EPR intensity (see Figure 50). Red spectrum in the Figure 50 represents the intensity of the spin label of the sample, for which Ac4GlcNAz was used for MGE treatment. Black spectrum from the Figure 50 represents the intensity of the spin label of the sample, for which Ac4GalNAz was used for MGE treatment. This spectrum appears to have a somewhat better signal-to-noise level, indicating the potentially better performance of the Ac4GalNAz for glycoprotein MGE. However, in the interest of consistency and

to avoid possible capture of the modified sugar by the cell membranes the work with the Ac4GlcNAz was continued throughout this thesis.

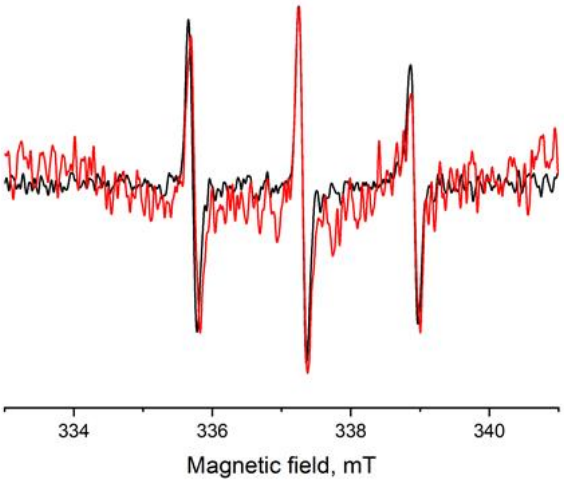


Figure 50. EPR spectra of O-GlcNAz modified p53 protein for which Ac4GlcNAz (red curve and Ac4GalNAz (black curve) were used for MGE treatment.

3.5. Intermediate conclusions regarding optimisation of the biological sample preparation protocol

▪ A handful of protein models was selected for the primary EPR methodology development in application to glycobiology of O-linked glycosylation modification;

▪ The DNA plasmid transfection process and protein expression conditions were optimized;

▪ To increase the protein concentration (and, as a result, the level of O-GlcNAcylated ptoteins in the final EPR sample) the isolation system at the washing step was optimized and prtotein expression in higher amounts of HEK 293t cells was utilised;

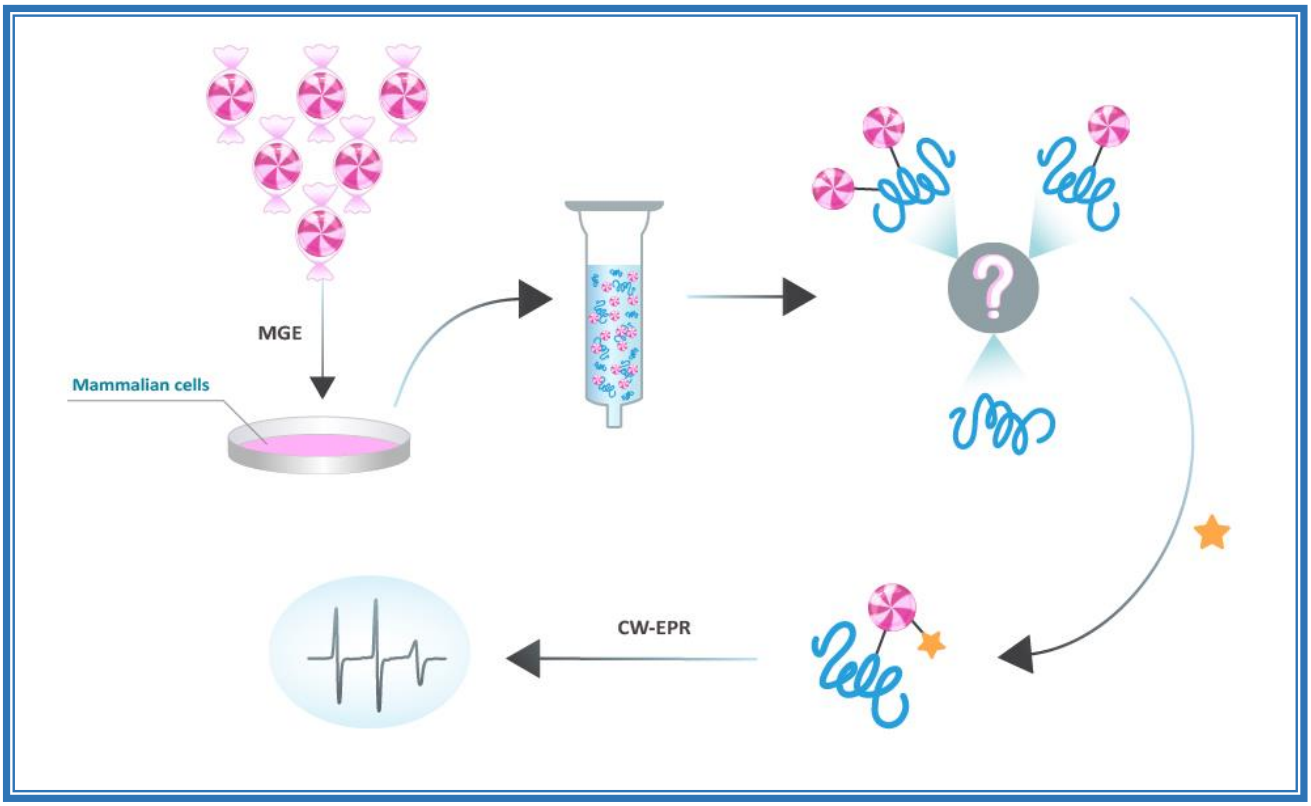
▪ After different experimental changes were established a reproducible protocol (the sequence of experimental steps: protein expression -> MGE -> protein purification on the responsible matrix -> spin labelling -> protein elution and EPR measurements) that for the first time gave an EPR signal from spin labelled O-GlcNacylated sites of OGT protein.

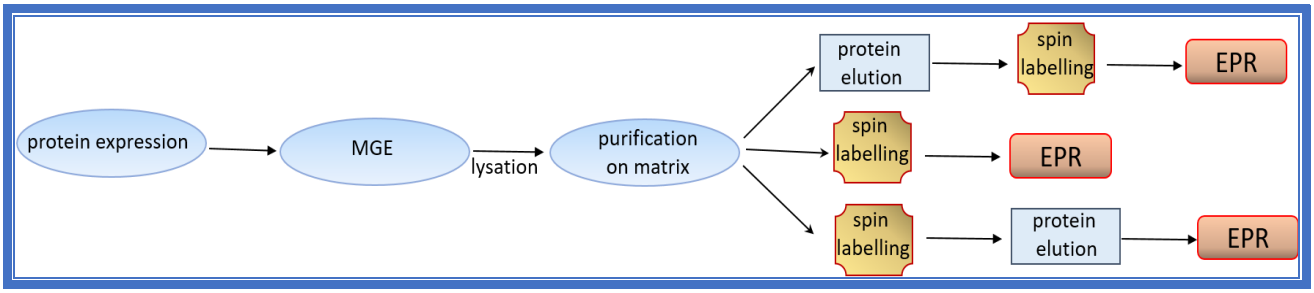
▪ Two types of modified sugars Ac_4GalNAz and Ac_4GlcNAz were tested to treat the expressed OGT protein in HEK 293T cells by MGE; Ac_4GlcNAz sugar was selected to be used for the HEK 293T cells treatments in the further experiments;

▪ An EPR signal was obtained as an evidence of the successful O-GlcNAzylation modification and glycosylation-specific spin labelling of the OGT protein model.

CHAPTER IV

Results and discussion





"To be, or not to be, that is the question."
by William Shakespeare

4. Results and discussion

4.1. Application of the developed methodology for another protein model

Above, in the chapter “Designing strategy for the biological sample preparation” a general strategy for the sample preparation was developed and the feasibility of a new analytical approach based on spin labelling and EPR for protein O-GlcNAcylation modification study was demonstrated for the first time. This is a novel methodology development for exploration of the O-linked glycosylation. While the potential application possibilities offered by this new methodology is a subject of further exploration, it is clear that the new technique will offer unique opportunities to add glycosylated sites into the structure and dynamics studies performed by bio-EPR spectroscopy. In the following sections some primary examples on possible routes for applying the new methodology to some model protein systems are given.

4.1.1. P53 protein model

As was mentioned above, the metabolic process of O-linked glycosylation is feasible mainly for mammalian cells. To implement the goal of EPR studies of O-linked glycosylation, it is necessary to take into account the following parameters: conditions and localisation of the protein of interest in cells, type of glycosylation and placement of the modification site in the protein structure, kinetics of the metabolic cycling, influence of additional PTMs which might have happened in the same time. One of perspective goals for EPR spectroscopy would be to identify an exact position of a glycosylation site. To achieve this goal DEER experiment can be used, which requires doubly spin labelled protein molecules. In case of glycosylation site determination, an ideal test protein model for the proof-of-principle experiments should have a single labelling site at a O-linked modified position and another site with a natural or introduced cysteine amino acid available for the standard SDSL. This arrangement appears optimal, as in this case one obtains data on a single intramolecular distance between the cysteine site and the glycosylated site, so that the interpretation of the distance distribution is rather unambiguous. Additionally, the well optimized and easier to perform standard SDSL technique often yields close to 100% labelling efficiencies. Thus, a possible reduction of the dipolar modulations in the DEER experiment would be only due to the reduced spin labelling efficiency at one site, but not at both sites.

O-GlcNAc modification is a mucin-type PTM, and it is targeting the serine or threonine hydroxyl moieties. As one of the model systems to test, the p53 protein was selected. For p53 the O-GlcNAc Database reports one O-GlcNAcylated site: Ser149. However, in another publication an O-glycnacylated site at Ser121 was considered. This site is associated with stability and activity of the protein.[175] In any case, even if two O-GlcNAcylated sites appear, p53 protein model can be good enough to continue the distance distribution study between two spin labelled sites.

Important information about p53 to take into account is that this protein is prone for tetramerization as was studied by X-ray (see Figure 51).[176]

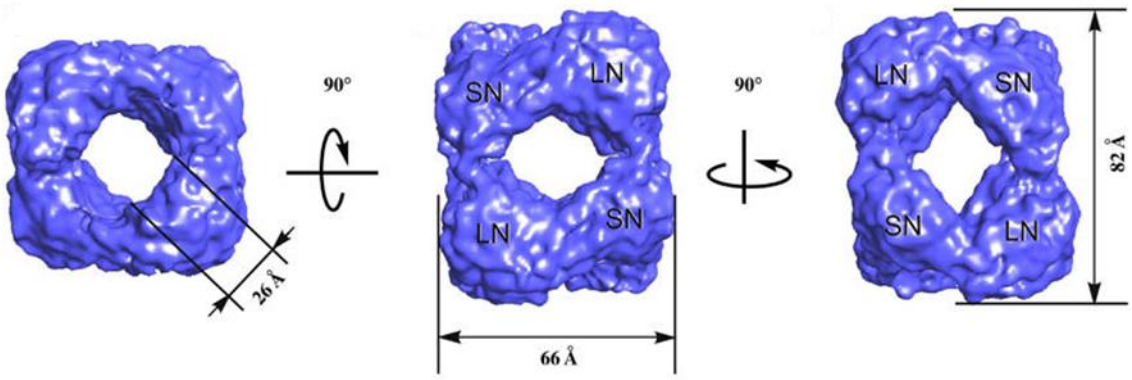


Figure 51. 3D reconstruction of p53. The skewed-cube-shaped p53 molecule side and top views.[176]

Due to the tetramerisation propensity of the protein, another crucial question arises: what is the predicted location of the glycan in the structure of the protein. To verify that aspect GlycoSHIELD simulation programm[75] was used to demonstrate localisation of the O-GlcNAcylation sites. The GlycoShield glycomodeling demonstrates a shielding area which is directly related to the glycan core: the larger the glycan core, the more shielding one will get. If one has a look at Figure 52 that is demonstrating the p53 protein model after glycosimulation, one can clearly observe one area of glycan shielding (left image). At the same moment, the right subplot demonstrates quite complex structure of several linked sugars. It means that attached GlcNAz sugar moiety is a part of a complex glycan core.

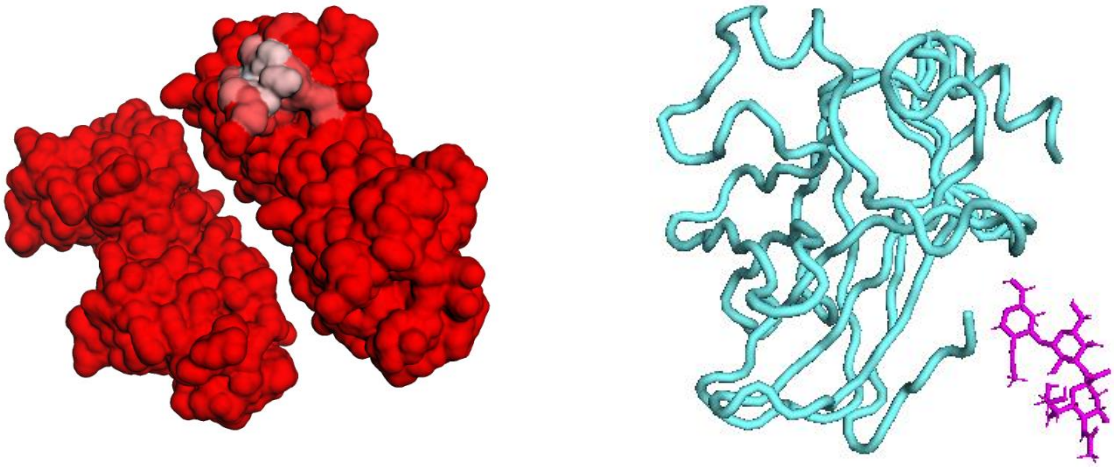


Figure 52. Demonstration of the glycan localisation in the p53 protein structure by using GlycoShield glycomodeling programm.[74]

One can only speculate, which kind of core this might be. Due to the ring numbers (of pyranose) in carbohydrate-based polymer, the glycan core might look as shown in Figure 53 (last image).
Opposite to the glycomodulated result, other variants from O-glycan repertoires on a mucin-type reporter proteins expressed in mammalian cells might be present (see Figure 34). In Figure 53, possible cores that contain GlcNAc site as the last sugar in the glycan chain are represented. Just in such cases one would be able to use the investigated chemical reporter for the further spin labelling reaction.

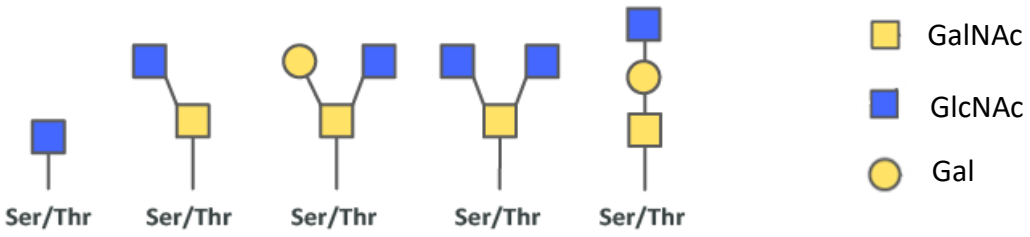


Figure 53. Possible cores that contain GlcNAc site as the last sugar in the glycan chain.

Further scenarios of extended core are possible, one of them is represented in Figure 54. In that case GlcNAc sugar is in the middle of glycan core and will be thus invisible for our spin labelling method.



Figure 54. Extended O-GlcNAcylation glycan core example with the GlcNAc sugar in the middle of the glycal chain.

Regarding the described aspects above one can conclude that detection of O-GlcNAzylation sites will be restricted to the cases where added GlcNAc sugar moiety is the last one in glycan core or is the single sugar at the Ser/Thr site (the latter is a very rare case).

Also, O-glycan heterogeneity is quite pronounced, resulting in the lack of a common glycan core. This is, of course, another complication for the EPR data analysis, still to be addressed in future works. In literature, chemical methods are reported that can liberate the O-glycans and thus make them available for spin labelling. Of course, such approaches detriment the glycan's structure due to "peeling" reactions. Hayden Wilkinson and Radka Saldova discussed in their review that "Universal enzyme exists to deglycosylate a variety of common-core structures from proteins for analysis using mass spectrometric and fluorescence techniques. Unfortunately, for their O-linked counterparts, no such enzyme exists".[177]

In the same way as it was previously described in case of OGT protein, p53 protein was expressed in HEK293T cells (six unis of 30 cm Petry dishes) and modified by Ac4GlcNAz sugar. After cells lysation p53 was purified via His-tag by His-NTA resin column. Spin labelling and washing steps were performed before protein was eluted from the solid state column. In Figure 55 one can observe the corresponding EPR signals of two spin labelled p53 samples differing by the overall protein concentration. The left spectrum was recorded with optimised EPR measurement parameters, protein concentration in the sample was 190 µM. The right one was recorded after an additional sample concentrating (250 µM).

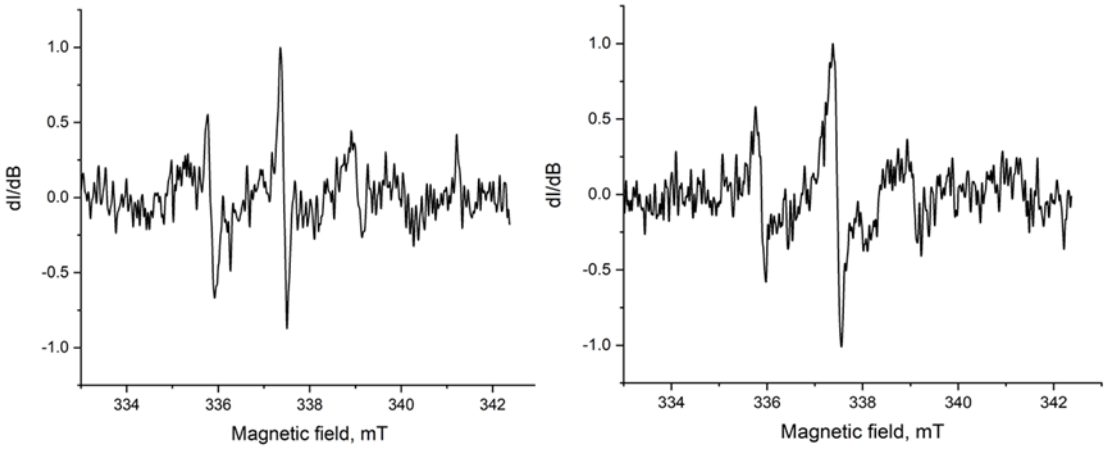


Figure 55. EPR spectra of O-GlcNAcylated p53 protein with protein concentration 190 µM (left spectrum) and 250 µM (right spectrum) obtained by Ac_4GlcNAz treatment of HEK293T cells, spin labelling on the Ni-NTA-beads and elution.

In the first part of the chapter, by the negative control experiments the absence of any EPR signal received from Ni-NTA resin and from other proteins that might be present as an impurity after the protein elution was proved. Here, an additional negative control by performing glycosylation-related spin labelling of the p53 in the absence of Ac_4GlcNAz treatment was performed. To achieve that task, tetraacetylated *N*-acetylglucosamine (Ac_4GlcNAc) was used as negative control (Figure 56). That experiment excluded possibility of any unspecific spin labelling binding.

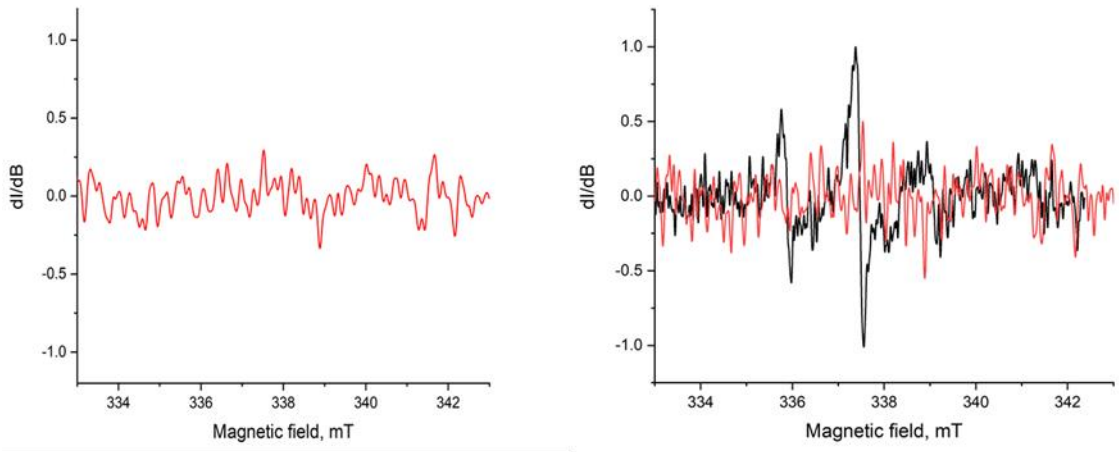


Figure 56. X-band EPR spectra of p53 upon in vivo MGE with Ac_4GlcNAz and CuAAC spin labelling on Ni-beads via His-tag and subsequent elution (black curve). Red (left and right spectrum): negative control using (Ac4GlcNAc) without an azido group (red curve).

4.2. Conclusions and outlook for the EPR methodology application to different protein models

▪ Using a second protein model for the verification of developed methodology for the EPR sample preparation demonstrated its universality as an application method for the general O-GlcNAc modification detection on the protein structure;

▪ It was proved that established approach is reproducible for different protein models;

▪ EPR analytic method might be used for the different protein models with multiple O-GlcNAcylated sites for their general detection, this would possibly require specific methodology developments, especially in the multi-site data analysis;

▪ In current section different glycan cores were considered. Their design is crucial for the prediction of protein structure modification by chemical reporter that was used for O-linked glycosylation.

4.3. Influence of the molecule's size and the solvent on the nitroxide radical's mobility

As mentioned previously, besides measuring distances between spin labels attached at two different sites of a protein, EPR spectroscopy is capable of measuring spin label tumbling times. The stochastic spin label tumbling can be used to characterize the properties of the spin labelled protein, in terms of its size and local rigidity at the labelling site – both these factors directly influence the spin label tumbling. In this section an example of EPR spectra characteristics understading for spin labelled rod-like molecules is given. This was accomplished in cooperation with Prof. Dr. Ulrich Steiner (Uni Konstanz) and Prof. Dr. Adelheid Godt (Uni Bielefeld). In the next section analogous treatment of the spin labels in the newly developed glycosylation/spin labelling technique is made.

In the given mini-project two rod-like (ruler) compounds with a length difference due to the extra alkine part present in the second ruler were studied. Compound №1 has a chemical formula $C_{43}H_{50}NO^{\bullet}_{4}$ and molecular weight 644.876 g/mol; Compound №2 has a chemical formula $C_{66}H_{80}NO^{\bullet}_{4}$ and molecular weight 951.369 g/mol (see Figure 57).

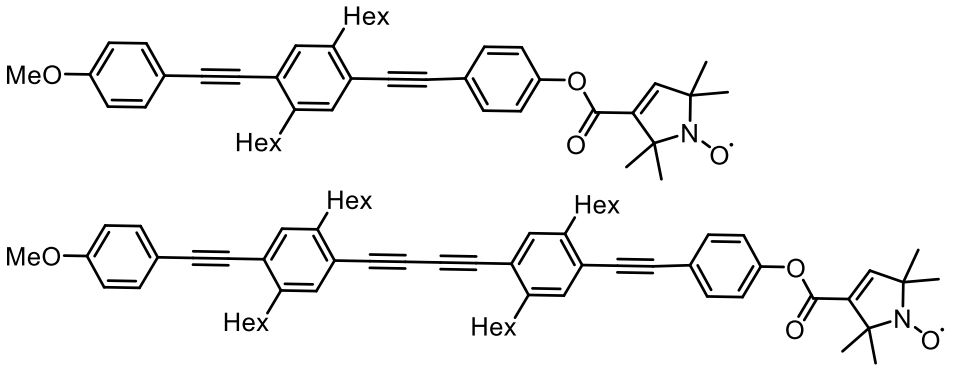


Figure 57. Chemical structures of the compound №1 $C_{43}H_{50}NO^{\bullet}_4$ (upper structure) and compound №2 $C_{66}H_{80}NO^{\bullet}_4$ (lower structure).

The goal of the project was to perform the nano-viscosity measurements by observing nitroxide radical's mobility via the EPR spectral simulations, and to possibly observe different behaviour in the solvents with different density. Such test solvents were tetrahydrofuran (THF) with a density ρ=0.889 g/cm^3 and polytetrahydrofuran (polyTHF) with a density ρ=0.982 g/cm^3.

EPR spectroscopy is a nice tool to implement that goal via detection of nitroxide radical rotational diffusion time in different environments. For each compound two samples with the same radical concentration of 125 μM were prepared in THF and polyTHF solvents. The EPR spectra of these samples were measured at X-band EMXnano Bruker spectrometer with the following settings: 29.5 dB attenuation, 0.113 mW power, 60s sweep time and 0.108 mT modulation amplitude. Four EPR spectra were obtained and analysed to obtain rotational diffusion times, which were then used to characterise the nano-viscosity related to the given compounds.

The EPR spectra were simulated using EasySpin library, with the obtained best fit parameters listed in Table 2.

Compound	Solvent	A-tensor, MHz	g-tensor	T_{corr} (ps)
№1 $C_{43}H_{50}NO^{\bullet}_4$	THF	[17.34 85.01]	[2.00906 2.00687 2.00300]	163.3
	polyTHF	[19.06 100.01]	[2.00906 2.00687 2.00300]	1259
№2 $C_{66}H_{80}NO^{\bullet}_4$	THF	[17.34 85.01]	[2.00906 2.00687 2.00300]	269
	polyTHF	[18.82 100.01]	[2.00906 2.00687 2.00300]	1585

Table 2. The value of simulated parameters (A- and g-tensors) and rotational correlation time (T_{corr}) of examined compounds in THF and polyTHF solvents.

The lengths of the two ruler compounds are different. From the beginning, a slower rotational diffusion of the nitroxide radical was expected in the second molecule that is larger and longer than the first one. To compare the viscosity parameters of the two solvents, the shapes of the corresponding EPR spectra of the same compound in different solvents were also simulated. For the following discussion in the glycosylation project, one should additionally note that the peak-to-

peak amplitude of the central EPR line and the relative broadenings of all three EPR lines are main tumbling-related characteristics of the CW EPR spectra of nitroxide radicals. Due to the anticipated anisotropy of tumbling for such elongated compounds, the exact fitting of their CW EPR spectra is a non-trivial task. Especially, for the compound №2 it was difficult to fit parameters (A- and g-tensors) that will simulate spectral shapes, when using a single rotational correlation time parameter. The final fits were performed, using the same g tensor for all cases, solvent-dependent axial hyperfine tensor and a single rotational correlation time (approximation of the isotropic tumbling). In Table 2. one can find the final rotational correlation time values for the examined compounds in the solvents with different viscosity.

The fitted EPR spectra demonstrated that the first molecule with lower MW and shorter ruler length shows faster mobility in both solvents as compared to the second compound (see Figure 58). However, the rotational correlation time in THF solvent, that has lower density, for both compounds is shorter than the corresponding rotational correlation time in polyTHF solvent. Such a result was anticipated in the beginning of that experimental series due to the higher density of polyTHF comparing to THF: higher viscosity of the solvent should lead to less mobility of the molecule in such environment and longer rotational correlation time.

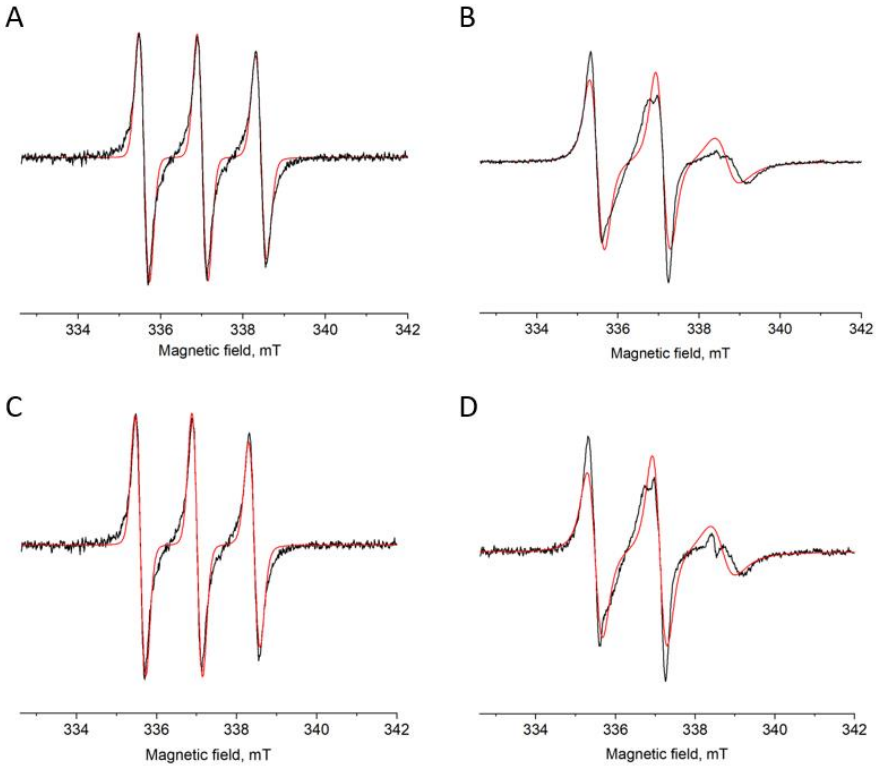


Figure 58. Simulation of EPR spectra of the next samples: compound №1 in THF (A) and polyTHF (B), compound №2 in THF (C) and polyTHF (D).

Note however that the spectra fitting provides more detailed information. For instance, it was observed that the relative change of the rotational correlation time between compounds 1 and 2 is smaller in the solvent with higher viscosity (~24% in polyTHF) and larger in the solvent with the lower viscosity (~65% in THF).

4.4. CW-EPR spectra analysis elucidates main steps in the glycan spin labelling

Qualitative and quantitative analysis of the CW EPR spectra is a convenient way to observe changes in the spin label mobility as it undergoes different steps in the spin labelling route. Figure 59 summarizes the CW EPR spectra of the free alkyne proxyl spin label, from a spin labelled O-GlcNAz, and from the O-GlcNacylated and spin labelled OGT and p53 proteins. One can see that a free label and a spin labelled sugar are characterized by narrow lines with relatively less difference in the peak-to-peak amplitude between the central line and the outer lines. These are clear indications of the so-called 'fast motion regime'. In the case of spin labelled O-GlcNacylated OGT and p53 the differences in the peak-to-peak amplitudes between the three EPR lines in the nitroxide spectrum are clearly larger, thus indicating the transition from the fast-motion to the slow-motion regime. However, one can recognize that individual lines are still relatively narrow. However, one can recognize that individual lines are still relatively narrow.

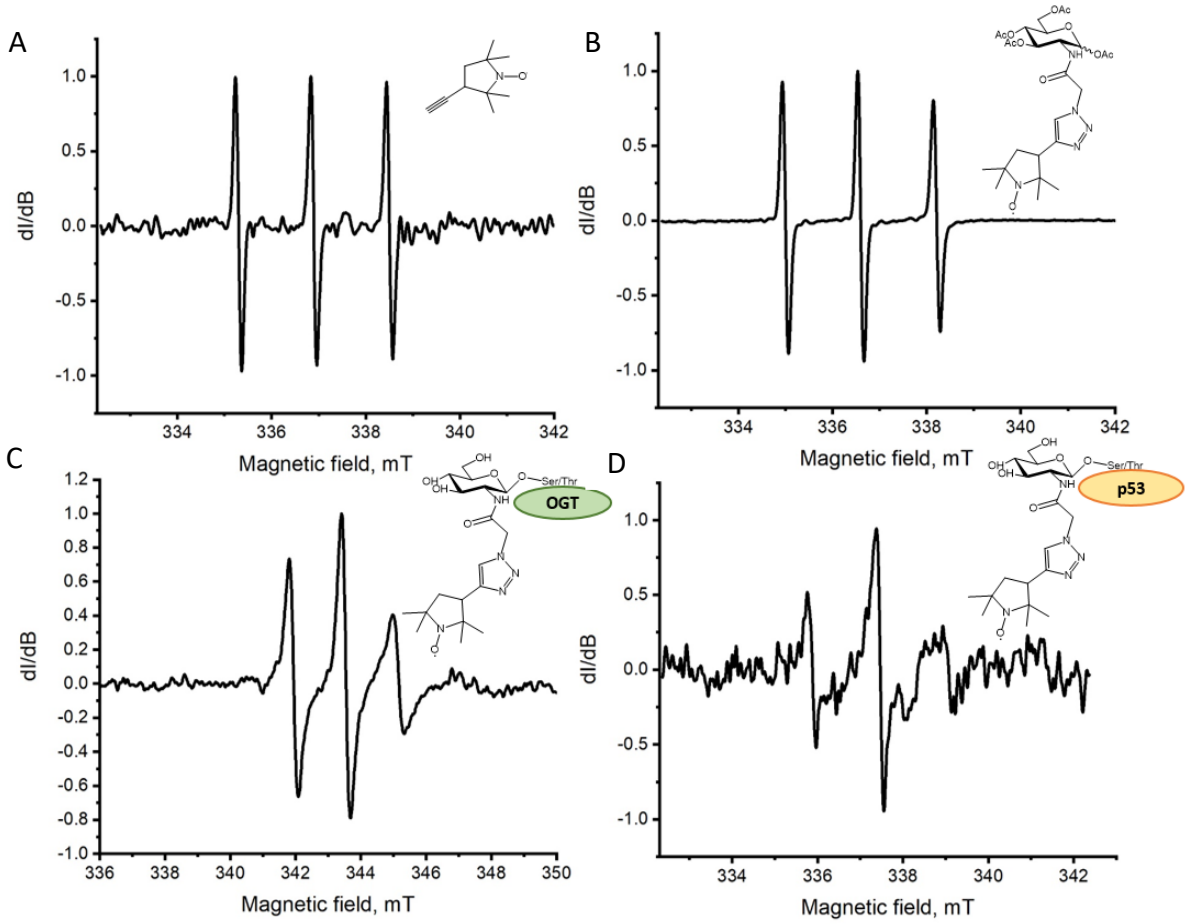


Figure 59. Experimental EPR spectra for (A) alkyne proxyl in solution (τ_r=10 ps), (B) alkyne proxyl attached to Ac_4GlcNAz in solution (τ_r=79 ps), (C) alkyne proxyl attached to GlcNAz incorporated in OGT (τ_r=1.0 ns) and (D) alkyne proxyl attached to GlcNAz incorporated in p53 (τ_r=1.2 ns). All EPR spectra are normalized to the height of the central peak.

This spectral shape might appear due to several reasons. In a spin labelled protein the overall rotational tumbling is a composition of spin label tumbling, which is fast but partially anisotropic due to the linker restrictions, and the overall approximately isotropic protein tumbling. This makes a complex rotational diffusion pattern that can be only approximately reproduced within the restrictions of an isotropic rotational diffusion. Another important point is that OGT has more than one glycosylation site and thus the observed CW EPR spectrum is actually a superposition of the spectra for all spin labelled sites, which might differ in the overall spin label mobility, for instance due to the different local crowding at different labelling sites. Importantly, as was discussed above, even at a single glycosylation site it is possible to have different glycan cores formed. Therefore,

even for a protein with a single glycosylation site like p53, the CW EPR spectrum can be a superposition of different spectral components due to different glycan cores present.

Spectral simulations were performed using the toolbox EasySpin 5.2.29..[178] Assuming an isotropic rotation of SL and only varying the rotational correlation time between the different spectra tumbling for the cases of free SL and SL attached to Ac_4GlcNAz worked well for describing the corresponding EPR spectra, the isotropic tumbling fits for SL attached to the proteins (Figure 59 C, D) are – as expected – slightly oversimplified but the isotropic rotational correlation time τ_r can be used as semi-empirical parameter. The fitted spectra are shown in Figures 60 and 61.

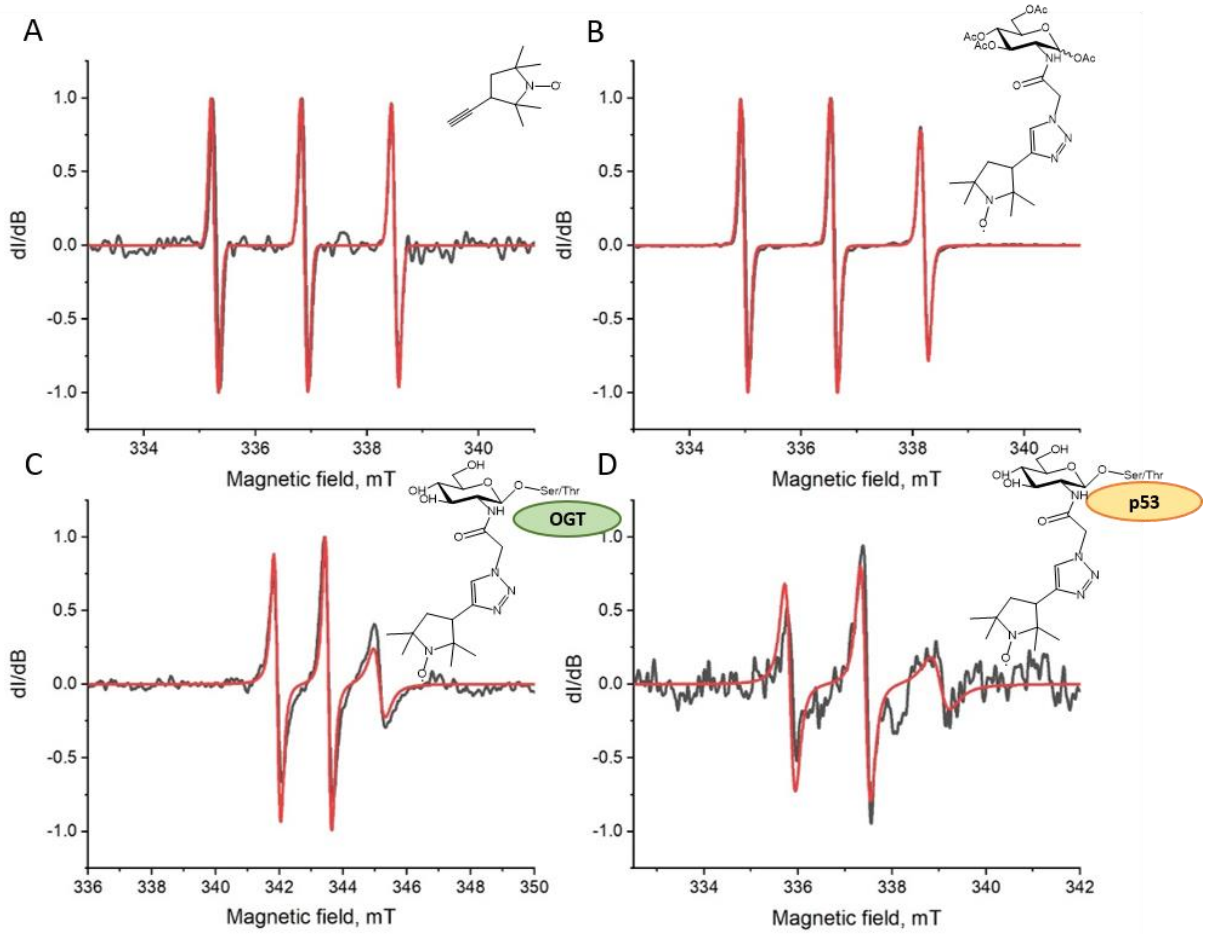


Figure 60. Experimental EPR spectra for (A) alkyne proxyl in solution, (B) alkyne proxyl attached to Ac_4GlcNAz in solution, (C) alkyne proxyl attached to GlcNAz incorporated in OGT and (D) alkyne proxyl attached to GlcNAz incorporated in p53. Spectra fitted with Lorentzian line form.

In the very fast tumbling regime the nitroxide CW EPR spectra tend to have Gaussian line shapes due to multiple weak unresolved proton couplings.

As soon as the characteristic rotational correlation time exceeds the inverted line widths, the Gaussian line shapes are substituted by Lorentzian line shapes, in accordance with the predictions of the general spin tumbling theory. Both fitting with intrinsic Gaussian and Lorentzian line forms was tested. In our case, Lorentzian fitting turned out to be slightly better than the fitting with Gaussian lines (compare Figures 60 and 61). Also, as just discussed, the heterogeneity of the glycan cores and labelling sites most likely contributed to a worse fit of the CW EPR spectra of the spin labelled O-GlcNacylated proteins. The spectra were best described by the rotational correlation times of τ_r = 10 ps for the free SL, τ_r=79 ps for the free label attached to the sugar Ac_4GlcNAz, τ_r= 1.0 ns, and τ_r=1.2 ns for the spin label attached to OGT or p53. The increase of τ_r with increasing complex size, i.e. from A) over B) to C) or D), respectively, reflects the slower tumbling of SL. In cases C) and D), τ_r does not reflect the overall molecular tumbling of the complex but rather the flexibility of the secondary structure element involved or steric hindrance from inter- and intra-molecular interactions.[179] Since p53 is prone to tetramerisation,[180] this might result in slightly slower tumbling of SL in p53 as compared to OGT.

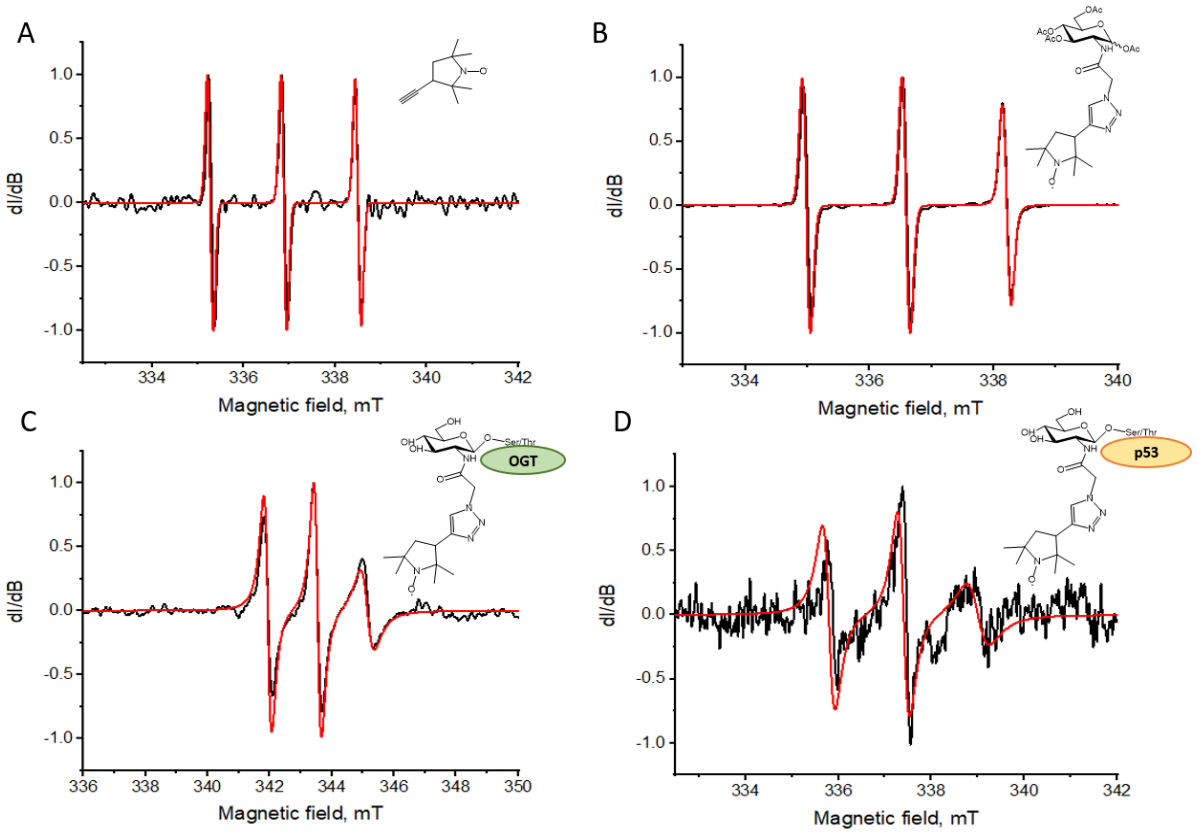


Figure 61. Experimental EPR spectra for (A) alkyne proxyl in solution, (B) alkyne proxyl attached to Ac_4GlcNAz in solution, (C) alkyne proxyl attached to GlcNAz incorporated in OGT and (D) alkyne proxyl attached to GlcNAz incorporated in p53. Spectra fitted with Gaussian line form.

4.4.1. EPR signal intensities and CuAAC reaction

In a control experiment under similar conditions, the spin labelling efficiency of Ac_4GlcNAz with Alkyne Proxyl in the CuAAC reaction was determine to be approximately 75%, which agrees (see Figure 38) with formerly reported CuAAC spin labelling efficiencies.[25,27] Assuming a labelling efficiency of 75% for the metabolically incorporated Ac_4GlcNAz, one can estimate that 0.4% (0.6%) of all available O-GlcNacylation protein sites in OGT (in p53) are occupied by Ac4GlcNAz.

For the OGT sample modified with Ac4GlcNAz, upon elution, buffer exchange and concentrating the protein solution to the original volume, a spin concentration of 0.87 µM at a protein concentration of 266 µM was found, thus corresponding to a relative spin concentration of approximately 0.3% per molecule.

4.4.2. Inhibitors can influence the interplay between O-GlcNAcylation and phosphorylation

In the literature overview the significance of O-GlcNAcylation process that play an important role in the regulation of the epigenome in response to the cellular metabolic status was described.[180] Through histone modifications and assembly of gene transcription complexes, OGT can impact expression of genes important for cellular metabolism.[181] Along with this, some O-GlcNAcylated site and Phosphorylation processes of the same proteins can interplay with each others. For instance, p53 protein undergoes intense regulation in cell which involves reversible glycosylation and phosphorylation[171]. The inhibitors Thiamet G and Streptozotocin inhibit OGA enzyme which is responsible for cleavage of O-linked sugar moieties.[182] The DNA damaging agent Doxorubicin induces cell stress. As a response, the cells upregulate the production of p53 and downregulate its phosphorylation.[171]

To quantify the effect of inhibitors on the interplay between O-GlcNAcylation and phosphorylation processes and on the stability of p53 protein the influence of Thiamet G, Streptozotocin and Doxorubicin on the MGE and O-GlcNAcylation level of p53 was thus studied. EPR spectroscopy gives the possibility to detect the concentration of the radicals in the sample. This is a convenient opportunity for O-GlcNAcylation level monitoring due to the high level of CuAAC spin labelling efficiency. It means that one can qualitatively and quantitatively identify the levels of O-linked glycosylation of the protein under different conditions.

In a control experiment without any inhibitors during MGE, the intensity of the corresponding CW EPR spectrum of the spin labels was determined. This intensity is in agreement with expectations based on formerly reported CuAAC spin labelling efficiencies.[24,29] (see Figure 62 and Figure 63). In case inhibitors Thiamet G, Steptozotocin or Doxorubicin were present during MGE, the intensity of the EPR spectra increased by 24%, 7.6% and 51%, respectively (see Table 3). This reflects the efficiency of these inhibitors in suppressing phosphorylation.

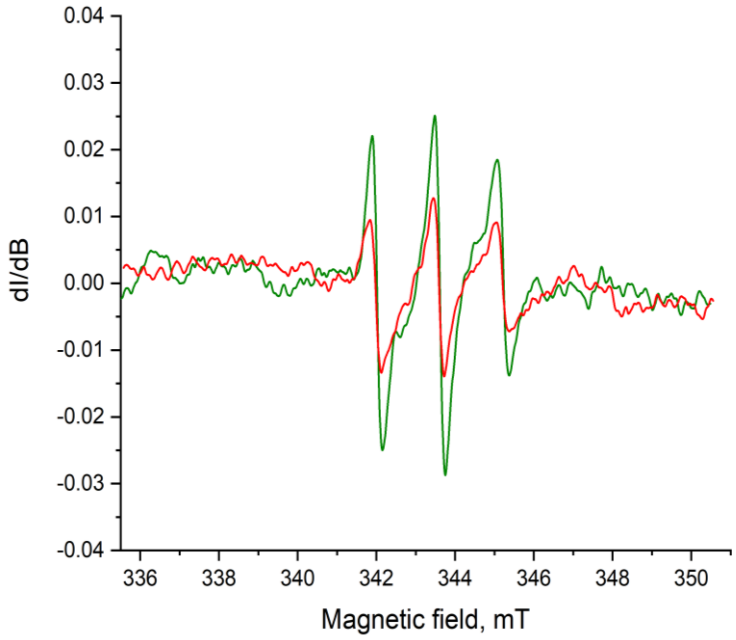


Figure 62. X-band CW EPR spectra of p53 protein (300 µM) extracted from HEK 293T cells after in vivo MGE with $Ac_4GlcNAz$, spin labelled by CuAAC on the Ni-beads and eluted. Red spectrum – no inhibitor at any stage; Green spectrum – with Thiamet G (100 µM) inhibitor during lysation and protein purification.

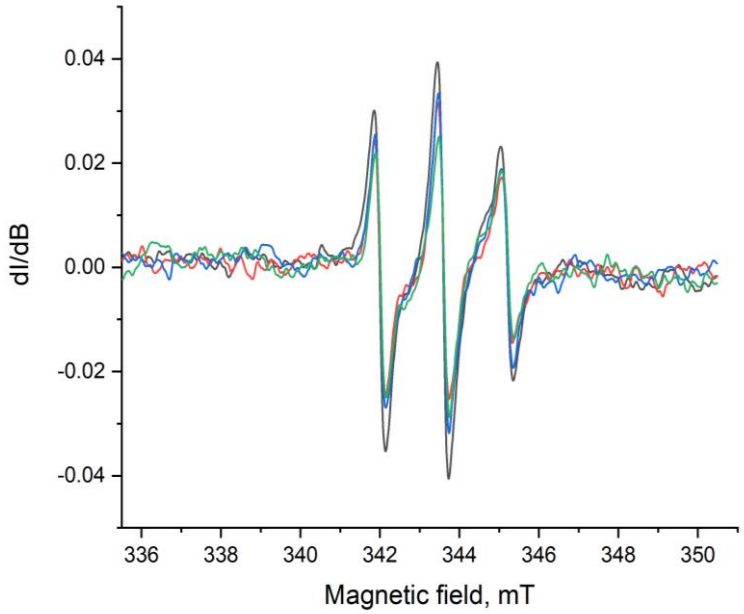


Figure 63. X-band CW EPR spectra of p53 protein (300 µM) extracted from HEK 293T cells after in vivo MGE with Ac_4GlcNAz, spin labelled by CuAAC on the Ni-beads and eluted. Green spectrum – with Thiamet G (100 µM) inhibitor during lysation and elution; Blue spectrum – with Thiamet G (100 µM) inhibitor during MGE, lysation and elution; Magenta spectrum – with Streptozotocin (2mM) inhibitor during MGE and Thiamet G (100 µM) inhibitor during lysation and protein purification; Black spectrum – with Doxorubicin (1 µg/ml) inhibitor during MGE and Thiamet G (100 µM) inhibitor during lysation and elution.

Sample	Spin concentration[1]	Intensity[2]	Intensity[3]	Labelling efficiency
Reference: no inhibitor at any steps	1.25 µM	100%	-----	0.6%
Thiamet G during lysation	2.46 µM	198%	100%	1.1%
Streptozotocin during MGE	2.66 µM	214%	107.6%	1.3%
Doxorubicin during MGE	3.75 µM	300%	151%	1.8%
ThiametG during MGE	3.08 µM	246%	124%	1.5%

Table 3. Relative EPR signal intensities for p53 (Peak to peak intensity comparison).
[1] Estimated error bars for the spin concentration and labelling efficiency were ±20%, except ±50% error bars for the p53 samples without any inhibitors.
[2] EPR spectra intensity increase as compared to the reference with no inhibitor at any steps.
[3] EPR spectra intensity increase as compared to the reference with Thiamet G during lysation.

4.5. Rapid scan measurements

The solution CW EPR measurements require high concentrations of the glycosylated and spin labelled protein on the one hand and need to be performed at ambient temperature on the other hand. Both these factors are risky for the protein's intrinsic structure stability, and might lead to the

protein unfolding, degradation or some other destructuring processes. To prevent the biological samples from undesirable effects, shorter EPR acquisition times might be favourable, as achieved with 'Rapid-scan EPR' (RS-EPR) (see Chapter 2). RS-EPR can potentially improve the signal to noise ratio and also significantly shortens the required measurement time (down to milliseconds) .[183]

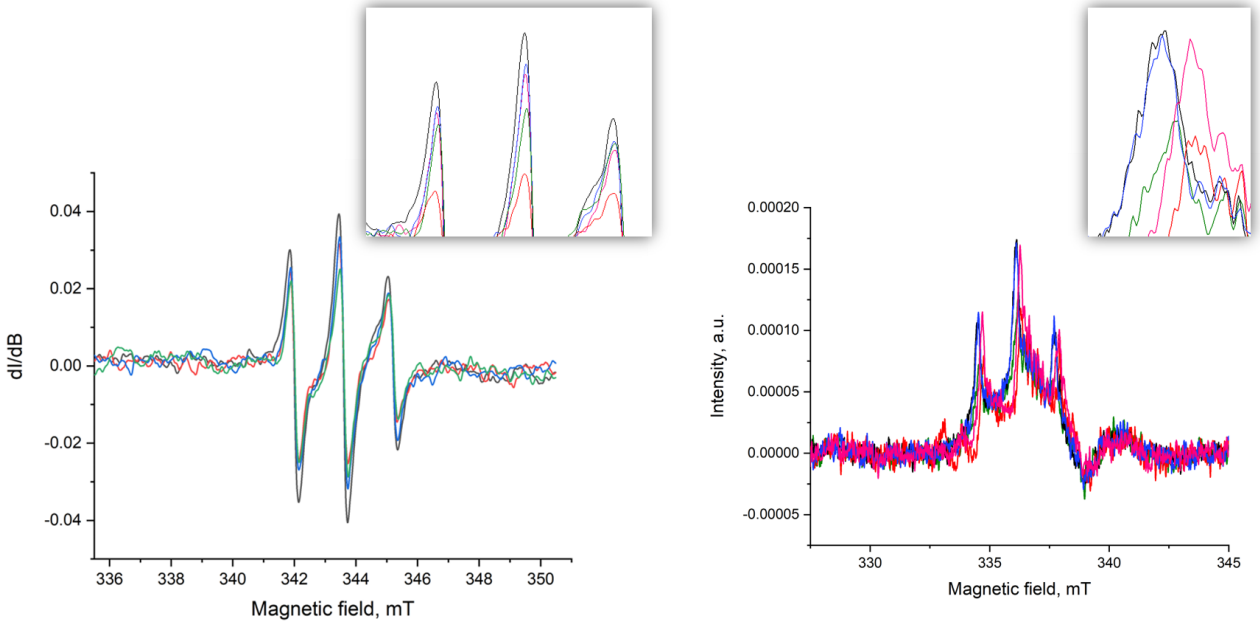


Figure 64. CW-EPR measurements (left spectra) and Rapid-scan (right spectra) of the samples spectra of p53 protein (300 µM) extracted from HEK 293T cells after in vivo MGE with Ac_4GlcNAz, spin labelled by CuAAC on the Ni-beads and eluted. Green spectrum – with Thiamet G (100 µM) inhibitor during lysation and elution.; Blue spectrum – with Thiamet G (100 µM) inhibitor during MGE, lysation and elution; Magenta spectrum – with Streptozotocin (2mM) inhibitor during MGE and Thiamet G (100 µM) inhibitor during lysation and protein purification; Black spectrum – with Doxorubicin (1 µg/ml) inhibitor during MGE and Thiamet G (100 µM) inhibitor during lysation and elution; Red spectrum – no inhibitor at any stage

Rapid-scan spectra shown in Figure 64 (right spectra) were measured in 60 seconds time comparing to the CW-EPR spectra that needed 30 minutes time. This means that for the same measurement time as in CW EPR the RS-EPR spectra would reveal the overall signal-to-noise ratio by approximately a factor of $\sqrt{30} \approx 5.5$ times better than for the RS-EPR spectra in Figure 64. This appears to be indeed better than for the presented CW EPR spectra (see rmsd and signal-to-noise values in the Table 4).

EPR method	Sample	Noise RMSD	Signal amplitude	SNR
CW-EPR	Reference: no inhibitor at any steps	0.0016	0.0532(66.5%)	33
	Doxorubicin during MGE, ThiametG during lysation and elution	**0.0011**	**0.08 (100%)**	**73**
	ThiametG just during MGE	0.0018	0.0539(67.4%)	30
	Streptozotocin during MGE, ThiametG during lysation and elution	0.0012	0.0569(71.1%)	47
	ThiametG during MGE, lysation and elution	0.0014	0.0653(81.6%)	47
RS-EPR	Reference: no inhibitor at any steps	8.17e-6	1.49 e-4(74.1)	18
	Doxorubicin during MGE, ThiametG during lysation and elution	**5.37e-6**	**2.01 e-4(100%)**	**37**
	ThiametG just during MGE	4.7e-6	1.64e-4(81.6%)	35
	Streptozotocin during MGE, ThiametG during lysation and elution	5.37e-6	1.90 e-4(94.5%)	35
	ThiametG during MGE, lysation and elution	5.07e-6	1.99 e-4(99%)	39

Table 4. Rmsd and signal-to-noise values for the samples of p53 protein (300 μM) extracted from HEK 293T cells after in vivo MGE with $Ac_4GlcNAz$, spin labelled by CuAAC on the Ni-beads and eluted. Percent value for the signal amplitude is given with respect to the sample with the maximum signal amplitude $I = I_{max}-I_{min}$.

However, the improvement of the signal to noise ratio would be anticipated within approximately a factor of two, or, possibly, even less, since in the present series the signal-to-noise ratios of the CW EPR spectra are systematically better than the corresponding values for the RS EPR spectra (see Table 4). Also, as one can see from the comparison of the two plots in the Figure 64, RS-EPR spectra reveal a somewhat worse spectral shape stability, which also can contribute to the accuracy of the EPR intensity quantification. From the spectral intensity of both measurements one can conclude that ratios of peak intensities (which are proportional to the ratios of the spin concentrations of O-linked glycosylated protein sites) are in a reasonable agreement for the two methods. For both types of EPR experiments, the p53 protein expressed without any inhibitors demonstrated the weakest nitroxide signal among all tested conditions. Likewise, the sample with the p53 protein exposed to the Doxorubicin (1 μg/ml) inhibitor during MGE and Thiamet G (100 μM) inhibitor

during lysation and elution has demonstrated highest signal intensity both in CW EPR and RS-EPR. Note however that the samples with the intermediate intensities cannot be ordered in the same way based on the CW EPR and RS-EPR data. While in the CW EPR series all samples reveal different intensities of the spin labels' signal, in the RS-EPR ordering of the 'intermediate' samples appear tricky. The latter might be related to the short measurement time and low signal-to-noise ratio, but this most likely also has some contribution from the worse spectral shape stability in the presented RS-EPR series. In any case, also in the case of spin labelled glycosylated protein studies RS-EPR is of potential high interest as a sensitivity improvement tool, since systematically low spin concentrations are anticipated.

4.6. Identification of the native O-GlcNAcylated sites and the O-GlcNAzylated sites

A possibly important process that can substantially contribute to the low spin labelling efficiencies in our approach for glycosylated proteins is the native O-GlcNAcylation modification that might (or might not) block the sites that would otherwise participate in MGE. To evaluate the level of native O-GlcNAcylation the following EPR-based strategy was developed. After isolation and purification of the MGE protein or natively glycosylated protein, in case 'free' (i.e. not O-GlcNAzylated) sites are left, the protein can be further enzymatically O-GlcNAzylated over the empty native O-GlcNAc sites and then additionally spin labelled. A comparison of samples with and without such in vitro MGE with spin labelling step would allow us to evaluate the balance between the O-GlcNAzylated sites and the glycosylation sites with terminal O-GlcNAc that were left free after the protein purification.
O-GlcNAc enzymatic labelling provides a highly sensitive and efficient method for the in vitro modification of O-GlcNAc-modified proteins. Proteins are enzymatically labelled with the permissive mutant of the β-1,4-galactosyltransferase (Gal-T1 (Y289L)), which transfers azide-modified galactose (GalNAz) from UDP-GalNAz to O-GlcNAc.[184] Chemoenzymatic labelling, one of the most widely used strategies for O-GlcNAc protein enrichment, specifically attaches an azido-derivatized galactose to the N- acetylglucosamine residue of the O-GlcNAcilated protein.[165] The wild type Gal-T1 exhibits very poor GalNAc-T activity, whereas Gal-T1 mutant exhibits higher GalNAc-T activity as compared to Gal-T.[185] The steady state kinetic analyses indicates that the K_m for GlcNAc has increased compared to the wild type about 3-5 times.[185] Thus, for the modification

and labelling of the native O-GlcNAc sites Gal-T1 (Y289L mutant) was used due to its high enzymatic efficiency.

In Figure 65 one can see the three sample preparation routes that allow us to detect native O-GlcNAcylation and to compare the level of O-linked glycosylation by using $Ac_4GlcNAz$ during MGE and sites that were occupied by native sugars that provide native O-GlcNAcylation.

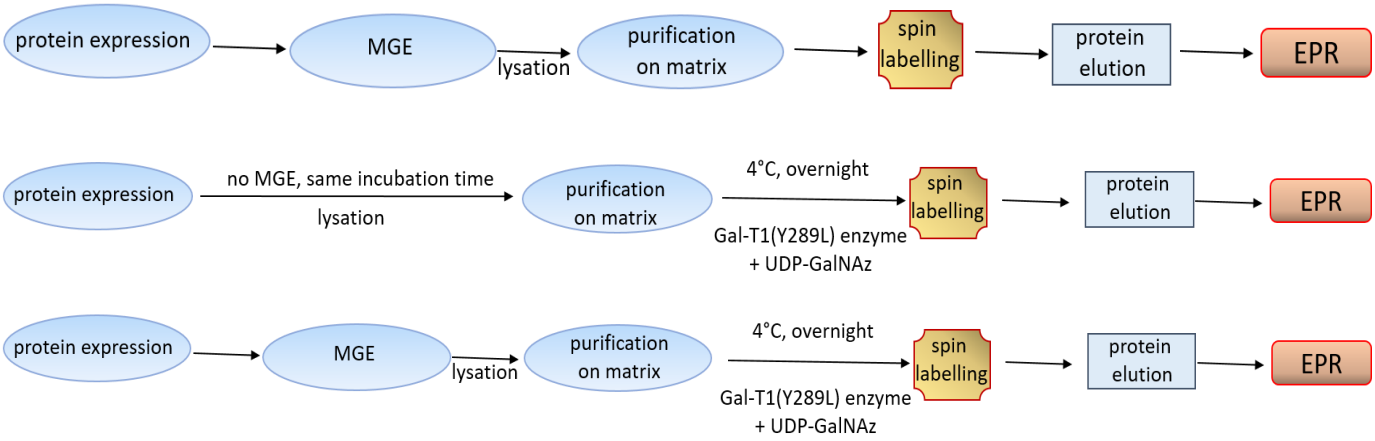


Figure 65. The scheme of three sample preparation routes for the study of native O-GlcNAcylation modification that is going in parallel to the MGE of the protein of interest. These experiments give us the possibility to evaluate the influence of native O-GlcNAcylation in the cells during the MGE process.

Three samples were prepared for the measurements, as indicated in Figure 65:

1. Sample of the p53 protein that was expressed in HEK293T cells and modified by MGE in usual way (as discussed in the previous sections). In the end, the EPR spectrum was measured to reveal the amount of the spin labelled sugars with azido group that isused for MGE.

2. Sample of the p53 protein that was expressed in HEK293T cells. In-cell MGE step was omitted and after p53 protein isolation the O-GlcNAc enzymatic labelling was performed using Click-IT O-GlcNAc Enzymatic Labelling system (ThermoFisher kit). The chemical reaction of the enzymatic labelling is given in Figure 66.

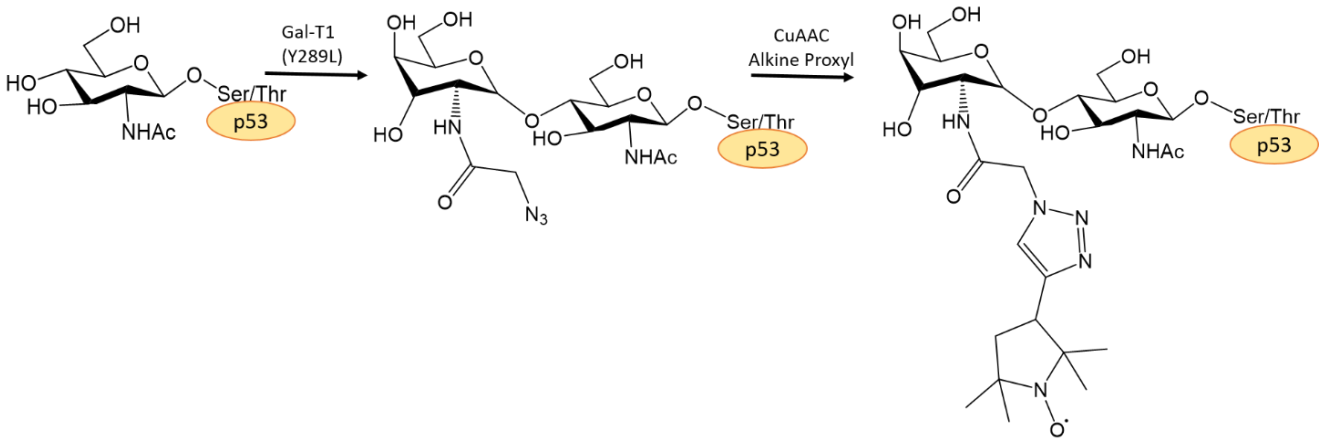


Figure 66. Chemoenzymatic labelling of O-GlcNAc proteins by Gal-T1(Y289L) enzyme and further CuAAC reaction with alkine proxyl spin label.

3. The last sample preparation combined both experimental strategies described for the two previous samples' preparation schemes: after p53 expression in HEK293T cells MGE was performed in a regular way but after the protein isolation O-GlcNac enzymatic labelling was made (how it was explained in the second strategy above) and only then the spin labelling reaction was performed. Alkyne Proxyl spin labelling of that sample is supposed to give the signal from natively O-GlcNAcylation after enzymatic reaction and O-GlcNAzylated sites during MGE.

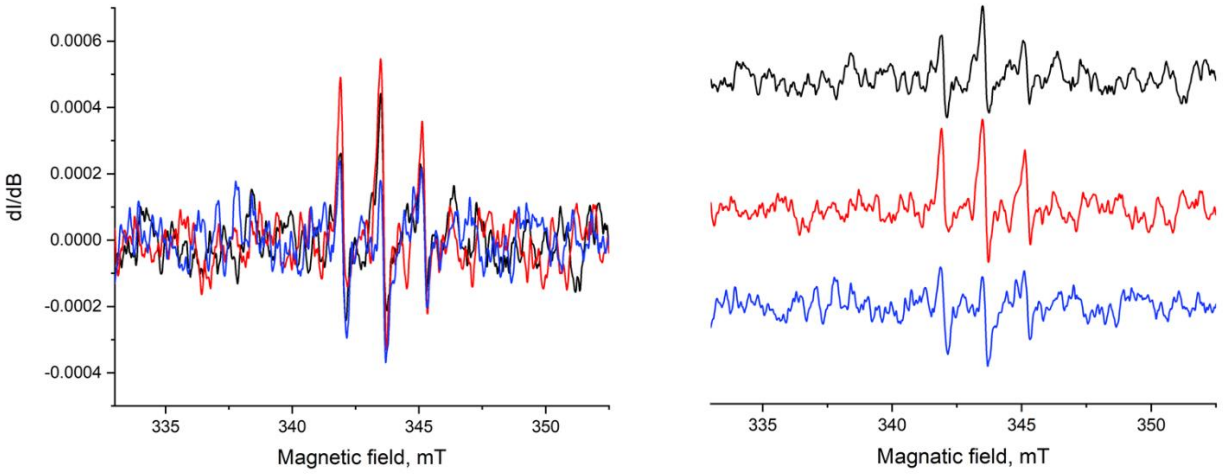


Figure 67. X-band CW EPR spectra of p53 protein (200 µM) extracted from HEK 293T cells after:
1. *in vivo* MGE with Ac_4GlcNAz, spin labelled by CuAAC on the Ni-beads and eluted (black spectrum);
2. no MGE was performed, after protein isolation on Ni-NTA O-GlcNac enzymatic labelling was done, spin labelled by CuAAC on the Ni-beads and eluted (blue spectrum);
3. *in vivo* MGE with Ac_4GlcNAz, , after pr otein isolation on Ni-NTA O-GlcNac enzymatic labelling was done, spin labelled by CuAAC on the Ni-beads and eluted (blue spectrum).

The EPR spectra in the Figure 67 correspond to these three sample preparations. When the spectrum has such a high noise level it is difficult to perform quantitative comparison and say about precise numbers of the labelling efficiency and as a result about O-GlcNAcylation levels. Still, one can appreciate the fact that the nitroxide EPR intensities of the labelled O-linked sites demonstrate (almost the same) contribution of native O-GlcNacylation of p53 (blue spectrum) and of p53 O-linked glycosylated via MGE using Ac4GlcNAz sugar modification (black spectrum). The red spectrum corresponds to the third sample preparation scheme, and it is the combination of the first and the second route for the O-GlcNAcylation modification (native and MGE). The intensity of the red spectrum is indeed,

within the present SNR level, about two times higher than for the black and the blue EPR spectrum.

Control reaction to prove the good selectivity in the route 2 (see Chapter V "O-GlcNAc enzymatic labelling) was performed also with the fluorophore TAMRA-5 instead of a spin label (Figure 68).

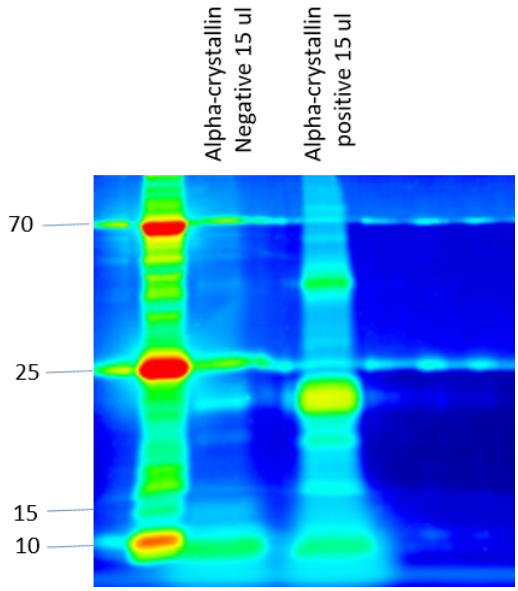


Figure 68. Fluorescence image in SDS-PAGE as a quality control of the Click-IT O-GlcNAc Enzymatic labelling reaction followed by the labelling with the fluorophore TAMRA-5.

Consistently with the EPR results, a substantial fluorescence signal is observed after using the Click-IT O-GlcNAc Enzymatic labelling kit followed by the fluorescent labelling, while only very weak fluorescence level is visible in case of labelling without preliminary treatment with this Gal-T1 kit.

Summarizing these results, one can conclude that our first methodology was developed just for the O-GlcNAcylation sites detection without taking into account native modification that is also going on in parallel to the MGE. To identify natively O-GlcNAcylated sites of the proteins,

chemoenzymatic approach was used as the selective labelling with GalNAz. This is achieved using Y289L GalT enzyme mutant that can recognize loose O-GlcNAcylated sites**.** This topic requires of course in future a deeper study including an optimization of the strategy established and tested here.

4.7. PaNDA spin labelling

CuAAC spin labelling reaction required many additional reagents that at some point are hazardous for the biological system. So far, click-chemistry was performed in vitro after the protein isolation from the cell lysate, but the final goal is to perform spin labelling in the living system – inside mammalian cells. For that reason another method has to be considered that is compatible with the *in cellula* spin labelling and can potentially lead to direct identification of O-GlcNAc modified proteins in their natural environment.

Ac_4GlcNAz has been chosen for MGE, because it does not significantly bind to the surface of HEK293 cells and features high cell permeability.[14,18,24] Ac_4GlcNAz is intracellularly processed and incorporated instead of its natural monosaccharide counterpart N-Acetylglucosamine (GlcNAc) and binds the hydroxy groups of serine and threonine, and for long time it was considered that it is not attached to cysteines.[186]

However, as was discussed in the thesis of F. Doll,[186] recent work by the Chen group indicated that peracetylated unnatural monosaccharides are not completely deacetylated inside cells and are also attached to cysteines instead of serine and threonine residues.[36] Using mass spectrometry, Chen and colleagues found that Ac4GlcNAz, Ac4GlcNAlk, and Ac36AzGlcNAc mainly and Ac4GalNAz partially end up on cysteine residues. Whether Ac_4GlcNCyoc, which was not investigated in the cited thesis, might also be linked to cysteine residues, was left as an important question to be answered. The experiments of F. Doll *et al*. revealed that Ac4GlcNCyoc is enzymatically attached to proteins and that OGT is involved in this process. New *O*-GlcNAc reporters with smaller chemical handles than Ac4GlcNCyoc, but also capable of being labelled in a DA_{inv} reaction, could be synthesized and used for protein-specific imaging.[144]

Introduction of the novel label (that was synthesized by Bjarne Silkenath in the group of Prof. Dr. Wittmann)[139] is achieved through the DA_{inv} reaction[187] of a 1,2,4,5-tetrazine with a strained alkyne (cyclooctyne, SCO)[188] or alkene (*trans*-cyclooctene, TCO).[189] This reaction forms the corresponding pyridazine and dihydropyridazine, respectively. Due to its excellent water compatibility, the DA_{inv}

reaction has proven to be suitable for a broad range of biochemical applications both *in vitro* and *in cellula*.[187,190] Strain-promoted DA_{inv} cycloadditions have been used for bioimaging of O-GlcNAcylated proteins previously in the group of Prof. Dr. Wittmann and Prof. Dr. Zumbusch at the University of Konstanz. Strain-promoted DA_{inv} cycloadditions for the photoactivatable nitroxide for DA_{inv} reaction (PaNDA) spin label have also been used for the stuctural study on the proteins with unnatural amino acids.[139] That reaction looks quite ptomising for the future perspective to try it *in vitro* and to attempt the glycosylation-related spin labelling *in cellula*.

In the Drescher group at the University of Konstanz, novel labelling strategy was presented for the site-directed spin labelling (SDSL) which is based on the spin label ligation to proteins by an inverse-electron-demand Diels–Alder (DA_{inv}) cycloaddition using genetically encoded noncanonical amino acids.[139,191] As described above, this is called PaNDA and its spin labelling ability was demonstrated on the example of TRX protein which contained non canonical amino acid site for the chemical conjugation.

After protein labelling, to address the nitroxide lability after irradiation, it was recommended to freeze the samples for EPR distance measurements directly after them being irradiated for minimising nitroxide degradation. Here, a series of time resolved (kinetic) CW EPR measurements was performed at room temperature, to quantify, how the intensity of the radical is evolving (see Figure 69).

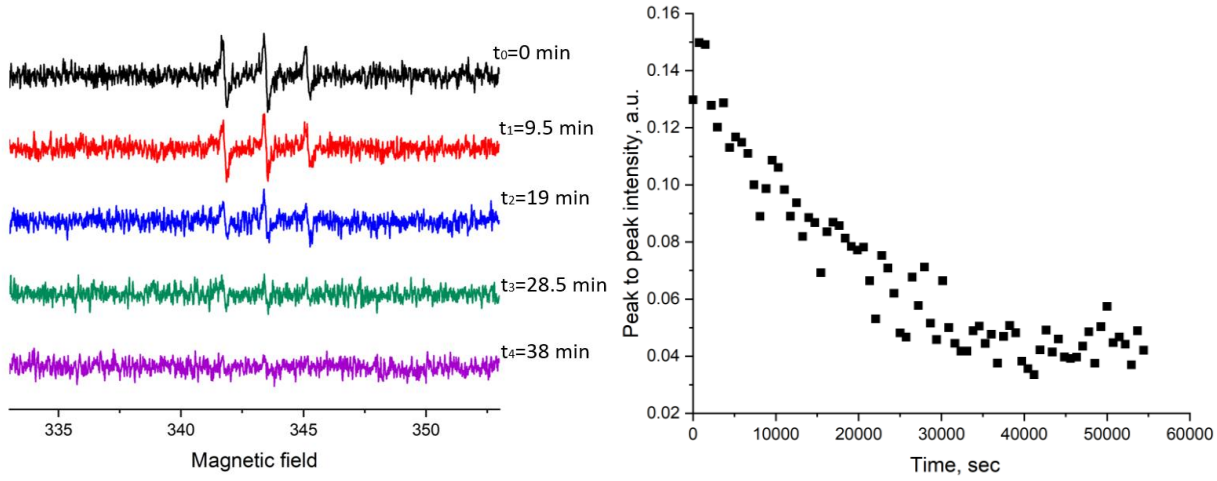


Figure 69. Kinetic monitoring of PaNDA nitroxide (attached to the O-lonked glycnylated site of p53 via MGE with Ac_4GlcNCyoc sugar treatments) signal decreasing.

Plotting spectra and peak-to-peak intensity dependence on time has shown that reliable CW EPR measurements took place during first 20 minutes after the sample was UV irradiated.

The UV irradiation, which is necessary for the PaNDA spin label activation, was done at 365 nm wavelength for 2 minutes. Activation of PaNDA only works in the presence of oxidation

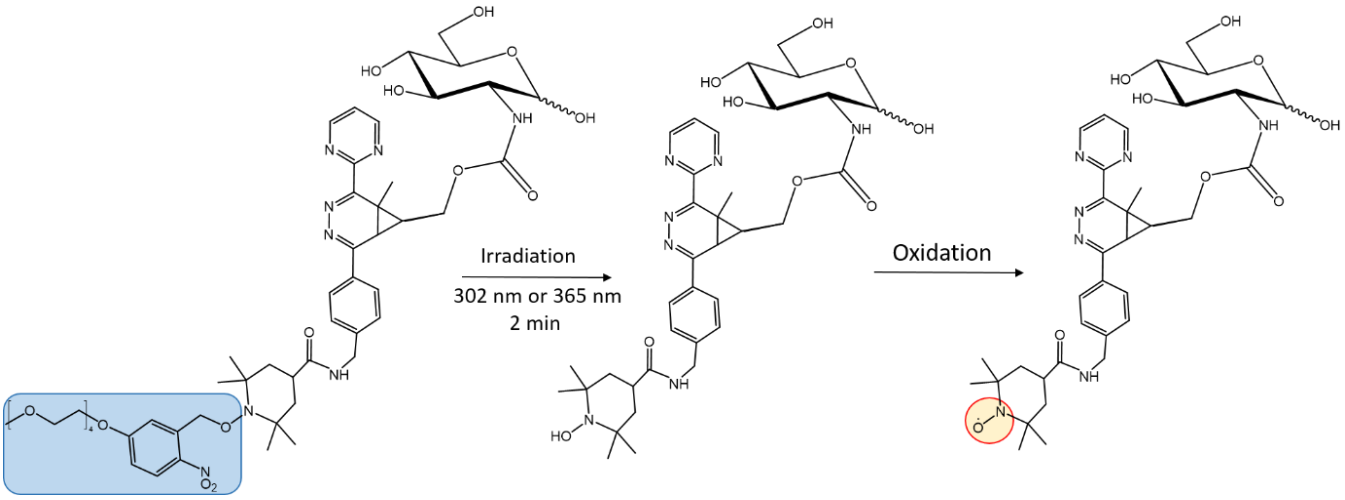


Figure 70. Deprotection of the photocage group with oxidation step for the nitroxide radical activization.[192]

agent, which is thus needed after the UV irradiation (see Figure 70). The EPR measurements were performed for the irradiated PaNDA free radical and for the sample of p53 that was O-linked glycosylated via MGE with Ac_4GlcNCyoc treatment and after protein isolation from HEK293T lysate was spin labelled by PaNDA, while being kept on the Ni-NTA beads. Negative control was performed with the Ni-NTA beads in the absence of p53 protein expression in HEK293T cells but with performing the protocol for MGE involving Ac_4GlcNCyoc on the p53-free cells. The aim of this control sample was to demonstrate the absence of the unspecific EPR signal (unspecific DA reaction) (see Figure 71).

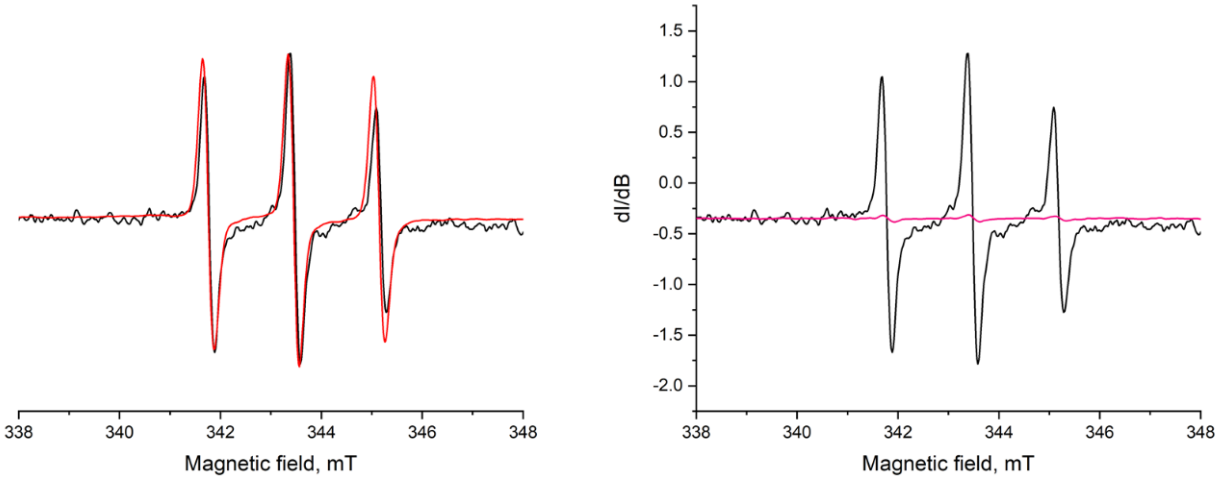


Figure 71. EPR spectra of p53 protein (280 µM) extracted from HEK 293T cells after *in cellula* MGE with Ac_4GlcNCyoc, spin labelled by photoactivatable nitroxide for DA_{inv} reaction (PaNDA) on the Ni-beads and eluted. Left spectrum demonstrates spin labelling intensity of free PaNDA label (red curve) and attached PaNDA label to the O-GlcNAcylated p53 protein. Right spectrum demonstrates paramagnetic signal of attached PaNDA label to the O-GlcNAcylated p53 protein (black curve) and negative control of experiment in absence of treatment with Ac4GlcNCyoc sugar.

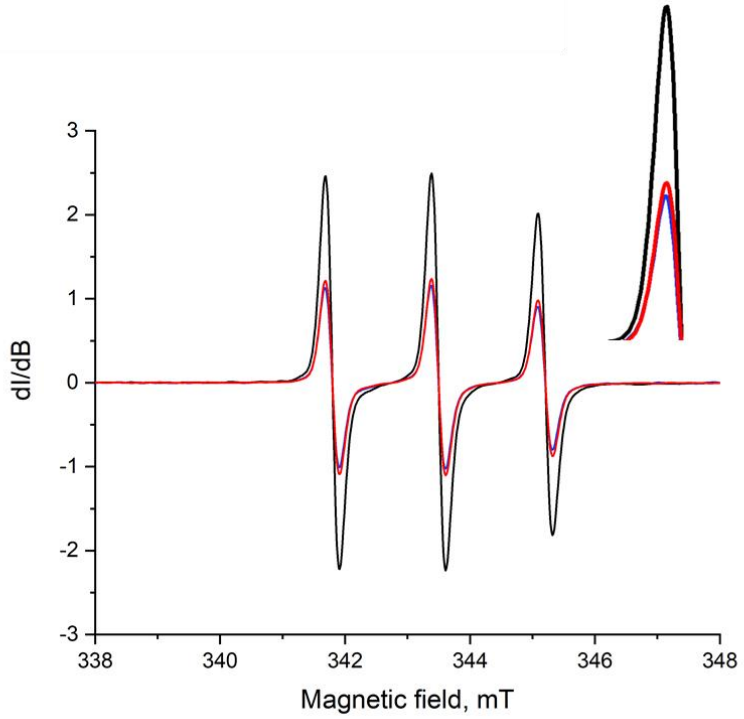


Figure 72. EPR spectra of p53 protein (300 µM) extracted from HEK 293T cells after in vivo MGE with Ac_4GlcNCyoc, spin labelled by DA_{inv} reaction with PaNDA spin label on the Ni-beads and eluted. Blue spectrum – with any inhibitor during MGE, lysation and elution; Red spectrum – with Streptozotocin (2mM) inhibitor during MGE; Black spectrum – with Doxorubicin (1 µg/ml) inhibitor during MGE.

For the experiment with Ac_4GlcNCyoc treatment during MGE Thiamet G inhibitor was not used during lysation and elution in contrast to how it was tested in the case of Ac_4GlcNAz sugar treatment during MGE. The reason for this choice is that experiments of F. Doll et al. have demonstrated no significant effect of Thiamet G on the stability of Cyoc-modified proteins in the HEK293T cells.[41] Using of Streptozotocin (2mM) inhibitor during MGE did not demonstrate significant signal increasing comparing to the sample prepared without inhibitors (Figure 72, blue and red EPR spectrum). Sample with Doxorubicin (1 μg/ml) inhibitor during MGE approximately doubled the EPR intensity signal of the PaNDA spin label attached of the O-GlcNAcylation site of p53 (Figure 72, black EPR spectrum).

4.8. DEER measurements for the detection of O-GlcNAc modification

The main part of this thesis was devoted to establishing the spin labelling strategy that can target glycosylated protein residues. Accordingly, so far, the CW EPR spectra of spin labelled proteins were only discussed in connection with the signal intensities and sometimes also spin label dynamics at ambient temperature in solution. A large number of the pulse bio-EPR spectroscopy applications deal with establishing different types of structural constraints, in order to determine biomolecular structural ensembles and a structure-function relationship. In this respect, it is important to study to which extent the spin labelling at the glycosylation sites can be utilized as one of the structure determination beacon sites.

Here, this research topic is initiated by demonstrating a simple proof-of-principle experiment that confirms that DEER measurements with spin labelled O-GlcNAcylation sites are feasible with to date pulse EPR setups.

A DEER experiment requires two spin labelled sites in the protein structure and allows to measure the distance distribution between the two spin labels. In our case at least one spin labelling position has to be at the sugar moiety of a O-GlcNAcylated site of the protein. DEER method is available for a restricted distance range somewhere between 1.8 nm and 10 nm.[100] As one knows from theory, in DEER experiments, the intramolecular spin label's separation distance is obtained by measuring the frequency of the dipolar interaction between the two electron spins.[193,194] Accordingly, the dipolar frequency is directly connected to the spin-pair distance.

Introduction of the paramagnetic centers in the biomolecule of interest is an important step, where the low spin labelling efficiency for sugar moieties in glycoprotein has to be considered as a serious

obstacle. Previously, for the spin labelling of O-GlcNAz chemical repoter alkyne proxyl spin label was used. For the measurement of intramolecular spin-pair distance as the second paramagnetic center (1-Oxyl-2,2,5,5-tetramethyl-3-pyrroline-3-methyl)methanethiosulfonate (MTSL) spin label was exploited due to its efficient attachment to the cysteine amino acid in the protein structure with high spin labelling efficiency (around 100%) after just two hours of labelling.[195] To address such a case, one needs to use a glycoprotein model with a single O-GlcNAcylated site and a single cysteine amino acid residue accessible for the spin labelling reaction with MTSL.

Considering these aspects, alpha-crystallin was selected as a glycoprotein model for DEER measurements, as it has in its structure a single O-GlcNAcylation site (Ser162) and two cysteine residues (Cys131 and Cys142, in the case of the A chain variant, see Figure 73). Cystein at position 131 was mutated out to exclude its participation in the MTSL labelling reaction. The alpha-crystalline amino acid sequence is given below:

MDVTIQHPWFKRTLGPFYPSRLFDQFFGEGLFEYDLLPFLSSTISPYYRQSLFRTVLDSGISEVRSDRDKFVIFLDVKH
FSPEDLTVKVQDDFVEIHGKHNERQDDHGYISREFHRRYRLPSNVDQSALS**C**SLSADGMLTF**(A)**GPKIQTGLDAT
HAERAIPV**S**REEKPTSAPSS.



Figure 73. Structural representations of alpha-crystallin protein structure generated by PyMol program (pdb code: 6t1r).[196] The O-GlcNAcylation site (S162) at C-terminus and the two cysteine residues (C131 and C142) are marked with red color.

According to the structure shown in the Figure 73 (PDB code: 6t1r), C-terminus is formed by a flexible loop, cystein 142 is located at the end of a beta-strand close to the C-terminus and serine 162 is placed directly in the C-terminal loop. Modelling of the spin label conformations using the

MATLAB plugin MMM 2018 under ambient (298 K) and cryogenic (175 K) temperature conditions did not give conclusive results, because for the given structure the C-terminal beta-strand is covered by the loop which results formally in a prediction of very low labelling efficiency, which is clearly a C-terminal loop conformation artefact. At the same time, currently, there is no rotamer library available for the alkine proxyl spin label attached to a flexible glycan tail, and most of the PDB files, including the referenced 6t1r structure of the alpha-crystalline, do not contain conformational distributions of the flexible protein regions. Thus, the spin label positions for the glycosylated site could only be very roughly estimated by adding *in-silico* a cysteine mutation to the position S162 and modelling the corresponding MTSL conformational distribution, which not surprisingly turned out to be quite broad (data not shown).

The sample for DEER measurement was prepared according to our standart protocol that was used above in case of OGT and p53 protein models, with few additions.

(i) Instead of calcium phosphate expression method lipofectamine transfection of DNA plasmid in the HEK293T cells was used.

(ii) After MGE, protein purification and spin labelling by alkine proxyl, CW EPR measurements were performed to make sure that labelling of O-GlcNAcylated site was successfully done. The EPR spectrum of alkine proxyl attached to the O-GlcNAz site of alpha-crystallin is shown in the Figure 74.

(iii) In the last step, after performing the spin labelling of O-GlcNAc modified site (Ser162), MTSL was used for cysteine residue C142 spin labelling according to the standard protocol established in our group for conventional protein SDSL (see Methods and Materials chapter for details).

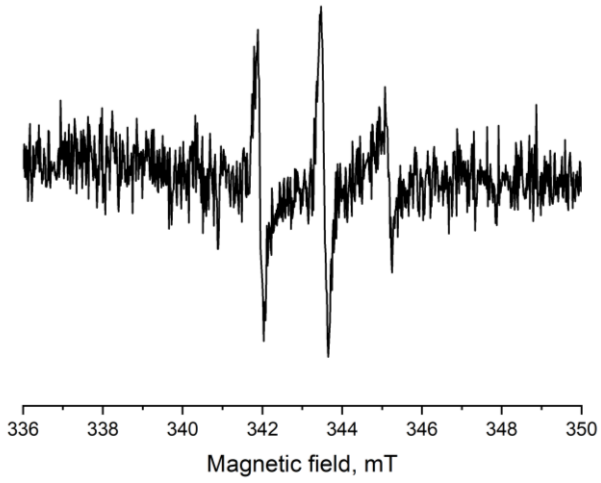


Figure 74. Experimental EPR signal of Alpha-crystallin extracted from HEK 293T cells after *in cellula* MGE with Ac_4GlcNAz, spin labelled by CuAAC reaction on the Ni-beads and eluted.

In the case of alpha-crystallin, due to its small molecular weight (≈20 kDa), MS proteomic analysis could be performed that confirmed O-GlcNAz modification at position Ser 162 (see Figure 75). Proteomic results are crucial for glycan core understanding, to uncover where GlcNAz (from Ac_4GlcNAz sugar used for MGE) site is ended. For alpha-crystalline it was demonstrated that HexNAcHexNAz modification is by far the most abundant at Ser 162 position.

					MS Id ← MS Alias name ←	1 Crystallin-oglucnaz (%)
Sequence (unformatted) ↑	Mod. Names ↑	Protein name ↑	Mod. AAs ↑	Var. Pos. Protein ↑	Labels ↑	
AIPVSR	HexNacHexNaz/447.1601	sp|P02470|CRYAA_BOVIN Alpha-crystallin A chain OS=Bos taurus OX=9913 GN=CRYAA PE=1 SV=1	S	162		100
EEKPSSAPSS	Glu->pyro-Glu/-18.0106	sp|P02470|CRYAA_BOVIN Alpha-crystallin A chain OS=Bos taurus OX=9913 GN=CRYAA PE=1 SV=1	E	164		0.298
HFSPEDLTVKVQEDFVEIHGK	Deamidated/0.9840	sp|P02470|CRYAA_BOVIN Alpha-crystallin A chain OS=Bos taurus OX=9913 GN=CRYAA PE=1 SV=1	Q	90		100
HNERQDDHGYISR	Deamidated/0.9840	sp|P02470|CRYAA_BOVIN Alpha-crystallin A chain OS=Bos taurus OX=9913 GN=CRYAA PE=1 SV=1	N	101		100
QDDHGYISR	Gln->pyro-Glu/-17.0265	sp|P02470|CRYAA_BOVIN Alpha-crystallin A chain OS=Bos taurus OX=9913 GN=CRYAA PE=1 SV=1	Q	104		19.1
VQEDFVEIHGK	Deamidated/0.9840	sp|P02470|CRYAA_BOVIN Alpha-crystallin A chain OS=Bos taurus OX=9913 GN=CRYAA PE=1 SV=1	Q	90		3.11

Normalize Column - {XIC area summed isoX normalized}, Level 1
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Figure 75. Mass Spectroscopy-based proteomics results for the studied alpha-crystalline sample.

Such glycan sequence can be interpreted in two ways: GlcNAcGlcNAz or GalNAcGlcNAz (see Figure 76).

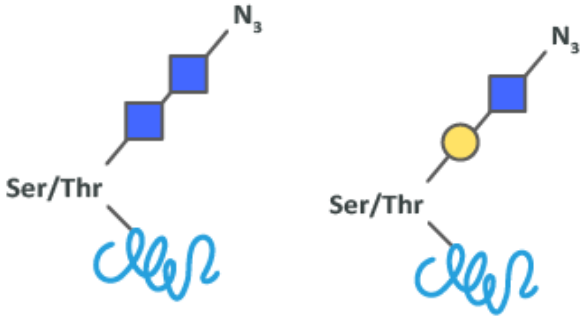


Figure 76. HexNAcHexNAz modification can be interpreted in two ways: GlcNAcGlcNAz (left image) or GalNAcGlcNAz (right image).

In the future, a more detailed distance distribution modelling might be feasible, if the glycnacylation chain, determined by the proteomic analysis, can be included in the modelling of the corresponding rotamer libraries.

The DEER trace for the doubly spin labelled alpha crystalline sample, and its standard processing results are shown in Figure 77. Unfortunately, in the course of this thesis work if was not possible to complete a detailed evaluation with different possible control experiments. However, one can notice that as expected the distance distribution is quite broad, and the modulation amplitude is low, in line with the low spin labelling efficiency at the glycosylation site. Thus, our results strongly suggest that despite obvious challenges, it is possible to perform also the pulse dipolar spectroscopy studies that would involve spin labelled glycosylation sites.

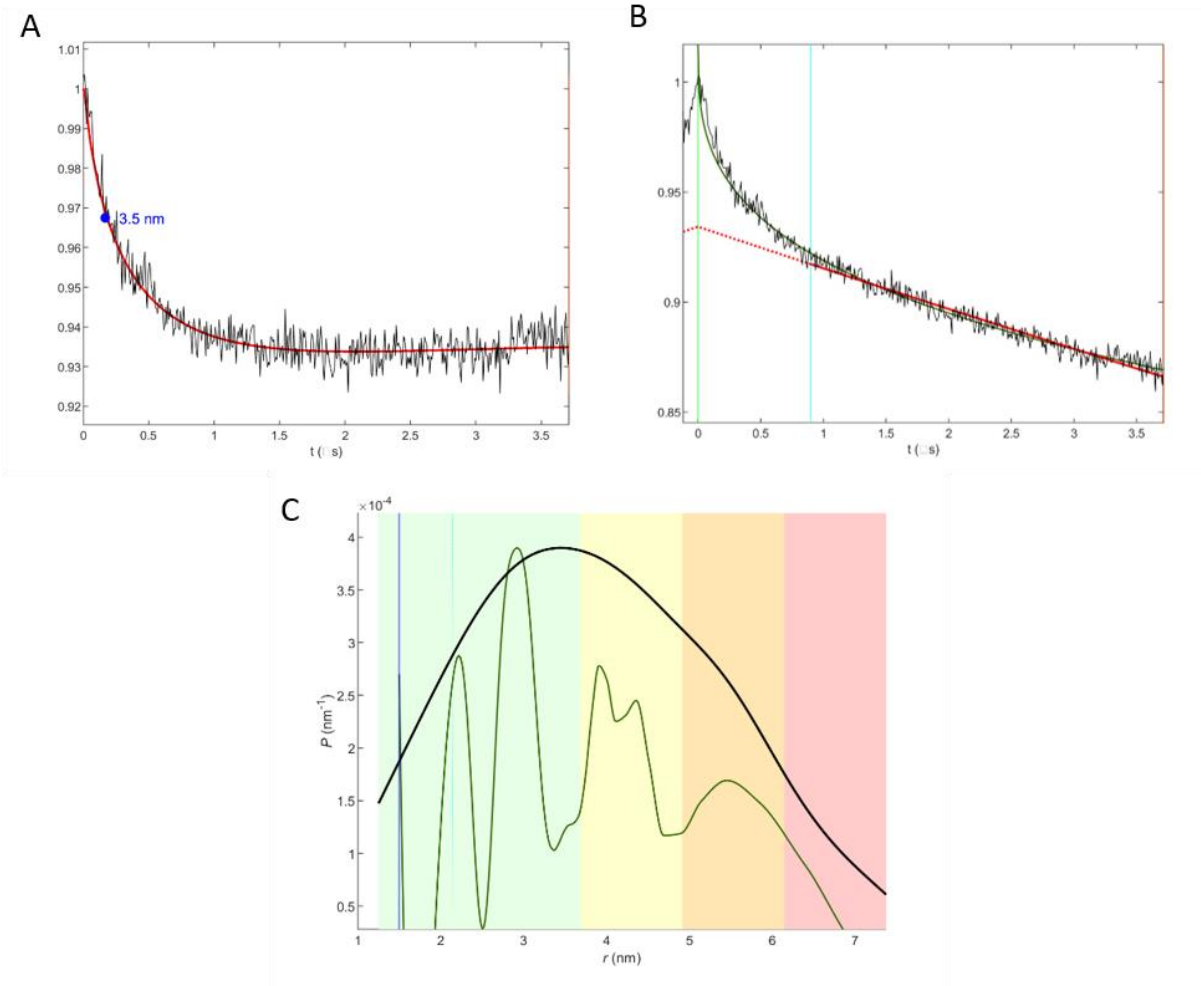


Figure 77. DEER experiment. Raw data (A), form factor (B) and corresponding distance distributions (C) for a doubly labelled Alpha-crystallin via cysteine site by MTSL and via attached sugar by CuAAC reaction.

4.9. DEER spectroscopy application for exploring the binding site, sugar relative, fucose-functionalized precision glycomacromolecules on the structure of human norovirus capsid protein.

In the final section of this chapter, another application of DEER spectroscopy to study structure of human norovirus capsid protein by spin labelling of fucose-functionalized precision glycomacromolecules is reported. This subproject is closely related to glycoprotein research and it also gives a short example of a relevant analysis of DEER data.

The project was done in collaboration with the groups of Laura Hartmann (University of Lübeck), Katharina Susanne Bücher (Heinrich-Heine-Inuversity Düsseldorf), and Sabrina Weickert (University of Konstanz). The described below subproject results were included in the publication: ***"Fucose-Functionalized precision glycomacromolecules targeting human norovirus capsid protein".***[197]

Biological background of the project.

Norovirus (NoV) characteristics: nonbacterial gastroenteritis in humans is mostly provoked by norovirus infection which belongs to the Caliciviridae family.[80] NoV assembles in 7 main genogroups which are divided into numerous genotypes. The GII genotype 4 (GII.4) is the clinically most relevant.[80] NoV particles contain 90 dimers and each have an icosahedral symmetry. VP (~60 kDa) is the major structure protein of NoV. VP1 has two domain types: a shell (S) and a protrudinal (P) domain. α-1,2 linked fucoside and histo-blood group antigens (HBGAs) are well studied binding determinants for NoV.

Binding sites of the P- dimer NoV: The P domain dimer contains two HBGA binding sites.[197] Two HBGA pockets (fucose site 1 and 2), also a third and fourth α-1,2-linked fucoside binding pocket (fucose sites 3 and 4) were discovered by X-ray crystallography. The X-ray structure shows distances between the L-fucose binding pockets are estimated to be 11 Å (fucose sites 1 and 3), 17 Å (fucose sites 1 and 4) and 27 Å (pockets 1 and 2).[197] The binding affinity to L-fucose is rather weak and theoretically, divalent glycomacromolecules could bridge the outer binding sites.

Aim of the project was:

- to provide the binding affinity of fucosylated glycomacromolecules to P-dimer as a multivalent model structure for the NoV;
- analysing the structural difference of glycomacromolecules in the absence and presence of the P-dimer using site-directed spin labelling and DEER EPR spectroscopy.

Synthesis of Glycomacromolecules was performed by solid phase polymer synthesis by our collaborators. The final compounds received for DEER experiments were the fucose glycomolecules made of alkyne-functionalized building block (TDS), fucose-azido (Fuc-N_3) and an ethylene glycol building block (EDS) (see Figure 78 and Figure 79).

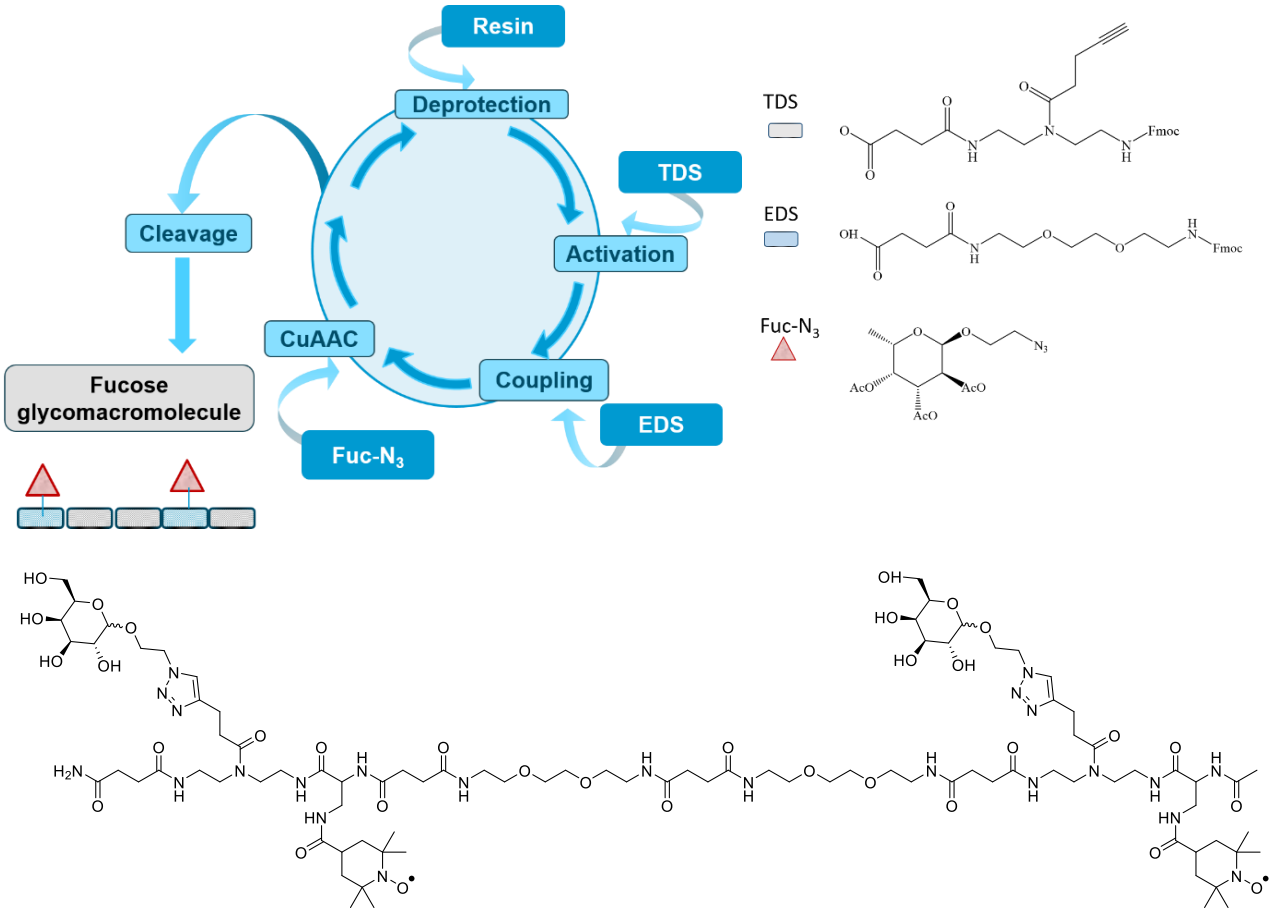


Figure 78. Synthesis scheme of the glycomacromolecule by stepwise assembly (upper left circular scheme) of tailor-made building blocks (upper right chemical structures) for solid support synthesis. Final chemical structure of glycomolecule that was used for the binding experiment represented on the down part of the figure.

Samples were prepared with two different glycomacromolecule concentrations: 75 µM, and 300 µM. Concentration of P-dimer was 75 µM so that resulting samples were with a molar ratio of P-dimer to glycomacromolecule fucose-ligand of 1:4 and 1:1, respectively. Sample with a pure ligand was measured at 90 µM concentration. The focus of EPR study was to detect fucose-ligand binding to NoV P-dimer at different sample concentrations.

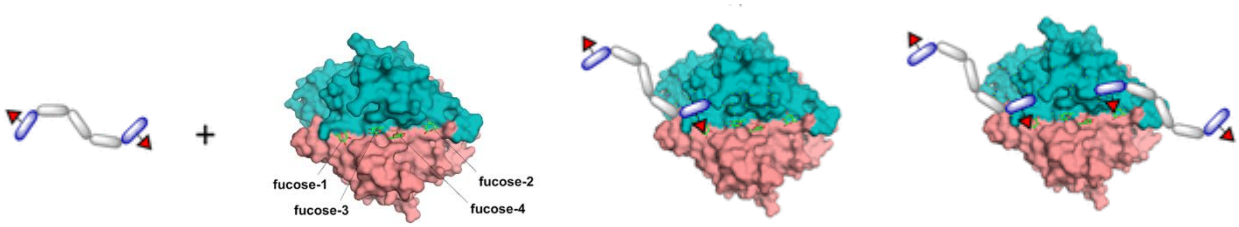


Figure 79. Structure of NoV GII.10 P-dimer with proposed binding of divalent glycomacromolecule with P-dimer (structures not to scale).[197]

Background-corrected data were analyzed by a worm-like chain model.[103] The DEER data in the absence of the Norovirus P-dimer can be well described by a worm-like chain model (contour length 3.8 nm, persistence length 2.0 nm, and standard deviation due to label flexibility 0.22 nm, Figure 80).

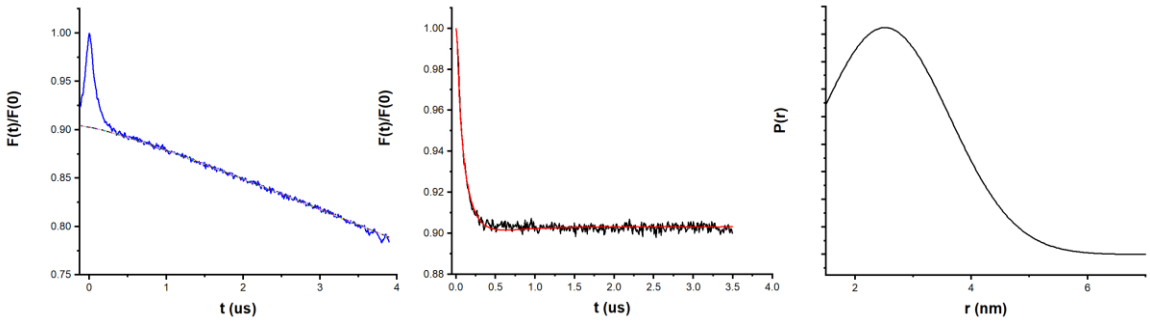


Figure 80. DEER distance measurements of the doubly spin-labelled ligand in the absence of the P-dimer: experimental EPR raw data (blue) and background fit (red) (left image); the form factor after background correction (black) and fit with a worm-like chain model (red) (middle image) and the corresponding distance distribution (worm-like chain model)(right image).

The DEER spectra represented in Figure 80 were recorded for the prepared samples with different fucose-ligand concentration.

The DEER data upon background correction for the spin-labelled fucose- ligand in the absence and presence of the P-dimer (see Figure 60) (molar ratio P-dimer/Fucose- ligand 1:4 and 1:1) is shown in Figure 81 (for the raw data, see also Figure 80). No significant differences are observed for the data obtained in these three experiments suggesting that the conformational ensemble of the fucose-ligand remains unaltered in the presence of the P-dimer. This indicates that despite four fucose binding sites, statistically, the binding happens only at one site per P-dimer (see Figure 79, middle

case). Presence of two fucose moieties bound would result in an appearance of more defined dipolar oscillations, which are not observed in the DEER data.

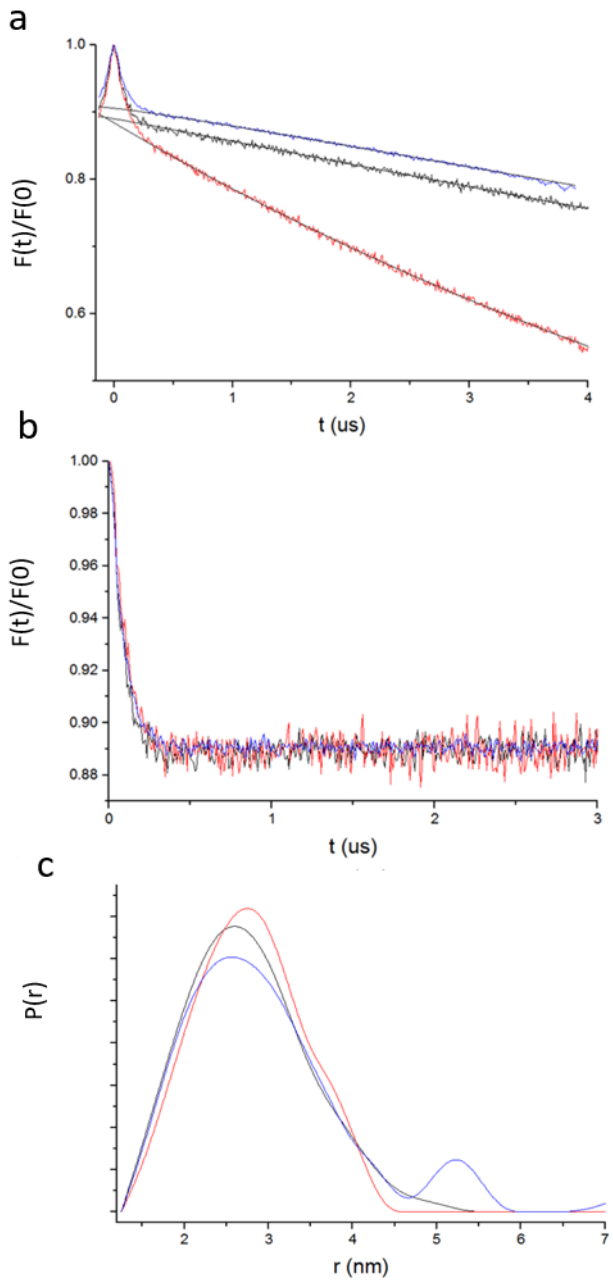


Figure 81. Raw data (a), form factor (b) and corresponding distance distributions (c) for a doubly labelled Fucose- ligand in the absence (blue) and in the presence (black and red) of the NoV P-dimer. The molar ratio NoV P-dimer/Fucose- ligand is 1:4 (red) and 1:1 (black).

3.2. Conclusions

There is a general interest to unveil the effects that PTMs may have on the structure, dynamics, and functions of proteins, and, at the same time, the variety of suitable experimental tools that are available for the study of the protein PTMs is limited. In this thesis a new approach for the PTM research - an EPR methodology for detection of O-linked glycosylation has been developed. This was assisted by an enrichment and validation of the MGE strategies originating from the fluorescent bioimage spectroscopy – such strategies were applied and partially improved to provide the best performance for EPR spectroscopy to identify, validate, and study the function of O-GlcNAcylati Ac4GalNAz on PTM process for a protein of interest.

The first part of the thesis demonstrated methodology development of *in cellula* sample preparation for its further O-linked modification study by EPR spectroscopy. Each step of the previously existing protocols from the similar bioimaging studies was improved with a goal to adopt it for the EPR technique application. For that goal HEK293T cell line was used to perform expression of the protein model for the MGE. O-linked GlcNAcylation was performed using chemical reporter for the CuAAC rection with alkine proxyl spin label. General detection of O-linked glycosylation modification was performed by CW EPR spectroscopy. Two different types of spin labelled reaction were used in this work: CuAAC and DA_{inv} with tetrazines group in the spin label structure. The two listed reactions required differen chemical reporter on the sugar moieties that were used for MGE.

For the first time EPR signal was obtained as well as an evidence of the successful O-GlcNAzylation modification and glycosylation-specific spin labelling of the OGT protein model. The first section of the Results and Discussion chapter demonstarated the capability of EPR spectroscopy to be applied to the study protein structure modification under carbohydrates modifications. Preliminary work was done to also assist the next move in the EPR methodology development and to establish the *in cellula* spin labelling strategy applicable to the glycosylation sites.

Many performed sample preparations on different protein models proved that established approach is reproducible and broadly applicable. The universality of the EPR spectroscopy as a method for the O-GlcNAc modification detection was demonstrated. Results obtained in this work promote EPR method as an analytical tool that might be used for the different protein models with multiple O-GlcNAcylated sites for their general detection, this would possibly require specific methodology developments, especially in the multi-site data analysis.

Different glycan cores were also considered and it was demonstrated that their design is crucial for the prediction of protein structure modification by chemical reporter that is used for O-linked glycosylation.

Comparing to other methods for O-GlcNAcylation study, EPR spectroscopy gave an opportunity to estimate the level of O-GlcNAcylation for OGT and p53 proteins. (One can estimate that 0.4% (0.6%) of all available O-GlcNacylation protein sites in OGT (in p53) are occupied by Ac4GlcNAz).

The direct determination of absolute glycosylation levels is not possible from the EPR spectra. Their intensities reflect i) the glycosylation statistics including the competition between the artificial GlcNAz and the native GlcNAc sugars (see Fig. 3), ii) the spin labelling efficiency in the CuAAC reaction, iii) the action of both OGT and OGA, and iv) interplay of phosphorylation and glycosylation. Due to these factors, EPR study of different inhibitors influence on the p53 O-GlcNAcylation level was performed. In that way one could find out the metabolic interplay between phosphorylation and *O*-GlcNAcylation at specific *O*-GlcNAc sites.

Two further EPR methods (Rapid scan and DEER) were tested and proved to be applicable for the O-linked glycosylation studies. The specific features of the Rapid Scan experiment suit well to the demand of a short measurement time of the *in cellula* biological samples. DEER technique was for the first time demonstrated to work on glycoproteins with a combined spin labelling of a cysteine residue and of a glycosylation site. DEER measurements can be potentially used *e.g*. to find exact position of O-GlcNAc modification in the protein structure and here the first preliminary results demonstrating the potential feasibility of such approach were given.

The results presented in the current thesis give promising perspectives for the future application of CW and pulsed EPR for the detection of O-GlcNacylated sites in the protein structures that at some point might get difficult to perform using other analytical methods.

4.3. Project perspectives

This thesis touches an almost completely unexplored field of EPR investigations of the protein PTMs. The following few bullet points can be considered as possibly the most important next steps in further developing this methodology.

-Amino acid sites mutations: one can obtain more valuable information by mutating out selected sites prone to PTM.

-using WizzArt program for the rotamer library might help predicting the spin label conformational distributions that include the conformational flexibility both of the sugar-chain and of the spin label.

-exploring the glycan core structures in detail and relating them to the spin labelling properties.

-spectra elucidation improvement for the MS results, involving novel O-GlcNAcylated site analysis programs.

- strategy establishment for in vivo EPR measurements by using more stable paramagnetic centers with short linkage. These can for instance Gd(III) paramagnetic centers chelated by dodecane tetraacetic acid (DOTA) also known as tetraxetan complex agent (see Figure 82).

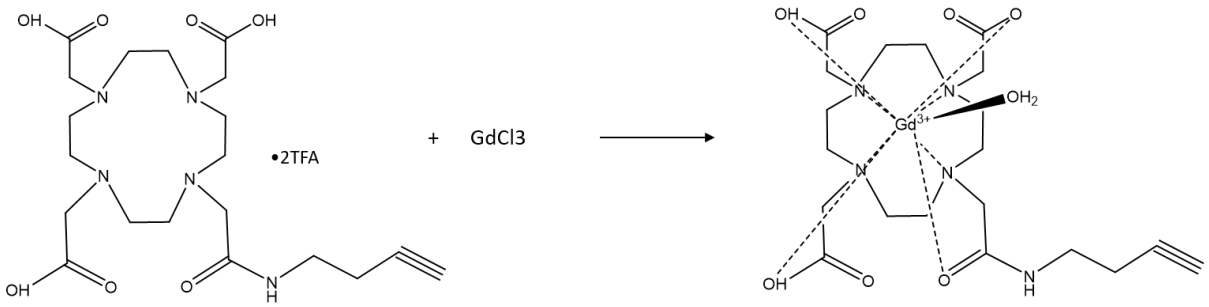


Figure 82. Generic structure of trivalent Gd-DOTA reagents described as the paramagnetic centers for in vivo spin SDSL.

The principle of spin labelling is then also the CuAAC reaction mechanism. Experiments with Gd(III) spin labels have to include two negative controls: in the absence of sugar treatment and in the absence of the target protein expression (see Figure 83).

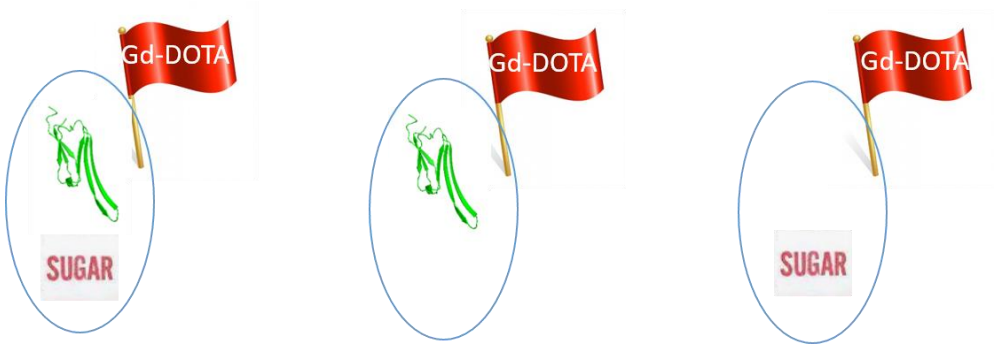
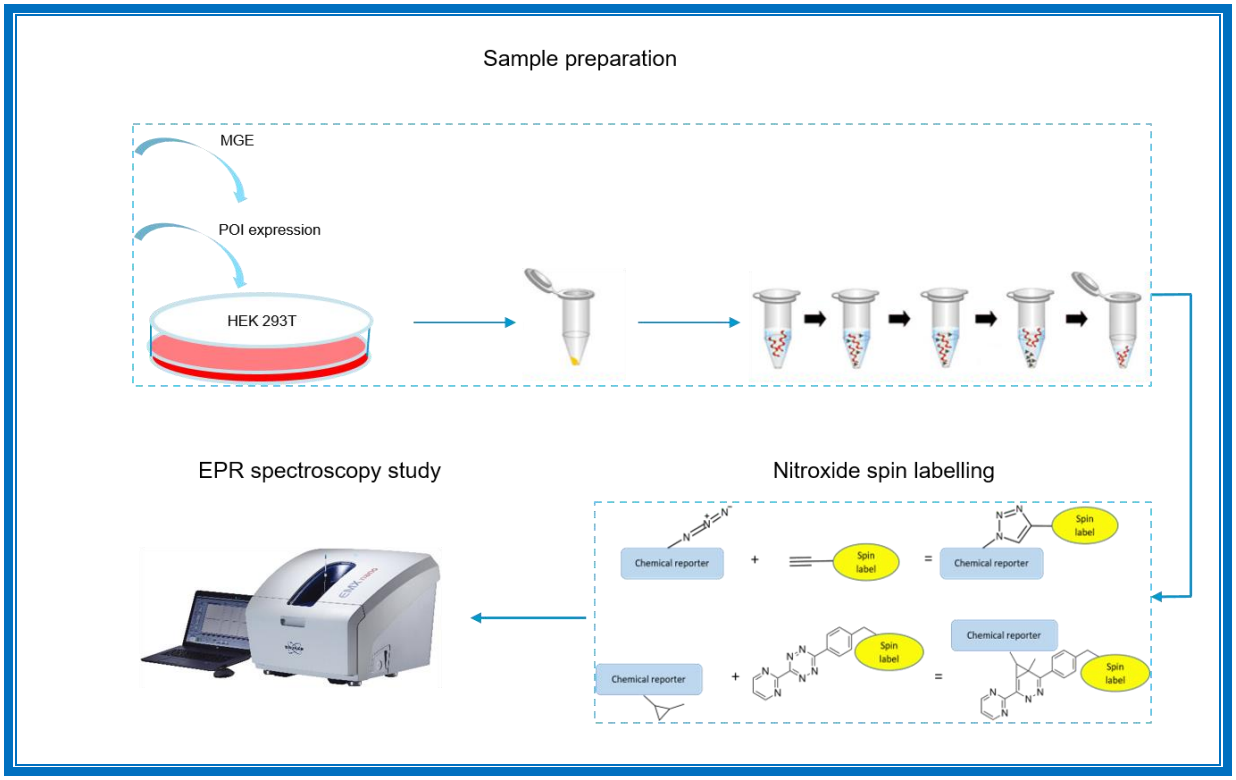


Figure 83. Experimental template for the future iv vivo study of O-GlcNAcylated protein via MGE in combination with SDSL.

CHAPTER V

Material and methods



"A scientist in his laboratory is not a mere technician: he is also a child confronting natural phenomena that impress him as though they were fairy tales..."
by Marie Curie

5. Matherial and methods

5.1. Chemical synthesis

▪Tetraacylated-N-azidoacetylglucosamine (Ac_4GlcNAz) was purchased from Jena Bioscience, Cat.No. CLK-1085-5.

▪Tetraacylated-N-Cyclopropeneacetylatedglucosamine (Ac_4GlcNCyoc) sugar synthesized by Markus Kufleitner, group of Prof. Dr. Wittmann, University of Konstanz.[198]

▪The spin label alkyne-proxyl was synthesized by Katharina Illenseer in our group according to the published procedure.[136]

▪Photoactivatable nitroxide label for DA_{inv} reaction (PaNDA) was synthesized by Bjarne Silkenath in the group of Prof. Dr. Wittmann, University of Konstanz. This photocaged nitroxide was previously used in our group in the work of Anandi Kugele for the protein spin labelling on non-canonical amino acids.[139]

5.2. Molecular biology

5.2.1. Prokaryotic cells

E.Coli strains:

▪ For DANN plasmid amplification the following cells were used:
GH371 cells with genotype - F-mcrA Δ(mrr-hsdRMS-mcrBC) ϕ80lacZΔM15 ΔlacX74 recA1 araD139 Δ(araleu)7697 galU galK rpsL (StrR) endA1 nupG fhuA:IS2 (confers phage T1 resistance), upp- (iGEM).

▪ For protein production, the following cells were used: BL21-Gold(DE3) cells with genotype - F- ompT gal dcm lon hsdS (rB– mB–) λ (DE3 [laci lacUV5-T7 gene 1 ind1 sam7 nin5) (Agilent).

5.2.2. Eukaryotic cells

▪ Human embryonic kidney (HEK 293T) cells were obtained from the group of Prof. Dr. Brunner, University of Konstanz;

▪ Cervical cancer cells taken from Henrietta Lacks (HeLa) were obtained from the group of Prof. Dr. Scheffner, University of Konstanz.

5.2.3. DNA plasmids

▪ pCMV3-N-GFPSpark-OGT (group cat.№95). Human OGT/O-Linked N-Acetylglucosamine transferase Gene ORF cDNA clone expression plasmid, N-GFPSpark tag from Sino Biological (Cat. № HG17892-ANG) was used. Vector name: pCMV3-N-GFPSpark. Vector type: mammalian expression vector. Bacterial resistance: kanamycin;

▪pCMV3-N-His-OGT (group cat.№96). Human OGT/O-Linked N-Acetylglucosamine transferase Gene ORF cDNA clone expression plasmid, N-His tag from Sino Biological (Cat. № HG17892-NH) was used. Vector name: pCMV3-N-His. Vector type: mammalian expression vector. Bacterial resistance: kanamycin;

▪p53_WT-His in pcDNA3 from AG Bürkle (group cat.№243);

▪pAAV asyn WT (group cat.№72) from AG Drescher;

▪pCMV-His-CRYAA (group cat.№240). Human Alpha A Crystallin (CRYAA) Gene ORF cDNA clone expression plasmid, N-His tag from Sino Biological (Cat. № HG17615-NH) was used. Vector name: pCMV3-N-His. Vector type: mammalian expression vector. Bacterial resistance: kanamycin;

▪pCMV-His-CRYAA_C141A (group cat.№249). Alpha-crystalline was mutated to introduce Alanine in position 141 instead of Cysteine. Successful cloning was verified by sequencing the primers: forward T7 primer (5’ TAATACGACTCACTATAGGG 3’) and reverse BGH primer (5’ TAGAAGGCACAGTCGAGG 3’).

5.3. Reagents

Reagent	**Manufacturer**
Ac4GalNAz	Jena Bioscience
Ac4GlcNAz	Jena Bioscience
Ac4GlcNCyoc	Jena Bioscience, synthesized by Markus Kufleitner, group of Prof. Dr. Wittmann, University of Konstanz
Ac4GlcNAc*HCl	Glycon Biochemiicals GmbH
Acetic acid	Carl Roth
NaCl	Carl Roth
KCl	Merck
KH_2PO_4	Sigma-Aldrich
$NaH_2PO_4x2H_2O$	Sigma-Aldrich

Methanol	Sigma-Aldrich
Ethanol	Sigma-Aldrich
SDS	Carl Roth
Tris-HCl	Sigma-Aldrich
Tris	Sigma-Aldrich
HEPES	Sigma-Aldrich
Glycerol	Carl Roth
ß-mercaptoethanol	Carl Roth
Bromophenol blue	Carl Roth
Glycine	Sigma-Aldrich
EDTA	Carl Roth
β-glycerophosphate	Sigma-Aldrich
NaF	Sigma-Aldrich
Na_3VO_4	Sigma-Aldrich
Acrylamide (Rotiphorese Gel 30)	Carl Roth
chloroquine	Sigma Aldrich
Lipofectamine™ 2000 Transfection Reagent	Thermo Fisher Scientific
isopropanol p.a.	Sigma Aldrich
TCEP	Sigma Aldrich
DTT	Sigma-Aldrich
THPTA	Sigma-Aldrich
BTTAA	Sigma-Aldrich
Triton X-100	Carl Roth
Imidazole	Sigma-Aldrich
$MgCl_2$	Carl Roth
Lysogeny Broth (LB) medium	LB medium obtained from Carl Roth and dissolved in milliQ to obtain the described composition
Agar	Carl Roth
IPTG	Carl Roth

Antibiotics (carbenicillin, kanamicine, ampiciline)	Carl Roth
Penicillin-Streptomycin (10,000 units/mL penicillin and 10,000 µg/ml streptomycin)	Thermo Fisher Scientific
Dublecco's modified eagle medium (DMEM) high glucose	Thermo Fisher Scientific
Fetal bovine serum (FBS)	Biochrom
Ni-NTA resin (HisPur™ Ni-NTA Resin)	Thermo Fisher Scientific
Capture select C-tag affinity matrix	Thermo Fisher Scientific

5.4. Enzymes

DNAse I (Thermo Fisher Scientific), protease inhibitors (complete mini, EDTA free, Roche® Life Science-Produkte), DnpI (NEB, 20 000 units/ml).

5.5. Kits

The following kits were used. Click-iT O-GlcNAc Enzymatic Labeling System (Thermo Fisher Scientific), Plasmid DNA Purification Mini Prep Kit (Thermo Fisher Scientific), Pierce Glycoprotein Staining Kit (Thermo Fisher Scientific), CaptureSelect C-tag affinity matrix (Thermo Fisher Scientific).

5.6. Equipments

BioPhotometer (eppendorf), camping stove (Camping Gaz), centrifuges 5810 R, 5424R, 5417R, and 5418 (Eppendorf), neolab 3-1810, ChemiDocTMTouch Imaging System (Bio-Rad), CO2 incubator for cell culture (Binder), confocal microscope TCS SP5 (Leica Microsystems CMS), contact thermometer (Heidolph), counting chamber (Neubauer), electronic balances (Sartorius and Kern), electroporator ElectroCell S20 (βtech), freezer -80 °C (Thermo Scientific, Binder), gel imaging system GelDocXR (Bio-Rad), incubator for E. coli (Infors HT Ecotron), magnetic stirrer with heating plate (Heidolph), mass spectrometer (micrOTOF, Bruker), microscope DMIL LED (Leica Microsystems CMS) with fluorescence lamp pE-300white (CoolLED), microwave (Severin), mini Trans-Blot Cell for Western

Blotting (Bio-Rad), mini-PROTEAN tetra vertical electrophoresis cell for SDS-PAGE (Bio-Rad), NanoDrop 1000 (PeqLab), overhead shaker (A. Hartenstein), polymerase chain reaction (PCR) cycler peqSTAR (PeqLab), pellet pestles (Thermo Fisher Scientific), PerfectBlue Gel System Mini for agarose electrophoresis (Peqlab), plate reader (BioTek), rocking shaker (stuart), shaker Polymax1040 (Heidolph), shaking incubator for E. coli (Edmund Bühler), small centrifuge MiniStar (VWR), spectrophotometer ND-1000 (PeqLab), sterile bench (Heraeus and Hera), thermomixer (Eppendorf), Typhoon FLA 9500 imager (GE Healthcare Life Sciences), ultrasonic bath (Bandelin Sonorex), UV/VIS spectrophotometer (Agilent), vortex MS2 (Ika) and vortex (Phoenix instrument), water bath (Benchmark), water bath, shaking (Thermo Fisher Scientific), wide-field frequency domain FLIM setup (diode laser LDM488.20.A350 (Omicron), DMI 6000B inverted microscope (Leica Microsystems CMS), multi-mode fiber with a numeric aperture (NA) of 0.25 (Thorlabs), PIMAX4:1024i RF CCD camera coupled to a GenIII intensifier (Princeton Instruments), EMXnano Bruker, sonicator (Q700-QSONICA), EPR Elexsys E580 Q-band spectrometer, pH-meter (VWR enomenal), ÄKTAprime plus (GE Healthcare), HiTrap Q HP column (2 x 1 mL, GE Healthcare).

5.7. Biochemical methods for prokaryotic cells

5.7.1. Preparing chemically competent *E.coli*

20 ml LB-medium was inoculated with a single *E. coli* colony picked from an agar plate and incubated overnight at 37 °C and 180 rpm in a shaking incubator. 15 ml of the overnight culture were added to 700 ml LB medium and incubated at 37°C and 180 rpm until the OD_{600nm} reached 0.5. The cells were then harvested by centrifugation (4000 g, 10 min) and the pellet was resuspended in cold 100 mM MgCl2 buffer (10 ml). The cell suspension was centrifuged at 4 °C (3320 x g, 10 min) and the pellet was resuspended in 75 mM CaCl2 buffer (10 ml) and incubated for 30 minutes on ice. The cell suspension was centrifuged at 4 °C (3320 x g, 10 min) and the supernatant was discarded. The pellets were resuspended in an overall volume of 10 ml storage buffer (50 mM CaCl2, 15 % glycerol v/v), aliquoted in 50 µl, and shock-frozen in liquid nitrogen. The cells were stored at -80 °C.

2 buffer (10 ml). The cell suspension was centrifuged at 4 °C (3320 x g, 10 min) and the pellet was resuspended in 75 mM CaCl2 buffer (10 ml) and incubated for 30 minut es on ice. The cell

suspension was centrifuged at 4 °C (3320 x g, 10 min) and the supernatant was discarded. The pellets were resuspended in an overall volume of 10 ml storage buffer (50 mM CaCl2, 15 % glycerol v/v), aliquoted in 50 µl, and shock-frozen in liquid nitrogen. The cells were stored at -80 °C.

5.7.2. Transformation of *E. coli*

100 µl Aliquots of chemically competent *E. coli* GH371 strain were thawed on ice. 10-100 ng of DNA plasmid were mixed with the competent cells and incubated 30 min on ice. Then, a 30s heat shock at 42˚C was performed and followed by a 2min incubation on ice. Afterwards 1ml of prewarmed at 37 ˚C SOC media was added and the whole mixture was shaked for 1h at 1400 rpm. Finally, cells were put onto LB agar plates containing 50 µg/ml antibiotic (carbenicillin for p53, ampicillin for OGT), and incubated at 37 ˚C overnight.

5.7.3. DNA plasmid preparation

To extract the plasmid DNA, a single colony from LB agar plate was picked up and grown in 10 ml LB medium containing 50 µg/ml of the appropriate antibiotic overnight. In the end, plasmids were isolated using GeneJet Miniprep Kit (Thermo Scientific) according to the recommended protocol. Briefly, 5 mL LB-medium containing the appropriate antibiotics were inoculated with GH371 *E. coli* transformed with the desired plasmid and incubated overnight at 37 °C and 180 rpm. Cells were harvested by centrifugation (4000g, 5 min). The cell pellet was resuspended in 250 µl resuspension solution and transferred into a microcentrifuge tube. 250 µl of lysis buffer were added, the solution was carefully mixed and 350 µl of neutralization buffer were then added. The mixture was centrifuged for 5 min (12 000 g). The supernatant was transferred into a purification mini-column and centrifuged for 1 min (12 000 g). DNA plasmid was washed by adding 500 µl of the wash buffer and centrifuging for 30 s (12 000 g). The washing step was repeated 3 times. Afterwards, the flow-through was removed and the column was centrifuged for 1 min (12 000 g). The column was then transferred to a sterile 1.5 ml Eppendorf tube and the DNA plasmid was eluted in 50 µl MQ water.

5.7.4. Cloning

Quick change PCR method for DNA cloning was used. For PCR reaction mixture was prepared the next mixture in PCR tube: 17 µl MQ water, 5 µl 5xHF Phusion buffer, 1 µl plasmid DNA template

(~40ng), 0.5 µl dNTP Mix, 0.5 µl primer forward (with concentration 10 µM), 0.5 µl primer reward (with concentration 10 µM), 0.5 µl phusion polymerase.

In another PCR tube mixture for the negative control was prepared: 18.5 µl MQ water, 5 µl 5xHF Phusion buffer, 1 µl plasmid DNA template (10 ng - 100 ng), 0.5 µl Dntp Mix. "Quick Change" PCR-cycler program was started with the next set up parameters:

Start: heat lid to 110°C, 30 s 95°C

16 cycles: 30 s 95°C (melting), 1 min 55°C (annealing), 8 min 68°C (elongation);

End: 8°C.

After that 0.5 µl of DpnI digest was used to remove methylated DNA with PCR-cycler "DpnI digest" program: heat lid to 110°C, 6 h (at least 1 h) 37 °C, store at 8°C.

5.7.5. Glycerol-stock preparation

After the transformation of *E. coli*, cells were grown overnight at 37°C on an LB-Agar plate (Lennox) containing the appropriate antibiotics. For the bacterial glycerol stock creation, LB-medium containing the appropriate antibiotics was inoculated with a single colony from the agar plate, incubated overnight at 37°C and 180 rpm, and afterward, 500 µL overnight culture were mixed with 500 µL 50 % (v/v) glycerol in a 2 mL cryovial and stored at -80°C.

5.7.6. Protein expression in *E.coli*

5 ml LB-medium containing the appropriate antibiotics were inoculated with cellular material taken from a glycerol stock and incubated at 37°C and 180 rpm overnight. For the protein expression, this culture was diluted 1:100 into 200 ml fresh LB medium supplemented with the same antibiotic and incubated at 37 °C and 180 rpm until an OD_{600nm} of 0.1 to 0.8 was reached. The protein expression was then induced with 0.8 mM IPTG (sterile in MQ water). The protein expression usually took place for 5 hours at 37 °C, 180 rpm. Afterwards, cells were harvested by centrifugation (4°C, 4000 rpm, 15 min) and the supernatant was discarded.

5.8. Buffers and solutions

5.8.1. Commercially available buffers and solutions

The following commercially available buffers/solutions were used. Dublecco's modified eagle medium (DMEM) high glucose, Opti-MEM/reduced Serum Medium (Gibco), DPBS (Thermo Fisher Scientific), Phusion HF buffer (Thermo Fisher Scientific).

5.8.2. Self-made buffers and solution

10x Phosphate-Buffer saline (PBS)

- 17.8 g/l of $Na_2HPO_4x2H_2O$
- 2.4 g/l of KH_2PO_4
- 80 g/l of NaCl
- 2 g/l of KCl

pH=7.4

PBS-T

- 0.5 V-% Tween-20 in 1x PBS

2x HEPES buffered saline (HBS)

- 280 mM NaCl
- 1.5 mM $Na_2HPO_4x2H_2O$
- 50 mM HEPES

pH=7.05

Lysis buffer used in glycosylation experiments

- 0.5 V-% Triton X-100
- 25 mM Tris pH 7.4
- 300 mM NaCl
- 5 mM EDTA (EDTA was omitted, if CuAAC was performed afterwards)
- 20 mM β-glycerophosphate
- 20 mM NaF
- 0.3 mM Na_3VO_4
- 10 U/ml DNAse I
- 1x protease inhibitors

Laemmli sample buffer (6x)

- 0.3 M Tris-HCl, pH 6.8,
- 30 V-% glycerol
- 0.21 M SDS
- 2.84 M ß-mercaptoethanol

- 0.29 mM bromophenol blue

Lysogeny Broth (LB) medium

- 5 g/l NaCl
- 10 g/l tryptone
- 5 g/l yeast extract

E. coli freezing medium

- 50 V-% liquid LB medium
- 50 V-% glycerol

E. coli LB agar plates

35 g/l of LB-agar (10 g L -1 tryptone, 5 g L -1 yeast extract, 5 g L -1 NaCl, 15 g L -1 agar-agar, pH 7.0)

Stacking gel buffer

- 0.5 M Tris-HCl, pH=6.8
- 4g/l SDS

Separation gel buffer

- 1.5 M Tris-HCl, pH=8.8
- 4 g/l SDS

SDS-PAGE Running buffer

- 3.03 g/l Tris
- 14.4 g/l glycine
- 1 g/l SDS

15 % separation gel (2 gels)

- 9 ml Rotiphorese® Gel 30
- 4.5 ml separation gel buffer
- 4.5 ml milliQ
- 60 µl APS 10 %
- 13.5 µl TEMED

10 % separation gel (2 gels)

- 6 mL Rotiphorese® Gel 30
- 4.5 mL separation gel buffer
- 7.5 mL milliQ
- 60 µL APS 10 %

- 13.5 µL TEMED

Coomacie solution

- 50% H_2O
- 40% methanol
- 10% glacial acetic acid
- 0.1% Coomassie Brilliant Blue R-250

Dilution buffer for CuAAC

- 1x PBS with 300 mM NaCl
- 1x protease inhibitors
- 0.5 V-% Triton X-100

Coomassie destaining solution

- 50% H_2O
- 40% methanol
- 10% glacial acetic acid

Western Blot running buffer

- 25 mM Tris pH 8.3
- 192 mM glycerol
- 20 V-% methanol

Blotto

- 2 w-% bovine serum albumin (BSA)
- 0.05 w-% NaAz in PBS-T

OGT buffer

- 25 mm HEPES pH=7.5
- 300 mM NaCl
- 5% glycerol
- 0.04% Triton X-100,
- 0.2 mM TCEP

p53 buffer (his-trap binding buffer)

- 50 mM $Na_2HPO_4x2H_2O$ pH=7.2
- 150 mM NaCl
- 2mM DTT
- 0.01% NP-40

p53 buffer (his-trap elution buffer)

- 1 M imidazole
- 50 mM $Na_2HPO_4x2H_2O$ pH=7.2
- 150 mM NaCl

- 2mM DTT
- 0.01% NP-40

p53 buffer (heparin column binding buffer)

- 50 mM $Na_2HPO_4x2H_2O$ pH=7.2
- 100 mM NaCl
- 2mM TCEP

p53 buffer (heparin column elution buffer)

- 50 mM $Na_2HPO_4x2H_2O$ pH=7.2
- 1 M NaCl
- 2mM TCEP

Alpha-synuclein buffer

- 10 mM Tris-HCl pH=7.4
- 150 mM NaCl
- 1mM DTT

Alpha-crystallin

- 20 mM HEPES pH=7.9
- 1% SDS

TRX buffer

- 20mM Tris-HCl buffer pH 8.0
- 10% glycerol
- 1mM DTT

GFP buffer

- 10 mM Tris-HCl pH 8.0
- 10 mM EDTA (EDTA was omitted, if CuAAC was performed afterwards)

LB medium (Lennox, LB Broth)

- 5 g/l NaCl
- 10g/l tryptone
- 5g/l yeast extract

Lysis buffer used in glycosylation experiments

- 5 mM EDTA
- 20 mM NaF
- 300 mM NaCl
- 0.5 V-% Triton X-100
- 20 mM β-glycerophosphate
- 0.3 mM Na_3VO_4
- 25 mM Tris pH 7.4

- 10 U mL-1 DNAse I

(1x protease inhibitors EDTA was omitted, if CuAAC was performed afterwards)

5.9. Biochemical methods for mammalian cells

5.9.1. Cultivation of mammalian cells

Handling of living cells was performed on a sterile bench. Cells were grown in 30 cm cell culture Petri dishes (Sarstedt®). The HEK 293T cell culture media was prepared by using Dulbecco's Modified Eagle Medium (DMEM, Thermo Fischer Scientific) with 10% Fetal Calf Serum (FCS, Sigma-Aldrich) and 1% Penicillin/Streptomycin (10,000 units/ml penicillin and 10,000 µg/ml streptomycin, Thermo Fischer Scientific). Cells were incubated at 37 ˚C and 5 % CO_2 and were each time splitted at 90% confluence (on average it took about 2-3 days for each passage). For cell culture splitting on 10 cm Petri dishes, old medium was aspirated and the cells were washed with 10 ml of Phosphate-Buffered Saline (PBS, Thermo Fischer Scientific), and detached by the addition of 1 ml trypsin solution (Sigma-Aldrich). The same procedure for the 30 cm Petri dishes was performed with 30 ml of PBS buffer and 3 ml trypsin. HEK293T cells were detached at room temperature rapidly without additional incubation at 37 °C. After that, 10 ml of a fresh cell culture medium were added, and the cells were resuspended by delicate pipetting up and down (3-5 times). If residual trypsin needed to be eliminated, the cells were pelleted by centrifugation at 550 g for 3 min, the supernatant was removed, and the cells were resuspended in the fresh cell culture medium. The density of the cells in the medium was counted using a Neubauer chamber and 900k cells were transferred into a new 30 cm Petri dish with 30 ml of the fresh media pre-warmed at 37˚C.

5.9.2. DNA plasmid transfection into HEK 293T cells

5.9.2.1. By calcium phosphate co-precipitation

Transient transfection of HEK 293T cells with a DNA plasmid was performed by calcium phosphate co-precipitation. 24 hours after splitting, 1-2 million cells were seeded per 10 cm cell culture dish or 3-5 million cells were seeded per 30 cm cell culture dish. For one EPR sample ten 30 cm Petri dishes were used. For microscopy control one additional 10 cm Petri dish was used for the consequent fluorescent labeling. For each Petri dish (30 cm) the added 8-10 µg of DNA plasmid were mixed with

1,5 ml MQ water, 1,5 ml 2xHBS buffer, and 150 µl 2,5 M $CaCl_2$ solution while vortexing. 8-10 hours after transfection the cells were washed with DPBS and new culture media was gently added, following by incubation at 37°C and 5 % CO_2 for 24 hours to express the protein of interest (POI).

5.9.2.2. Lipofectamine-mediated transfection

For Lipofectamine-mediated transfection, 1 µL Lipofectamine 2000 Transfection Reagent were diluted in 50 µL Opti-MEM medium. 0.25 µg to 2 µg plasmid DNA was added to 50 µL Opti-MEM medium. Both solutions were vortexed separately and incubated at room temperature for 3 min. They were combined by pipetting and incubated at room temperature for further 15 min. This complex was transferred to cells seeded in a well of an 8-well µ-slide the day before transfection (20,000 in 200 µL cell culture medium). Protein expression was sufficient after one to two days.

5.9.3. Metabolic Glycoengineering of the protein *in cellula*

MGE was performed after the protein expression in the HEK 293T cells. 10 µl of 100 mM Ac_4GlcNAz stock solution in DMSO was added per each 10 ml of the cell culture. The final concentration of Ac_4GlcNAz for O-GlcNAcylation modification in the media was thus 100 µM. After Ac_4GlcNAz addition cell cultures were incubated at 37°C and 5 % CO_2 for 24 hours. After incubation, the cells were carefully washed with PBS, and 1ml of trypsin was added per 10 ml of cell culture to detach cells from Petri dishes. The amount of 5 ml of PBS per Petri dish was used to wash all cells from the dishes and collect them in a falcon tube. The collected cell culture was centrifuged at 550g for 5 min and the supernatant was carefully aspirated. The cells from 10 ml cell culture were resuspended in 1 ml of PBS and transferred to 1.5-mL reaction tube. The tube was centrifuged at 550g for 5 min and the supernatant was carefully aspirated from the cells. All further procedures were conducted on ice or at 4°C.

Negative control: Ac_4GlcNAc was used for the cells MGE treatment instead of Ac_4GlcNAz at the same concentrations.

5.9.4. HEK293T cell lysis

Cells growing in 30/10 cm dishes (6-well plates) were washed once with DPBS and detached by addition of 3/1 mL (200 µL) trypsin. Next, 15/5 mL (800 µL) DPBS were added, cells were transferred into a 15 mL falcon (1.5 mL reaction tube), and were centrifuged at 600 g for 3 min. The

supernatant was aspirated, and the cell pellet was resuspended in 1 mL DPBS. Cells were transferred into a 1.5 mL reaction tube and pelleted by centrifugation. The supernatant was aspirated, the cell pellet was resuspended in 1 mL DPBS, and tubes were centrifuged once more. The supernatant was completely removed, 350 µL to 600 µL lysis buffer (35 µL to 50 µL) were added, and samples were mixed by pipetting up and down. Reaction tubes with samples were incubated on ice for 30 min and centrifuged at 4 °C and 18,000 g for 30 min. The supernatant, i.e., the lysate, was transferred into a new 1.5 mL reaction tube. Lysates were further processed or stored at -80 °C. In glycosylation experiments, lysis buffer was usually supplemented with 100 µM PUGNAc to inhibit the activity of OGA. In acetylation and methylation experiments, HEK293T cells were detached by pipetting their medium up and down. Cells were collected, transferred into a 15 mL falcon and further handled as mentioned above.

5.9.5. Protein purification

5.9.5.1. Purification of OGT

The lysis buffer was prepared with 25 mM Tris pH 7.4, 300 mM NaCl, 5 mM EDTA, 0.5 V-% Triton X-100, 20 mM β-glycerophosphate, 20 mM NaF, 0.3 mM Na3VO4, 10 U mL^{-1} DNAse I, 1x protease inhibitors. If CuAAC was performed afterwards then EDTA was omitted. After centrifugation, 400 µl of lysis buffer was added to the HEK 293 T cells collected from 10 ml of cell culture and the cells were resuspended carefully. The lysated cells were incubated on ice for 30 min and after that were centrifuged for 20min at 18,500g. After centrifugation supernatant was transferred to a new 1.5-mL reaction tube, and 30 µl of the supernatant was stored at -80°C for Western blot analysis. To purify the OGT protein from the supernatant 3 µl Ni-NTA resin and 400 µl of dilution buffer were added to the supernatant (≈350 µl) and incubated for 6 hours with overhead rotation at 4 °C. After incubation, 50 µl of the Ni-NTA resin precipitate was stored at -20°C for SDS-PAGE analysis. Next, the Ni-NTA resin was precipitated by centrifugation for 3 min at 8600 g. The supernatant was carefully aspirated and another 500 µl of dilution buffer was added followed by another centrifugation. The dilution/centrifugation cycle was repeated 3 times. All steps of washing required to be especially careful to not lose the beads with adsorbed protein on the surface during aspiration of the supernatant. To elute the post-translationally modified protein from the Ni-NTA resin, 100 µl of 300 mM imidazole solution in dilution buffer was added, the resulting suspension

was filtered by the column previously equilibrated with imidazole solution, and additionally washed twice with 100 µl of the imidazole solution, after which 1 V-% protease inhibitor (cOmplete^{TM}Protease Inhibitor Cocktail, Rosche) was added to the filtered protein solution. The eluted and filtered protein solution was neutralized using dialysis membrane and dilution buffer. After 10-12 hours of dialysis, the protein solution was replaced from dialysis membrane to the membrane MWCO with Ultracel 3K or 10K and concentrated to the 100 µl volume by centrifugation at 8600 g during 10 min at 4°C.

The final protein concentration was determined by optical absorption at λ=280nm with a BioPhotometer (Eppendorf).

5.9.5.2. Purification of p53

His-tagged p53 expressed in HEK 293T cells was lysed following the same conditions used for OGT. Inspired by a recently published protocol[4] the p53 was purified as detailed below.After the lysate was centrifuged for 20 min at 18,500 × *g* and the supernatant was loaded onto a Ni-NTA column, His-tagged p53 was eluted with a gradient of 25 column volumes from buffer A (50 mM phosphate buffer pH 8, 150 mM NaCl, 0.01% NP-40, 2 mM DTT, 25 mM imidazole) to 100% buffer B (50 mM phosphate buffer pH 8, 150 mM NaCl, 0.01% NP-40, 2 mM DTT, 1 M imidazole) and dialyzed overnight (50 mM phosphate buffer pH 7.2, 150 mM NaCl, 0.01% NP-40, 2 mM DTT) in the presence of thrombin (Sigma-Aldrich). Cleaved p53 was separated from the His-tags using a heparin column. p53 was eluted from the heparin column with a gradient over 25 column volumes from buffer A (50 mM phosphate buffer pH 7.2, 150 mM NaCl, 0.01% NP-40, 2 mM DTT) to 100% buffer B (50 mM phosphate buffer pH 7.2, 1 M NaCl, 0.01% NP-40, 2 mM DTTThe final protein concentration was determined by optical absorption at λ=280nm with a BioPhotometer (Eppendorf).

5.9.5.3. Purification of Alpha-synuclein

The CaptureSelect C-tag affinity matrix shows good compatibility with high throughput purification set-ups based on e.g. spin columns or microtiter filter plates. The below protocol describes the use of spin columns and serves as a guideline for similar formats, which can easily be adapted to optimize performance. − Spin columns (or microtiter plates) can be used with filters having a pore size of less than 100 µm (e.g. MoBiTec cat.no.: M1002S or M105035F). Carefully apply 100 - 200 µl of the C-tag affinity matrix to the spin column and wash the resin 3 times by adding a suitable

binding buffer (e.g. 500 µl PBS, pH 7.2 - 7.4) and subsequently spin for 1 minute at ≈ 700 g (3000 rpm in eppendorf centrifuge). − Load the sample to the affinity matrix spin column and incubate for 30 - 60 min (e.g. on a spinning wheel). − Binding conditions for samples are buffers around pH 6-8 with NaCl up to 150 mM. Ultimately, suitable binding conditions need to be determined empirically. Binding of the target molecule can also be performed in the presence of Urea (up to 8.0 M) and Guanidine HCl (up to 1.0 M). − Collect flow-through in a 2 ml collection tube: spin for 1 min at 700 g − Wash the resin 2 - 4 times as described above − Elute the captured protein of interest by adding 200 - 400 µl elution buffer (see Product Info table) and collect elution fractions by spinning. − The affinity matrix can be stripped with e.g. 0.1 M Glycine pH 2.

5.9.5.4. Purification of Alpha-crystallin

Purification of Alpha-crystallin was performed in similar way as for OGT exception was working buffer protein: 1% SDS, 20 mM HEPES pH=7.9.

5.9.6. SDS-PAGE and Western-Blot analysis

For the protein denaturation, the samples were kept at 98 °C during 10 min with 4xSDS-buffer in ration 1:3 (4xSDS-buffer: sample).

10 w % acrylamide SDS gels were prepared. Separation gel was prepeared (volume for 2 gel plates): 6.0 ml rotiphores gel 30, 4.5 ml separation gel buffer (1.5M Tris-HCl, 0.4% SDS, pH=8.8), 7.5 ml deionized water, 60 µl 10%·APS, 13.5 µl TEMED (Sigma-Aldrich).

Stacking gel (volume for 2 gel plates): 1.0 ml rotiphorese gel 30, 1.7 ml stacking gel (0.5 M Tris-HCl, 0.4% SDS, pH=6.8), 4.0 ml deionized water, 20 µl 10%·APS, 12 µl TEMED (Sigma-Aldrich).

After pouring the separating gel between two glass plates, the liquid form of the gel was covered with a small amount of isopropanol. The alcohol was removed prior to pouring the collecting gel. Typically, combs with 15 teeth were used. Gels were transferred into the electrophoresis chamber filled with SDS-PAGE running buffer. The protein ladder and lysates were pipetted into gel bags. SDS-PAGE was performed at 30 mA or 50 mA for approximately 60 min until bromophenol blue and excess fluorescence dyes left the gel.

A nitrocellulose membrane (0.2 µm (Bio-Rad)) was placed on filter papers (thick blot paper from Bio-Rad Laboratories, Inc). The gel was transferred to the top of the nitrocellulose membrane and covered with another piece of the filter paper. This “sandwich” was caged in a gel holder cassette

and installed in the electrophoresis chamber (mini Trans-Blot Cell for Western Blotting (Bio-Rad)) filled with ice-cold Western Blot running buffer (25 mM Tris pH 8.3, 192 mM glycerol, 20 V-% methanol). An ice pack and a stirring bar were added. Blotting was performed at 120 V for 60 min under constant stirring. Afterwards, membranes were laid in PBS-T buffer. For immunoblotting, membranes were covered with 5 w % milk powder in PBS-T and shaken for 60 min. Two quick washing steps with PBS buffer contains the non-ionic detergent Tween^{TM} 20 (PBS-T) removed residual milk. Membranes were sealed with the desired primary antibody in welding foil and incubated at 4 °C overnight or at room temperature for 2 h under constant shaking. Most primary antibody solutions were recycled and stored at 4 °C. Membranes were washed three times with PBS-T for 10 min each. The secondary antibody was diluted in 5 w % milk powder in PBS-T. Approximately 25 mL of this solution were added to each membrane. Membranes were subsequently incubated at room temperature for 60 min. Secondary antibody solutions were not reused.

Membrane was washed 3 times for 10 min each with PBS-T. ECL western blotting substrates were combined and pipetted on top of membranes. Equal loading was assessed by incubation of membranes in Ponceau S solution for some minutes. To remove excess Ponceau S, membranes were quickly washed with PBS-T or MQ water.

5.9.7. Glycoprotein Staining

For staining glycosylated proteins the sensitive colorimetric “Pierce Glycoprotein Staining Kit” from ThermoScientific was used.[199] The principle of work of the kit is based on the periodic acid-Schiff (PAS) method in polyacrylamide gels or nitrocellulose membranes.

The reaction of periodic acid oxidizes the vicinal diols in these sugars, usually breaking up the bond between two adjacent carbons not involved in the glycosidic linkage or ring closure in the ring of the monosaccharide units that are parts of the long polysaccharides, and creating a pair of aldehydes at the two free tips of each broken monosaccharide ring. The oxidation condition should be sufficiently regulated so as to not oxidize the aldehydes further. These aldehydes then react with the Schiff reagent to give a purple-magenta colour.[200]

The polyacrylamide gels for the glycoprotein sugar moieties visualisation were used. The acrylamide gels were prepared in the regular way as described above. Kit contains positive and negative control standards. Horseradish Peroxidase was a positive standard and Soybean Trypsin

Inhibitor used as a negative standard with a final concentrations 1 mg/ml for the SDS-PAGE injections.

Staining of SDS-PAGE (all steps were performed under a fume hood):

1. After electrophoresis, immerse the gel completely in 100 mL of 50% methanol and fix for 30 minutes;

2. Immerse the gel in 100 mL of 3% acetic acid, then wash with gentle agitation for 10 minutes. Repeat this step once. (Note: this step can be a stopping point. The gel can be left in water overnight at 4°C.)

3. Transfer the gel to 25 mL of Oxidizing Solution and gently agitate for 15 minutes.

4. Wash the gel by gently agitating with 100 mL of 3% acetic acid for 5 minutes. Repeat this step two more times.

5. Transfer the gel to 25 mL of Glycoprotein Stain and gently agitate for 15 minutes. Note: If crystals form in the Glycoprotein Stain, centrifuge the solution, and use the supernatant. Do not heat to dissolve crystals. Only use stain after crystals have been removed.

6. Transfer the gel to 25 mL of Reducing Solution and gently agitate for 5 minutes.

7. Wash the gel extensively with 3% acetic acid and then with ultrapure water. Glycoproteins appear as magenta bands. Store gel in 3% acetic acid.

5.9.8. In vitro O-GlcNacylation

Protocol №1. Labelling buffer contained 25 mM Tris pH 7.5, 10 mM of MnCl2 and 150 mM NaCl. For OGT pretreatment, a sample of 30 mL cellular extract was mixed with 10 mL labelling buffer supplemented with 25 nM UDP-GlcNAc and 1 mg of OGT and incubated at 37°C for 20 minutes. For labelling with OGT, a mixture of 4-10 mg of protein, 5 nM of UDP-GlcNAz and 1 mg of OGT, in 50 mL of labelling buffer was incubated at 37°C for 30 minutes.[51]

Protocol №2. 2 μg of recombinant C/EBPβ was incubated with 1 μg of recombinant OGT in buffer containing 50 mM sodium cacodylate (pH 6.5), 1 mg/ml bovine serum albumin, 35 mM NaF, 2 μM 2-acetamido-2-deoxy-D-gluconohydroximolacetone, 2.5 mM AMP, and 1 mM UDP-GlcNAz at 25 °C for 60 min.

5.9.9. O-GlcNAc enzymatic labelling

The Click-iT™ O-GlcNAc Enzymatic Labeling System from ThermoScience was used for the labelling of already existed O-GlcNAc modifications in the protein structure. This system provides a highly sensitive and efficient method for the in vitro modification of O-GlcNAc modified proteins. Proteins are enzymatically labeled utilizing the permissive mutant β-1,4-galactosyltransferase (Gal-T1 (Y289L) which transfers azido-modified galactose (GalNAz) from UDP-GalNAz to O-GlcNAc residues on the target proteins.[185,201]

Additional materials were required: 1% SDS in 20 mM HEPES buffer, pH 7.9 10 mM HEPES buffer, pH 7.9 1% SDS in 50 mM Tris-HCl, pH 8.0 Methanol Chloroform 18 megaOhm purified water.

Preparing the 0.5mM UDP-GalNAz stock solution:

To prepare 0.5mM solution of UDP-GalNAz dissolved 50 µg of UDP-GalNAz in 144 µl of 10 mM HEPES buffer, pH=7.9 and mixed well to ensure complete reconstitution. The solution should be on ice during usage time and any used portion has to be stored at -80˚C.

Preparing the 5 µg/µl Alpha-crystallin stock solution: 250 µg of α-Crystallin has to be dissolved in 50 µl 1% SDS, 20 mM HEPES pH=7.9 and mixed well. Any used portion has to be stored at -20˚C.

Preparing the sample: protein aliquot of 80–200 µg was attached to the purification matrix and placed into a 1.5 mL microcentrifuge tube. The volume of the sample was brought up to 200 µL, if necessary, with 1% SDS, 20 mM HEPES pH 7.9.

Enzymatic labelling of the protein:

1. In a separate 1.5 mL tube, 4 µL of α-crystallin control protein were pipetted.
2. The test reaction and the positive control reaction were set up according to the instructions below. It is important to add the various reaction components to each tube in the correct order and perform any necessary mixing steps as specified.
3. To each tube the following components in order were added: the 18 megaOhm water, followed by the labelling buffer, and the MnCl2 as described in Table 4.
4. Each sample tube was vortexed briefly to mix, and then briefly centrifuged to collect the contents of the tube.
5. UDP-GalNAz was added to each sample and pipetted up and down to mix.
6. A volume of 50 µL was removed from the test reaction and pipetted into a separate 1.5 mL micro centrifuge tube for use as an unlabelled (negative) control.

7. Gal-T1 (Y289L) enzyme (Component B) was added to each sample. Each sample was pipetted up and down to mix.

8. All reaction mixtures including the unlabelled control(s) were incubated at 4˚C overnight (14 to 24 hours).

9. Samples were stored at –20˚C until they were analysed using the Click-iT™ Protein Analysis Detection Kits.

Reaction components	**Test reaction**	**Positive control reaction**
Protein of interest: 2-5 µg/µl in 1% SDS, 20 mM HEPES pH 7.9	40 µl	–
18 megaOhm water	49 µl	4.5 µl
Labelling buffer	80 µl	8 µl
MnCl2, 100 mM	11 µl	1.5 µl
UDP-GalNAz	10 µl	1 µl
50 µl of prepared mixture have to separate for the negative control		
Gal-T1 (Y289L)	7.5 µl (to the remainder of the reaction)	1 µl
Mix after the addition of the components		

Table 4. Protocol of the volumes and concentrations for Click-IT enzymatic labelling reactions. Taken from the protocol of ThermoFischerScientific applied to the kit.

Fluorescence SDS-PAGE was imaged using the Odyssey imaging system. The general scheme of experiment represented on the Figure 84.

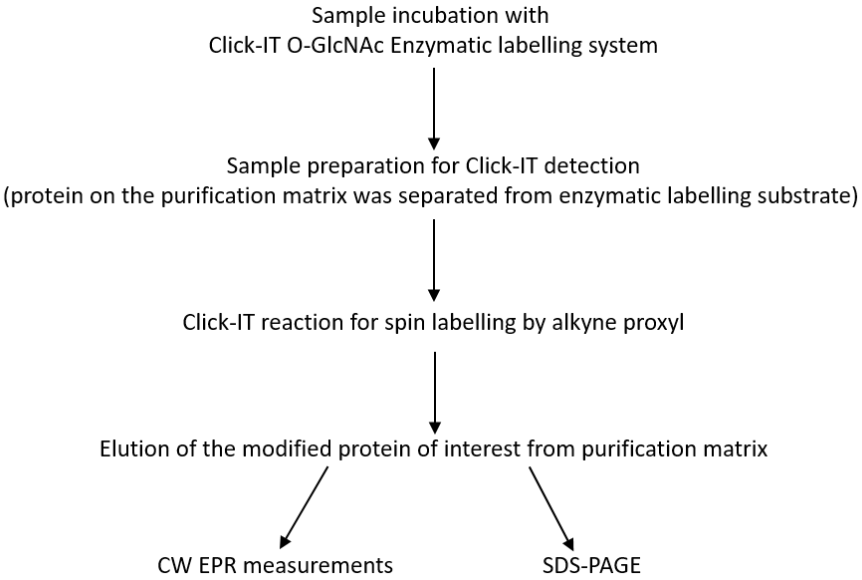


Figure 84. Schematic representation of the Click-IT O-GlcNAc enzymatic labelling and its detection by EPR spectroscopy.

5.9.10. Spin labelling

5.9.10.1. *In vitro* CuAAC spin labelling reaction

For spin labeling, the reaction mixture was prepared with 1 mM $CuSO_4$, 1,25 mM THPTA, 5mM sodium ascorbate and 0.5 mM of the Alkyne Proxyl spin label. Per one EPR sample, 1.7 µl of freshly prepared reaction mixture was added to 100 µl of 50 µM post-translationally modified protein (PTM protein) in working buffer, vortexed briefly, and incubated at room temperature while shaking in a thermomixer at 400 rpm for 60 min. After completion of the CuAAC reaction for spin labels, the excess of the label was removed by multiple cycles of dilution using a dilution buffer followed by centrifugation in molecular weight cut-off membrane (MWCO) 3kDa. After each filtering the presence of spin labels in filtrate was controlled by CW EPR measurement.

To perform the CuAAC labelling reaction *in vitro*, copper(II)-sulfate (CuSO4) and the ligand 2-(4-((bis((1-(tert-butyl)-1H-1,2,3-triazol-4-yl)methyl)amino)methyl)-1H-1,2,3-triazol-1-yl) acetic acid (BTTAA) were premixed in water in a 1:3 ratio. Ascorbic acid was added in a 1:1 ratio to copper to reduce Cu(II) ions into the catalytically active Cu(I) species. The labelling reagents were then diluted with PBS buffer pH 7.5 and added to the solution of purified protein. Finally, the spin labelling

reagent was added from concentrated stock solutions (100 mM proxyl label (L) in DMSO), resulting in a final concentration of 1 mM copper(II)-sulfate, 3 mM BTTAA, 1 mM sodium ascorbate, 50 µM protein, and 1 mM spin label. The CuAAC reaction took place at 25°C and 800 rpm in an Eppendorf ThermoMixer C for 1-2 h (1 h in the case of proxyl label, 2 h for the remaining spin label). To remove excess reagents, samples were loaded onto spin desalting columns (Zeba TM Spin Desalting column, 7k MWCO, 2 mL column material, Thermo Fisher Scientific). Afterwards, additional washing steps were performed *via* ultrafiltration in centrifugal filter units (Amicon Ultra-0.5 mL Centrifugal Filters, 3.5 K MWCO, Merck; 15 min, 12 000 rpm, 4°C) to remove excess reagents and concentrate the protein sample in the process: Protein samples were washed 4x with 400 µL PBS buffer containing 1 mM EDTA to remove remaining copper ions. Depending on the planned experiment, the buffer was then exchanged to the MQ water (samples prepared for MS measurements), 1.25x PBS pH 7.5 (samples prepared for CW EPR measurements), or deuterated 1.25x PBS (samples prepared for DEER measurements) by additional washing steps.

5.9.10.2. *In vitro* strain-promoted inverse-electron-demand Diels–Alder cycloaddition (DA_{inv}) spin labelling reaction

The labelling protocol was based on a reaction mixture of 200 µL of 40 µM protein (in PBS, pH 7.4) containing a 10-fold molar excess of PaNDA spin label (from a 10 mM stock solution in DMSO, which was stored at -20 °C). The mixture was incubated for 30 min at 20 °C and 0 rpm in an Eppendorf ThermoMixer C. The excess reactants were then removed by using HisPur Ni-NTA resin (Thermo Fisher Scientific). 120 µL of bead slurry were washed three times with MQ in a centrifuge tubebefore being incubated together with the labelling mixture for 30 min at 4 °C under constant rotation. In a centrifuge column (Pierce Centrifuge Columns, 0.8 mL, Thermo Fisher Scientific) the beads with adherent proteins were washed approx. ten times with washing buffer (50 mM NaH2PO4, 300 mM NaCl, pH 8). For elution of the spin-labelled proteins, the beads were incubated for 20 min with 60 µL of washing buffer containing 500 mM imidazole, and then proteins were collected by centrifugation. All washing steps and elution were performed in a benchtop centrifuge. The proteins were also incubated with the PaNDA spin label as described above to exclude unspecific labelling.

5.9.10.2.1. Deprotection of the PaNDA spin label by irradiation

A typical sample of 30 µL was filled into a glass capillary (HIRSCHMANN® ringcaps®; inner diameter 1.02 mm) and sealed with tube sealing compound (Kimble Cha-Seal) on one end. The capillary was placed on a table and covered with a 302 nm handheld UV lamp (UVLM-28 EL Series UV Lamp, 8 Watt, 302/365 nm, analytikjena). Samples were irradiated for 2 min (or longer, if indicated; Figure 70). The sample in the capillary was directly used for subsequent EPR spectrometric measurements.

5.9.11. In cell fluorophore labelling of PTM proteins

For microscopic controls, cell seedling was performed on the microscopy chambers (8-well ibiTreat µ-Slides). The chambers were pre-coated with Poly-L-lysine 20V-% solution in MQ water at 37 °C for 60 min or at 4 °C overnight. After that, the Poly-L-lysine coating was aspirated and each well was washed with DPBS. Around 30 000-35 000 cells were seeded per well in 200 µl of cell culture medium. Cells were incubated at 37 °C and 5 % CO_2.

As soon as cells reached 70% - 80% of growing, transient transfection of HEK293T cells was performed by calcium phosphate co-precipitation to express the cells with a protein of interest. For each well of microscopy chambers added 0,5 µg of plasmid DNA were mixed with required amount of MQ water, HBS buffer, and $CaCl_2$ solution while vortexing as it was described above. 8-10 hours after transfection washed cells with DPBS and gently added new culture media and incubated at 37°C and 5 % CO_2.

Before performing the CuAAC, cells were fixed and permeabilized. Cells were washed twice with DPBS and fixed with 4 w-% PFA in DPBS for 20 min. Afterwards, they were washed with DPBS and with 50 mM NH4Cl in DPBS for 2 min. Following another DPBS washing step, cells were permeabilized with 0.5 V-% Triton X-100 in DPBS for 15 min. Cells were washed once with DPBS, treated with 3 w-% BSA in DPBS for 2 min, and washed with DPBS. A CuAAC reaction mixture containing 1 mM CuSO4, 1.25 mM THPTA, 5 mM Na-ascorbate, and 10 µM 5-Carboxytetramethylrhodamine (5-TAMRA) in DPBS was freshly prepared from stock solutions in MQ water. The compounds for the CuAAC reaction mixture were added one after the other to ensure proper reaction conditions. Cells were incubated with this mixture at 37 °C in the dark for 15 min followed by two DPBS washing steps. Finally, they were covered with DPBS and stored at 4 °C in the dark. The microscopy measurements were taken within a day. Confocal fluorescence microscopy was performed on a Zeiss CellObserver microscope equipped with Yokogawa CSU-X1

Spinning Disk Unit using an oil immersion objective (40 x 1.3 Plan-Apochromat, Zeiss Microsystems), PvCamEvolve and the ZEN 2.6 software. For experiments with fixed cells the temperature was kept at room temperature within the microscope box. 5-TAMRA was excited at 561 nm (diode pumped solid state (DPSS) laser). Its fluorescence was detected with a photomultiplier tube (PMT) between 561 nm and 700 nm. Transmission images were acquired with a transmitted light detector. Images were taken with the following parameters: scanning speed: 100 lines s^{-1}, pixel size: 170.64 x 170.67 microns (512 x 512 pixels), image size: 1024x1024 pixels (70.3 µm x 70.3 µm), bit depth: 16 bit. ImageJ software was used to processed the images.

5.10. Spectroscopic methods

5.10.1. Bio photometry

The concentration of the purified protein was determined via light absorption at 280 nm with an Eppendorf BioPhotometer texttextsuperscript® D30 in an Eppendorf µCuvette® G1.0.

5.10.2. EPR spectroscopy

5.10.2.1. CW EPR spectroscopy

CW EPR spectra of nitroxides were recorded at room temperature, at X-band microwave frequencies (9.645 GHz) with the table-top spectrometer EMX-Nano from Bruker, equipped with a cylindric cavity mode TM110. Typically, 40 µL sample volume was filled into a glass capillary (HIRSCHMANN® ringcaps®; inner diameter 1.02 mm) which was sealed with Hemato-Seal™ capillary tube sealant (Fisherbrand™). Spectra were recorded with modulation amplitude of 1 G and microwave attenuation of 15 dB corresponding to a power of 3.162 mW (to avoid saturation or overmodulation effects), sweep width of 200 G, modulation frequency of 100 kHz, and sweep time of 30 – 60 s. For reduction-insensitive samples, spectra were averaged over several scans to improve the signal-to-noise ratio.

5.10.2.1.1. Quantitative determination of the spin concentrations

Quantitative determination of the spin concentration was performed with the built-in EMXnano reference-free spin counting module (Xenon software, Bruker) and was used to determine the labelling efficiency by calculating the ratio of the spin concentration to the product of the protein concentration and the number of labelling sites per protein molecule.

5.10.2.1.2. Rapid Scan EPR spectroscopy

RS spectra were recorded on an Elexsys 500 spectrometer (Bruker) equipped with the Rapid-Scan Accessory (Bruker) at X-band frequency (9.426 GHz). Sinusoidal rapid magnetic-field scans at a frequency of 20 kHz with a scan width of 200 G were applied using a 1D field experiment. The center field was set to 3364 G, the attenuation to 20 dB (2 mW incident microwave power), and the background correction function TwoTone (with reduction factor 5) included in the software was chosen. To increase the signal-to-noise ratio, the measurement time was set to 60 s resulting in averaging of 2,405,296 scans (65008 on board averages, 37 off-board averages).

5.10.2.2. DEER experiment

The samples were transferred into 3 mm outer diameter quartz tubes (Fused quartz tubing, Technical Glass Products) and flash frozen in liquid nitrogen. DEER experiments were performed in a commercial Q-band resonator (ER5106QT2, Bruker) at Q-band frequency (34 GHz) and 50 K with an Elexsys E580 spectrometer (Bruker), equipped with an arbitrary waveform generator (AWG) unit (Bruker) and a 150 W traveling-wave tube (TWT) amplifier (Applied Systems Engineering). The temperature was controlled by a cryogen-free helium recirculation system with a ColdEdge cryocooler (CE-FLEX-4K_0100, Bruker), a F-70H helium compressor (SHI cryogenics), and a Mercury ITC temperature controller (Oxford Instruments). Pulsed experiments were performed with the standard four-pulse DEER sequence ($(\pi/2)_{obs} – \tau_1 – \pi_{obs} – (\tau_1 + t) – \pi_{pump} – (\tau_2 - t) – \pi_{obs} – \tau_2 –$ echo) with rectangular pump and observer pulses. The pump frequency was set to 34 GHz. A frequency offset of 70 MHz was chosen for the observer pulse. The length of the pump and observer pulses were chosen individually for each experiment. Nuclear modulation averaging was done by increasing τ_1 in 8 steps by 16 ns each. Artefacts caused by interfering echoes were removed with an 8-step on pump {(x) x [xp] x} phase cycling. Processing and analysis of raw DEER data was

performed with Matlab R2019b (The MathWorks, Inc.), the toolbox EasySpin 6.0.0165 and DeerAnalysis 2016.[178,106]

5.11. Spin labelling efficiency identification (TLC/MS)

The reaction between Ac_4GlcNAz and Alkyne Proxyl spin label was verified by mass spectroscopy measurements with Advion's expression Compact Mass Spectrometer (CMS) on the thin-layer chromatography (TLC) plates (TLC/MS technique). Aluminium plates (5*5 cm, Merck, Germany) with silica gel 60 F_{254} were used for TLC chromatography. 2-µL aliquot of the crude reaction mixture from the stirred round-bottom flask were spotted onto the TLC plate at a position 1 cm above the bottom of the plate. The spotted TLC plate was placed into the developing chamber and sealed with the glass lid. Separation was concluded when the development solvent front reached a position 1 cm from the top of the TLC plate. The TLC plate was then transferred to an oven and allowed to dry at 80 °C for 5 min. The separated "spots" were observed on the TLC plate under UV light at 254 nm.[7]

5.12. Data analysis and spectra simulation

Gel or membrane images were read out using the ChemiDocTMTouch Imaging System. All images were processed and analyzed using the software Image Lab, AdobePhotoshop, and Microsoft Office 2010.

All recorded EPR spectra were processed using MatLab2019B (MatWorks, Inc.) which included, if necessary, a baseline correction, adjusting the center field position of the spectra to correct for small deviations in the microwave frequency between different samples, smoothing data with a Savitzky-Golay filter function as well as normalization of the intensity.

For the CW-EPR line shape simulations the EasySpin toolbox (5.0.16) was used. Simulations were performed using EasySpin function "*chili*" and the following parameters: *A* [18 100] and $g = [g_x\ g_y\ g_z] = [2.0091\ 2.0069\ 2.0030]$ and linewidth lw [0.15 0] (for simulations in Figure 60 A and B) and linewidth lw [0.0 0.2] (for simulations in Figure 60 C and D). Systematic errors of the magnetic field measurements were compensated by a frequency shift.

5.13. Mass spectrometry analysis

5.13.1. In-gel digestion procedure of Coomassie stained proteins

The gel pieces were processed for mass spectrometric analysis as described before (Shevchenko et al., 2006).[202] Briefly, cysteines were reduced with DTT and alkylated using chloroacetamide (Nielsen et al., 2008) (CAA).[203] Finally, proteins were digested overnight with trypsin and desalted by C18 Spin Tips (Thermo Scientific) prior to mass spectrometric analysis.

5.13.2. Mass spectrometry measurements

All digested samples were analysed by reversed phase liquid chromatography nanospray tandem mass spectrometry (LC-MS/MS) using a Q Exactive HF mass spectrometer (Thermo Fisher) and an Easy-nLC 1200 (Thermo Fisher). The dimensions of the reversed-phase LC column were 2 µm, 100 Å pore size C18 resin in a 50 µm i.d. × 15 cm long piece of fused silica capillary (Acclaim PepMap100, Thermo Scientific). After sample injection, the column was washed for 5 min with 94% mobile phase A (0.1% formic acid) and 6% mobile phase B (0.1% formic acid in acetonitrile/water (80:20, v/v)), and peptides were eluted using a linear gradient of 6% mobile phase B to 36% mobile phase B in 45 min, then to 100% B in an additional 5 min. The Q Exactive HF mass spectrometer was operated in a data dependent mode in which each full MS scan (resolution 120 000 at m/z 200) was followed by ten MS/MS scans where the ten most abundant molecular ions were dynamically selected and fragmented by higher energy collision dissociation (HCD) using a normalized collision energy of 28%. Dynamic exclusion was allowed.

Tandem mass spectra were analysed using Byonic (Protein Metrics demo version) and pGlyco3.

5.14. DEER measurements for exploring the binding site of glycomacromolecules on the structure of human norovirus capsid protein.

Compound	Conc. of ligand (μM)	Conc. of NoroP-dimer (μM)	V(deer sample)	Pump frequency			Observer frequency	
				per Ange(ns)	EDFS B0(G)	V(GHz)	V_{obs} (GHz)	Nutation $_{obs}$ (ns)
Fucose ligand	90	–	12.5	14/28	12042.0	33.839	33.839	20/40
Fucose+Noro P-dimer №1	300	75	12.5	12/24	12079.6	33.995	33.945	14/28
Fucose+Noro P-dimer №1	75	75	12.5	28	12008.0	33.795	33.745	40
Galactose ligand	300	–	12.5	6/12	12066.8	33.959	33.909	18/36
Galactose+ Noro P-dimer	300	75	12.5	14/28	12019.1	33.802	33.752	30/60

Table 4. Samples preparation for DEER measurements.

Samples for the CW EPR measurements containing spin labelled Fucose ligand (100μM) in presence and absence of NORO P-dimer (100 μM), were prepared in the 1M sodium phosphate buffer (pH 7.3). Upon incubation (from 0 to 60 min), samples were loaded into quartz capillaries (50 μl Hirschmann ringcaps) and CW EPR experiments were carried out at X-band using a Bruker EMXnano spectrometer at room temperature. Measurement parameters were adjusted in a way that the spectral line shape was not distorted by overmodulation or saturation effects.

For DEER experiments, samples containing spin labelled Fucose ligand in the absence and presence of the NORO P-dimer were prepared in a deuterated sodium phosphate buffer solution with 20% (v/v) of deuterated glycerol. The samples were transferred into quartz tubes (ER 221 TUB-Q-10; Bruker), shock-frozen in liquid nitrogen to trap annealed conformations and subsequently measured.

The pulse EPR experiments were performed using Bruker ELEXSYS E580 spectrometer equipped with an EN5107D2 Q-band EPR probe head (both Bruker Biospin) and a 10 W solid state amplifier. A CF935 helium gas flow system (Oxford Instruments) was used for temperature control. Experiments were performed at T = 50 K. The DEER experiment was performed using a dead-time free four-pulse sequence.[100] The echo amplitude was recorded as a function of the dipolar evolutions time t. The corresponding EPR spectrum was obtained by field-swept echo acquisition and the pump and observer pulses were positioned on the global maximum and close to the most intense local

maximum (shifted by 50 MHz) of the spectrum, respectively. The pump pulse length was adjusted to obtain a flip angle of π resulting in a typical pulse length of 24 ns. The pulse separation time τ_1 was 400 ns and the dipolar evolution time τ_2 was 5000 ns. Nuclear modulation artifacts were suppressed by variation of the interpulse delay. An eight-step phase cycle was used. For a complete DEER experiment, typically 41 individual scans were recorded, subjected to phase correction individually, and subsequently summarized.

DEER data sets were analyzed using the DeerAnalysis 2016 software package for MATLAB.[103] Extraction of the dipolar evolution function was achieved by background correction with a 3-dimensional homogeneous background function. Modulation depth scaling was used to take different pump pulse lengths into account. Background-corrected data were analyzed by model-free Tikhonov regularization and a worm-like chain model.

List of abbreviations

Ac4GalNAz	Tetraacetylated N-azidoacetylgalactosamine
Ac4GalNCyoc	Tetraacetylated methylcyclopropene-tagged N-acetylgalactosamine
Ac4GlcNAc	Tetraacetylated N-acetylglucosamine
Ac4GlcNAz	Tetraacetylated N-azidoacetylglucosamine
Ac4GlcNCyoc	Tetraacetylated methylcyclopropene-tagged N-acetylglucosamine
ADP	Adenosine diphosphate
AMP	Adenosine monophosphate
ATP	Adenosine triphosphate
Az	Azide
BSA	Bovine serum albumin
BTTAA	2-(4-((Bis((1-(tert-butyl)-1H-1,2,3-triazol-4-yl)methyl)amino)methyl)-1H-1,2,3-triazol-1-yl)acetic acid
BTTES	2-(4-((Bis((1-(tert-butyl)-1H-1,2,3-triazol-4-yl)methyl)amino)methyl)-1H-1,2,3-triazol-1-yl)ethyl hydrogen sulfate
CoA	Coenzyme A
CuAAC	Cu(I)-catalyzed azide-alkyne-cycloaddition
CW	Continuous wave
Cy	Cyanine
DA_{inv}	Inverse-electron-demand Diels-Alder
DAPI	4’,6-Diamidino-2-phenylindole dihydrochloride
DEER	Double electron-electron resonance
DMEM	Dublecco’s modified eagle medium
DMSO	Dimethyl sulfoxide
DNA	Deoxyribonucleic acid
DOTA	Dodecane tetraacetic acid
DPBS	Dulbecco’s phosphate buffered saline
DTT	DL-Dithiothreitol

E. coli	*Escherichia coli*
ED	electron donating substituent,
EDTA	Ethylenediamine-N,N,N',N'-tetraacetic acid
ESI	Electrospray ionization
EPR	Electron paramagnetic resonance
ER	Endoplasmic Reticulum
EW	electron withdrawing substituent
FBS	Fetal bovine serum
FLIM	Fluorescence lifetime imaging
FRET	Förster resonance energy transfer
Fuc	Fucose
GalNAc	N-acetylgalactosamine
GFAT	glutamine-fructose-6-phosphate transaminase
GFP	Green fluorescent protein
GlcNAc	N-acetylglucosamine
HEK	Human embryonic kidney cell line
HeLa	Cervical cancer cells taken from Henrietta Lacks
HEPES	2-[4-(2-Hydroxyethyl)-1-piperazinethanesulfonic acid
HOMO	Highest occupied molecular orbital
IPTG	Isopropyl-beta-D-thiogalactopyranoside
LUMO	Lowest unoccupied molecular orbital
MGE	Metabolic glycoengineering
MOE	Metabolic oligosaccharide engineering
MS	Mass spectrometry
MTSL	Methanethiosulfonate spin label
MQ	Milli-Q
Na4P	Sodium 4-pentynoate
ncAA	non-canonical amino acids

OGA	O-GlcNAcase
OGT	O-GlcNAc transferase
PaNDA	photoactivatable nitroxide for DA_{inv} reaction
p53	Tumor protein p53
PBS	Phosphate buffered saline
PCR	Polymerase chain reaction
PTM	Post-translational modification
RNA	Ribonucleic acid
SDS-PAGE	Sodium dodecylsulfate-polyacrylamide gelelectrophoresis
SL	Spin label
SDSL	Site-directed spin labelling
TAMRA	Tetramethylrhodamine
TBTA	Tris((1-benzyl-1H-1,2,3-triazol-4-yl)methyl)amine
TCEP	Tris(2-carboxyethyl)phosphine
TEMED	Tetramethylethylenediamine
TFA	Trifluoracetic acid
THPTA	Tris(3-hydroxypropyltriazolylmethyl)amine
TLC	Thin-layer chromatography
UDP	Uridine diphosphate
V-%	Volume percentage
WT	Wild type

Appendix

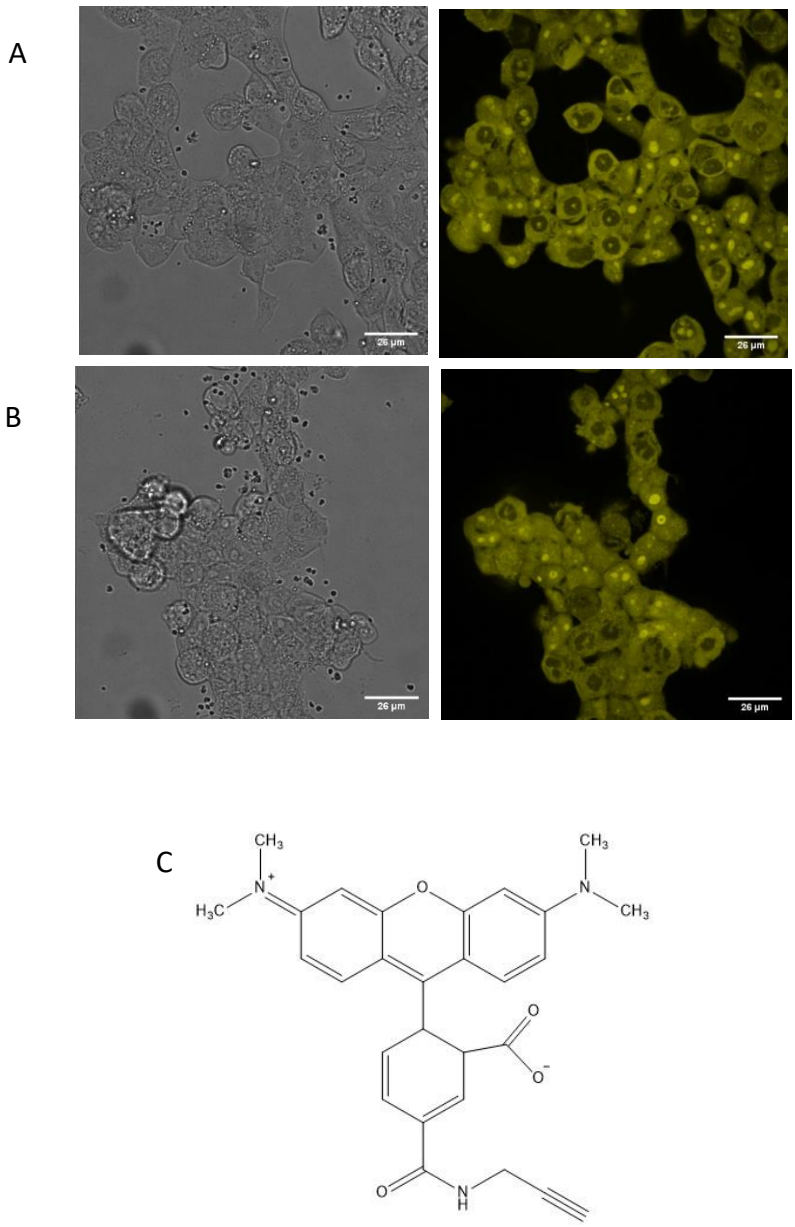


Figure A1. Labelling of metabolically engineered proteins in vivo. HEK 293T cells were grown for 48 h with 100 µM Ac4GlcNAz 1 (A, B), and subsequently incubated with TAMRA fluorophore with CuAAC reaction substrate (1 mM, 1 h, 37 °C). Nuclei were stained with Hoechst33342. Scale bar: 30 µm. (C) TAMRA fluorophore reagent used in CuAAC labelling reaction.

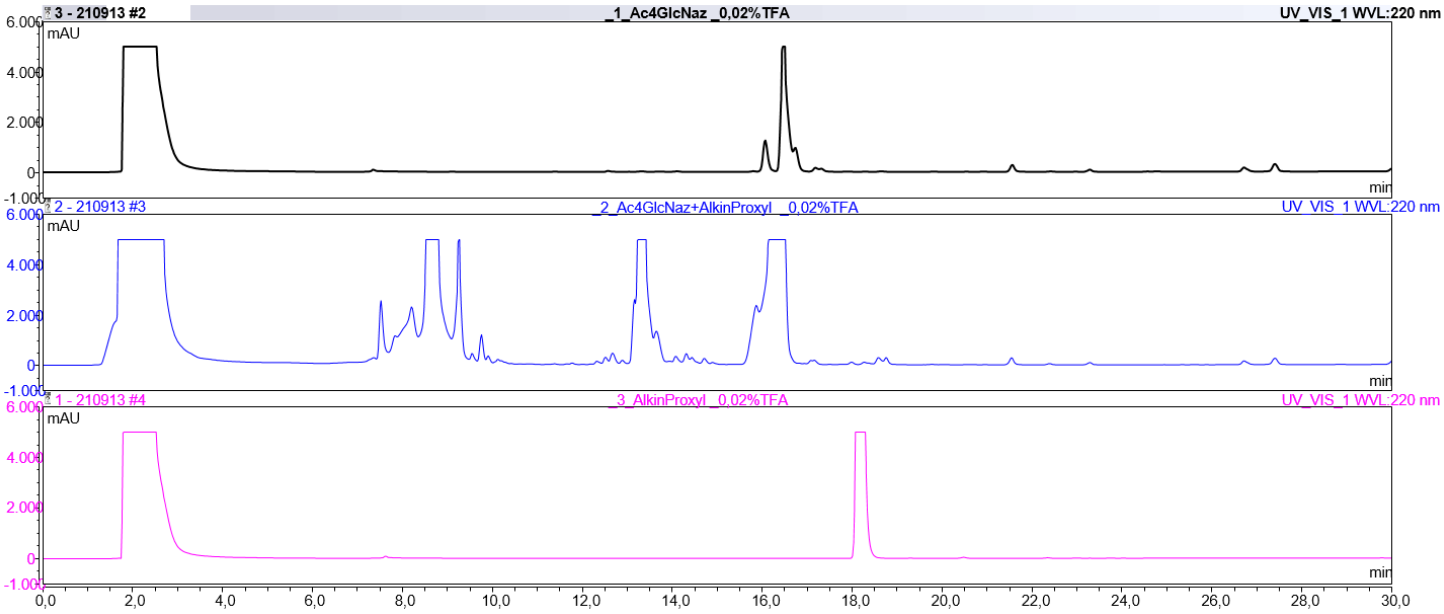


Figure A2. The following figure provides one presentation of the HPLC data that demonstrates signals of $Ac_4GlcNAz$ sugar (black), free Alkyne-Proxyl spin label (purple), $Ac_4GlcNAz$ spin labelled by Alkyne-Proxyl in a CuAAC reaction for 1h (blue) at λ = 220 nm.

```
Seq_1   420    AAACTTTAATACGACTCACTATAGGGGCCGCCACCAAGCTTGGTACCATGCACCATCACC   479
                                                              ||||||||||||
Seq_2    24    --------------------------------------------------TGCACCATCACC    35

Seq_1   480    ACCATCATCACCACCATCACGGTGGAGGCGGTAGCGACGTGACCATCCAGCACCCCTGGT   539
               ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Seq_2    36    ACCATCATCACCACCATCACGGTGGAGGCGGTAGCGACGTGACCATCCAGCACCCCTGGT    95

Seq_1   540    TCAAGCGCACCCTGGGGCCCTTCTACCCCAGCCGGCTGTTCGACCAGTTTTTCGGCGAGG   599
               ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Seq_2    96    TCAAGCGCACCCTGGGGCCCTTCTACCCCAGCCGGCTGTTCGACCAGTTTTTCGGCGAGG   155

Seq_1   600    GCCTTTTTGAGTATGACCTGCTGCCCTTCCTGTCGTCCACCATCAGCCCCTACTACCGCC   659
               ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Seq_2   156    GCCTTTTTGAGTATGACCTGCTGCCCTTCCTGTCGTCCACCATCAGCCCCTACTACCGCC   215

Seq_1   660    AGTCCCTCTTCCGCACCGTGCTGGACTCCGGCATCTCTGAGGTTCGATCCGACCGGGACA   719
               ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Seq_2   216    AGTCCCTCTTCCGCACCGTGCTGGACTCCGGCATCTCTGAGGTTCGATCCGACCGGGACA   275

Seq_1   720    AGTTCGTCATCTTCCTCGATGTGAAGCACTTCTCCCCGGAGGACCTCACCGTGAAGGTGC   779
               ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Seq_2   276    AGTTCGTCATCTTCCTCGATGTGAAGCACTTCTCCCCGGAGGACCTCACCGTGAAGGTGC   335

Seq_1   780    AGGACGACTTTGTGGAGATCCACGGAAAGCACAACGAGCGCCAGGACGACCACGGCTACA   839
               ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Seq_2   336    AGGACGACTTTGTGGAGATCCACGGAAAGCACAACGAGCGCCAGGACGACCACGGCTACA   395

Seq_1   840    TTTCCCGTGAGTTCCACCGCCGCTACCGCCTGCCGTCCAACGTGGACCAGTCGGCCCTCT   899
               ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Seq_2   396    TTTCCCGTGAGTTCCACCGCCGCTACCGCCTGCCGTCCAACGTGGACCAGTCGGCCCTCT   455
```

```
Seq_1   900   CTTGCTCCCTGTCTGCCGATGGCATGCTGACCTTCTGTGGCCCCAAGATCCAGACTGGCC  959
              ||||||||||||||||||||||||||||||||||  ||||||||||||||||||||||
Seq_2   456   CTTGCTCCCTGTCTGCCGATGGCATGCTGACCTTCGCTGGCCCCAAGATCCAGACTGGCC  515

Seq_1   960   TGGATGCCACCCACGCCGAGCGAGCCATCCCCGTGTCGCGGGAGGAGAAGCCCACCTCGG  1019
              ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Seq_2   516   TGGATGCCACCCACGCCGAGCGAGCCATCCCCGTGTCGCGGGAGGAGAAGCCCACCTCGG  575

Seq_1   1020  CTCCCTCGTCCTGAATCTAGAGCGGCCGCCGAATTCGGGCCCGTTTAAACCCGCTGATCA  1079
              ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Seq_2   576   CTCCCTCGTCCTGAATCTAGAGCGGCCGCCGAATTCGGGCCCGTTTAAACCCGCTGATCA  635

Seq_1   1080  GCCTCGACTGTGCCTTCTAGTTGCCAGCCATCTGTTGTTTGCCCCTCCCCCGTGCCTTCC  1139
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Seq_2   636   GCCTCGACTGTGCCTTCTAGTTGCCAGCCATCTGTTGTTTGCCCCTCCCCCGTGCCTTCC  695

Seq_1   1140  TTGACCCTGGAAGGTGCCACTCCCACTGTCCTTTCCTAATAAAATGAGGAAATTGCATCG  1199
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Seq_2   696   TTGACCCTGGAAGGTGCCACTCCCACTGTCCTTTCCTAATAAAATGAGGAAATTGCATCG  755

Seq_1   1200  CATTGTCTGAGTAGGTGTCATTCTATTCTGGGGGGTGGGGTGGGGCAGGACAGCAAGGGG  1259
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Seq_2   756   CATTGTCTGAGTAGGTGTCATTCTATTCTGGGGGGTGGGGTGGGGCAGGACAGCAAGGGG  815

Seq_1   1260  GAGGATTGGGAAGACAATAGCAGGCATGCTGGGGATGCGGTGGGCTCTATGGCTTCTGAG  1319
              ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Seq_2   816   GAGGATTGGGAAGACAATAGCAGGCATGCTGGGGATGCGGTGGGCTCTATGGCTTCTGAG  875

Seq_1   1320  GCGGAAAGAACCAGCTGGGGCTCTAGGGGGTATCCCCACGCGCCCTGTAGCGGCGCATTA  1379
              ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Seq_2   876   GCGGAAAGAACCAGCTGGGGCTCTAGGGGGTATCCCCACGCGCCCTGTAGCGGCGCATTA  935

Seq_1   1380  AGCGCGGCGGGTGTGGTGGTTACGCGCAGCGTGACCGCTACACTTGCCAGCGCCCTAGCG  1439
              ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Seq_2   936   AGCGCGGCGGGTGTGGTGGTTACGCGCAGCGTGACCGCTACACTTGCCAGCGCCCTAGCG  995
```

Figure A3. Alignment of pCMV3-His-CRYAA sequence with sequence of pCMV3-His-CRYAA C141A, where:

```
CRYAA              : [515 : 1033]
Start              : [467 : 469]
Stop               : [1031 : 1033]
10xHIS-Tag [3']    : [471 : 499]
4xGly [3']         : [500 : 511]
T7promotor         : [426 : 444]
```

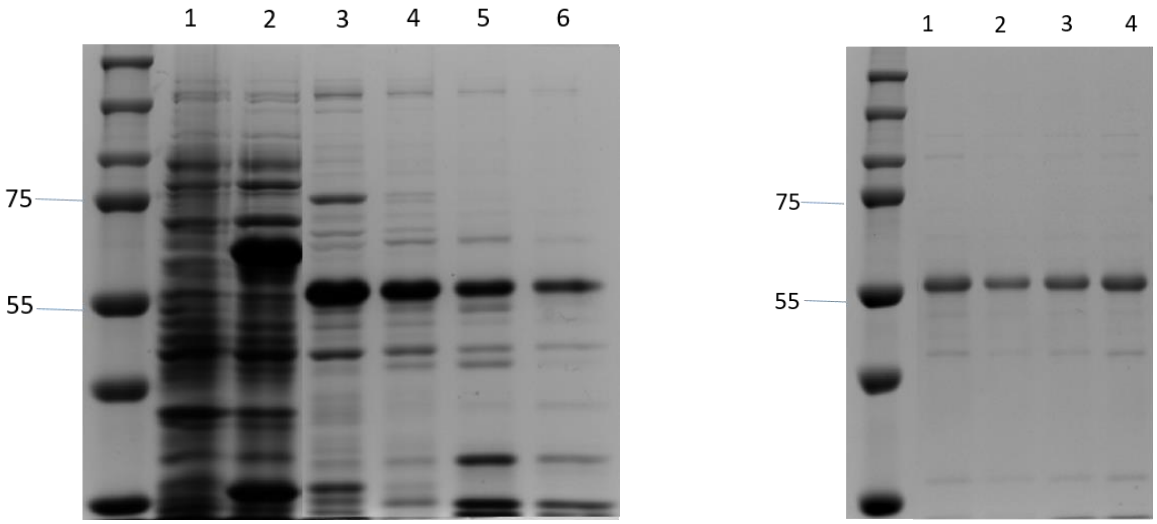


Figure A4. SDS-PAGE analysis of lysated HEK293T cells before (left image injection №1) and after expression (left image injection №2) with p53 DNA plasmid. Also the fractions of p53 protein purification by His-trap column (left image) and after an additional purification by Heparin column (right image).

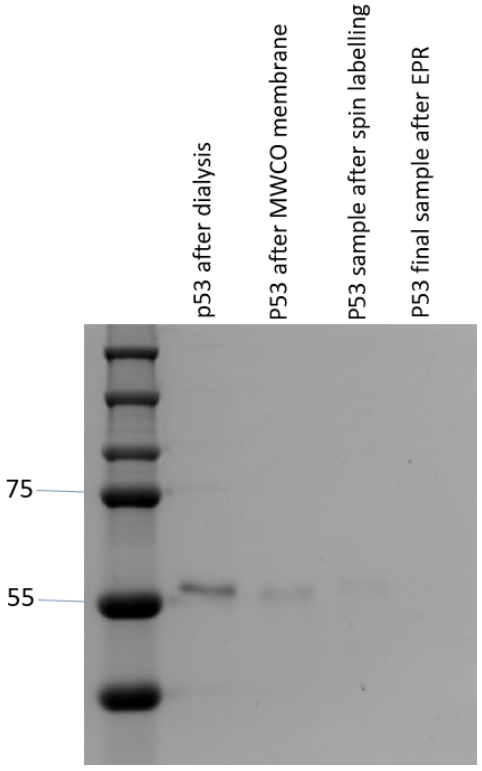


Figure A5. SDS-PAGE analysis of purified p53 protein at different steps of the sample preparation.

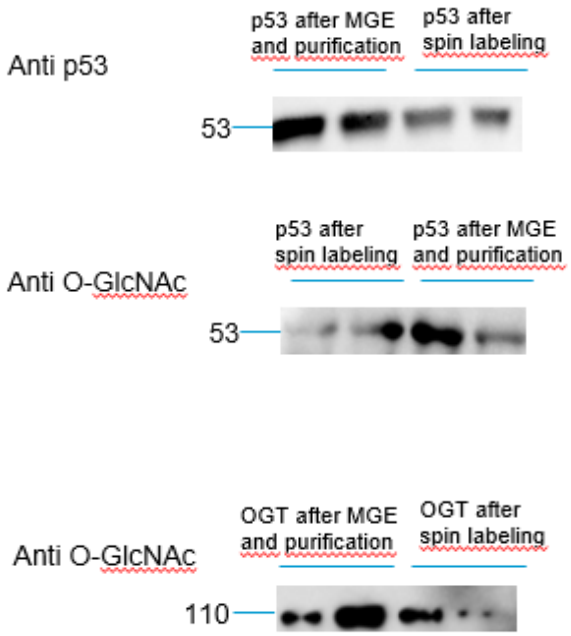


Fig.A6 WB images as a quality control of the protein O-GlcNAcylation of p53 (two first images) and OGT (last image).

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